Group schemes in *La Longue Marche*, sections 48–50: equations, corrections, and exact remaining questions

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Abstract

We compute equations, integral Groebner bases, fibres and point groups for the final modular group scheme in sections 48–50 of *La Longue Marche*. It is flat of relative dimension six and smooth exactly away from two. We prove pinning and parametrization results, correct the integral sheaf presentation, compute geometric central extensions and the vanishing manuscript Brauer class, and construct exponential counterexamples and the virtually abelian envelope. Every statement carries a proof status. Remaining comparison, extension and universality questions have proved cases and executed computations. No priority is claimed.

Conventions and scope

All rings are commutative with identity. Unless a discrete or profinite group is expressly named, a group is a group scheme and exactness means exactness of fppf sheaves. An identity on all geometric fibres does not assert an identity on a nonreduced base. We write $[a,r]=ara^{-1}r^{-1}$ and use conjugation, rather than multiplication, for the manuscript's $r(a)$. Our fixed matrices and principal notation are as follows.

Our notation	Meaning	Manuscript location
r, s	$\left(\begin{smallmatrix}0&1\\-1&1\end{smallmatrix}\right), \left(\begin{smallmatrix}0&1\\-1&0\end{smallmatrix}\right)$	ρ, σ, C3:479, (8)
$s_- = -s$	Earlier lift of the same projective σ	C3:264, 327
$L_1(z), L_0(z)$	$\left(\begin{smallmatrix}1&0\\z&1\end{smallmatrix}\right), \left(\begin{smallmatrix}1&z\\0&1\end{smallmatrix}\right)$	$\varepsilon_1(z)$; L_0, C3:414
Q_3, Q_2	$c^2+cd+d^2,\ t^2+u^2$	Determinants in C3:479, (9)
$\mathcal{M}, \mathcal{H}$	Final signed group; its quotient by SL_2	$\mathcal{M}_{\rho,\sigma}$, $\mathcal{T}_{\rho,\sigma}$; C3:399,416
$\mathcal{M}^0$	Kernel of the two sign characters	$\mathcal{M}^\circ$; C3:400; not a fibrewise identity component at 2
$\widehat{\Gamma}$	Profinite completion of a discrete group	Never identified with a congruence completion without a map

In particular $r^3=s^2=-I$, whereas their projective images have orders $3,2$. The phrase "elements of orders $3,2$ in $\mathrm{SL}_2(\mathbb{Z})$" is not used for these lifts. The earlier convention satisfies $r+s_-=\mathrm{diag}(0,1)$; the final convention satisfies $rs^{-1}=L_1(1)$ and $s^{-1}r=L_0(-1)$.

We cite the manuscript (Grothendieck, 1981) by cote, scan and autograph folio where legible, distinguishing transcription from ink. Every numbered statement appears in the results CSV. "Open" means unresolved here. Partial tasks have proved cases, executed computations and exact residuals; the report identifies any unfinished task. Verified references were opened at the recorded passages. The attached LM3, BISITE and GALOIS returns and reviews are leads to check, not authorities. We claim no priority.

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A Definitions, normalizations and the modular triple

A.1. The modular dictionary and its polynomial identities [Proved]

For the matrices fixed above, the following table defines affine group schemes over $\mathbb{Z}$. In entries involving solutions of an equation, the group law is transported through the indicated isomorphism; it is not asserted to be componentwise multiplication in (α, β).

Symbol	Definition or corrected formula	C3 scan; folio
G, SG, PG	$\mathrm{GL}_2, \mathrm{SL}_2, \mathrm{PGL}_2$	323;462
B_0, B_0'	Upper triangular invertible matrices; $b_0(m,l) = \left(\begin{smallmatrix} m & l \\ 0 & 1 \end{smallmatrix}\right)$	368;485
T_0, L_0, L_1	$D(m) = \mathrm{diag}(m,1)$; $L_0(z)$; $L_1(z)$	414;508–509
T_r	$\{cr + dI : Q_3(c,d) \in R^\times\}$	479;(9)
T_s	$\{ts + uI : Q_2(t,u) \in R^\times\}$	479;(9)
ST_r, ST_s	Kernels of the respective determinant (norm) maps	376;489
N_s'	$\{b \in G : bsb^{-1} = es,\ e^2 = 1\}$	325,479
$\mathcal{X}$	(ζ, η, μ) with determinants 1, $\mathrm{tr}(\zeta r) = 1$, $\mathrm{tr}(\eta s) = 0$, $\zeta = \eta L_1(\mu - 1)$	268,370;435,486
$\mathcal{G}_{r,s}$	$[\alpha, r] = [\beta, s]L_1(\mu - 1)$, $\alpha, \beta \in G$; $\simeq B_0' \times T_r \times T_s$	368;(1)–(2)
$\mathcal{G}^*_{r,s}$	$[\alpha, r] = e[\beta, s]L_1(\mu - 1)$, $e^2 = 1$; $\simeq B_0' \times T_r \times N_s'$	371;(9)–(10)
$S\mathcal{G}^*_{r,s}$	Previous row with $(\det\alpha)^2 = 1$, $\det\beta = 1$	373;487
$\widetilde{S}\mathcal{M}$	$B_0' \times T_r \times N_s' \times \mathrm{GL}_2$ in the coordinates of B.4	393;(43)–(44)
$\widetilde{\mathcal{M}}$	$\mathbb{G}_m \times T_r \times N_s' \times \mathrm{GL}_2$; quotient of the preceding row by L_0	393;(43)–(44)
$\mathcal{H}$	$\{(a,b) \in T_r \times N_s' : (\det b/\det a)^2 = 1\}$	375,416;509
$\mathcal{M}$	$\{(a,b,U) : \det U = \det b = e' \det a,\ (e')^2 = 1\}$	399;(13);479;(3)
$S\mathcal{M}$	$B_0' \times_{\mathbb{G}_m, \det, \chi} \mathcal{M}$	479;(4)
χ, e, e'	$\det U = \det b$; $bsb^{-1} = es$; $\det b/\det a$	480;(12bis)
$\mathcal{M}^0, \mathcal{H}^0$	Kernels of (e, e'); equal-norm triples or pairs	400,416
$\mathcal{M}(0)$	$(a,b) \mapsto (a, b, D(\det b))$	402–403,417
$\mathcal{M}[0]$	$\mathcal{M}(0)L_0 = \{(a,b,U) : U \in B_0'\}$	417–418
$\mathcal{M}(s), \mathcal{M}(r)$	$U = b$; $U = a$, the latter only over $e' = 1$	476,491
Σ	Scalar sign factors in $ST_r \times ST_s \times \mathrm{SL}_2$; the scheme is μ_2^3, whereas global discrete signs form $(C_2)^3$	401–402
I^0, J^0	$\ker(\mathcal{M}^0 \to G)$; $\ker(\mathcal{M}^0 \to PG)$	400;(17)–(18)

Proof. Solving $Ar = rA$ gives $A = cr + dI$; solving $As = sA$ gives $A = ts + uI$. Their determinants are the displayed quadratic forms. Solving $bs = esb$, with $e^2 = 1$, gives

$$b = \begin{pmatrix} u & t \\ -et & eu \end{pmatrix}, \qquad \det b = e(t^2 + u^2).$$

Thus the sign is unique, including over nonreduced rings. It is a character because conjugation is multiplicative. The subgroup N_s' is $T_s \rtimes \mu_2$: a splitting is $e \mapsto \mathrm{diag}(1, e)$, and its action on $ts + uI$ sends t to et. All remaining determinant conditions are equalities between characters and hence define closed subgroups of the indicated affine groups. Representability of inverses uses the explicitly inverted norms. The parametrization assertions are proved in B.1; the subgroup and exact-sequence assertions are proved in B.4–B.6.

The program `AB_identities.py` works over universal polynomial rings and their indicated localizations. It verifies matrix powers, both norms, commutator formulas, the parametrizing

identity and the general (v, w) normalization. It does not replace proofs of representability or exactness by finite testing. □

A.2. Exactly when a regular unipotent lift is unique [Proved]

Let R be a ring and let $U \in \mathrm{GL}_2(R)$ have trace 2, determinant 1 and a non-scalar image in every residue field. The matrices with those properties and the same image as U in $\mathrm{PGL}_2(R)$ are exactly

$$(1+\tau)U, \qquad \tau \in R, \quad 2\tau = 0, \quad \tau^2 = 0.$$

Here "same image" is equality of sections of the fppf quotient; the scalar relating two global matrix lifts is global. Consequently uniqueness on R-points is equivalent to $\{\tau \in R : 2\tau = \tau^2 = 0\} = 0$. For a scheme S the analogous statement uses $\Gamma(S, \mathcal{O}_S)$; uniqueness after every base change is equivalent to 2 being invertible on S.

Proof. If $U' = \lambda U$, determinants give $\lambda^2 = 1$ and traces give $2\lambda = 2$. With $\tau = \lambda - 1$ these equations are precisely $2\tau = \tau^2 = 0$. Conversely these equations preserve trace and determinant, and Cayley–Hamilton gives $(U' - I)^2 = 0$. They preserve regularity because $\lambda = 1$ in every residue field. The scalar is global since the kernel sheaf of $\mathrm{GL}_2 \to \mathrm{PGL}_2$ is $\mathbb{G}_m$. If 2 is invertible the assertion persists under all base changes. If a characteristic-2 point exists, base change to $k[\delta]/(\delta^2)$ supplies the nonzero $\tau = \delta$.

The ambiguity is represented by the nonflat finite group scheme

$$H = \operatorname{Spec} \mathbb{Z}[\tau]/(2\tau, \tau^2), \qquad \Delta\tau = \tau \otimes 1 + 1 \otimes \tau + \tau \otimes \tau.$$

This is exactly the manuscript's μ_2', not a finite flat replacement for it (C3:358–360, fols.480–481). Over $\mathbb{Z}/4$, the two matrices $\left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 3 & 0 \\ 3 & 3 \end{smallmatrix}\right)$ are a counterexample to unrestricted uniqueness. Both rings of cardinality 2 and 3 are fields, so cardinality 4 is minimal. The computation checks all stated equalities, not only their reductions modulo 2. □

A.3. Pinning with specified lifts, and its projective ambiguity [Proved]

Suppose $R = \mathbb{Z}$, a field, or a local ring, and let (E, R_0, S_0) be a triple in $\mathrm{GL}_2(R)$ satisfying

$$E = -R_0 S_0^{-1}, \quad \det R_0 = \det S_0 = 1, \quad \operatorname{tr} R_0 = 1, \quad \operatorname{tr} S_0 = 0.$$

Assume E is regular unipotent and its fixed line and its inverse image under S_0 are complementary. There is a basis, unique modulo a common scalar, carrying the triple to $(L_1(1), r, s_-)$. Therefore the corresponding inner identification of group schemes is unique. For the pair of projective images alone, the stabilizer of the standard pair in $\mathrm{PGL}_2(R)$ is

$$\{[I + \tau r] : 2\tau = \tau^2 = 0\}.$$

Thus projective data alone need not determine a unique pinning.

Proof. Write $N = E - I$. A regular nilpotent endomorphism of a rank-two free module has image and kernel the same invertible direct summand L_1. Locally this follows by choosing a unit entry of N and solving $N^2 = 0$; the kernel and image constructions then glue. Put $L_0 = S_0^{-1} L_1$. Complementarity implies that $N : L_0 \to L_1$ is an isomorphism. Over the listed rings both lines are free. Choose e_0 generating L_0 and put $e_1 = Ne_0$. Then $E = L_1(1)$ and S_0 has form $\left(\begin{smallmatrix} 0 & b \\ c & d \end{smallmatrix}\right)$. Its trace gives $d = 0$; from $R_0 = -ES_0$ and its trace we get $b = -1$; the determinant gives $c = 1$. Only the common scalar remains. The same proof for a rank-two vector bundle gives $V \simeq L \otimes \mathcal{O}_S^2$, hence an identification of its endomorphism algebra even when L is not trivial. For an Azumaya algebra the argument is made fppf locally and descends

because the identification just obtained is unique. This proves the assertion for specified nilpotent/root-subgroup data on a general base as well.

For projective data, a matrix representative of a conjugator satisfies $grg^{-1} = \lambda r$. Taking traces gives $\lambda = 1$, so $g = cr + dI$. Write $gs_- = es_-g$; determinant comparison gives $e^2 = 1$. The entries give $c = (e-1)d$ and $(e-1)c = 0$. Since g is invertible, d is a unit, so $\tau = e - 1$ satisfies $\tau^2 = 2\tau = 0$ and $[g] = [I + \tau r]$. Conversely these equations suffice. Over $\mathbb{Z}/4$ the matrix $\left(\begin{smallmatrix} 1 & 2 \\ 2 & 3 \end{smallmatrix}\right)$ is a nonidentity projective stabilizer; this is checked in the code. For the listed rings every projective conjugator has a matrix representative because their Picard groups vanish. On a general base the calculation describes the stabilizer sheaf fppf locally. The qualification in the margin of C3:323, fol.462, is therefore essential. The later inclusion of a based additive subgroup in C3:413, fol.508, repairs the ambiguity. □

A.4. Classification of normalized modular triples [Proved]

Over each of $R = \mathbb{Z}$, an arbitrary field, $\mathbb{Z}/4$, $\mathbb{Z}/8$ and $\mathbb{Z}/9$, the triples of A.3 form one $\mathrm{GL}_2(R)$ conjugacy class. Their $\mathrm{SL}_2(R)$ conjugacy classes are naturally indexed by $R^\times/(R^\times)^2$. In particular there are respectively $2, 2, 4, 2$ such classes over $\mathbb{Z}, \mathbb{Z}/4, \mathbb{Z}/8, \mathbb{Z}/9$; for a field the square-class formula applies in every characteristic, including imperfect characteristic 2.

Proof. A.3 gives transitivity. The common centralizer of r, s_- is the scalar group: commuting with their sum $\mathrm{diag}(0, 1)$ forces a matrix to be diagonal, and commuting with s_- makes the diagonal entries equal. Two conjugators yield the same SL_2 orbit exactly when their determinants differ by the determinant of a scalar, namely a square. Every unit is the determinant of $\mathrm{diag}(a, 1)$. This proves both completeness and separation of the invariant.

For finite rings the program enumerates every matrix, every pair with the prescribed traces and determinants, and the regularity and transversality conditions in every residue field. Independently it forms every conjugacy orbit. The results are

R	$\mathbb{F}_2$	$\mathbb{F}_3$	$\mathbb{Z}/4$	$\mathbb{F}_5$	$\mathbb{F}_7$	$\mathbb{Z}/8$	$\mathbb{Z}/9$
triples	6	24	48	120	336	384	648
GL_2 orbits	1	1	1	1	1	1	1
SL_2 orbits	1	2	2	2	2	4	2

Complete representatives and orbit sizes are in `AB_enumeration.json`. This classification concerns the normalized lifts, not arbitrary matrices merely having powers $R_0^3 = S_0^2 = -I$. The latter conditions alone permit additional triples; transversality and the based unipotent datum cannot be deleted. □

A.5. The regular conjugacy condition and the two norm obstructions [Proved]

For $z = r$ or s, let $\mathcal{A}_z$ consist of matrices $Y \in \mathrm{SL}_2$ such that $\mathrm{tr}(Yz) = \mathrm{tr}\, z$, and let $\mathcal{A}_z^{\mathrm{reg}}$ be the open locus where Yz is non-scalar in every geometric fibre. The map

$$\mathrm{GL}_2 \longrightarrow \mathcal{A}_z^{\mathrm{reg}}, \qquad A \longmapsto [A, z]$$

is an fppf T_z-torsor. The same map from SL_2 is an fppf ST_z-torsor. Over a ring R, once a $\mathrm{GL}_2(R)$ conjugator exists, its $\mathrm{SL}_2(R)$ obstruction is exactly its determinant in $R^\times/N(\mathcal{O}_z \otimes R)^\times$.

Proof. The trace and determinant conditions say that Yz and z have the same characteristic polynomial. A matrix that is non-scalar over every residue field is locally cyclic: a cyclic vector in a residue field lifts to one after localizing at its cyclic determinant. Relative to a cyclic basis its matrix is the companion matrix of that polynomial. Consequently Yz and z are Zariski locally conjugate, and two conjugators differ on the right by T_z. This proves the first torsor

assertion. The norm $T_z \to \mathbb{G}_m$ is fppf surjective: a unit has a square root after the finite faithfully flat extension $R[T]/(T^2 - a)$, and scalar elements have square norm. Adjusting a conjugator by such an element gives determinant 1, proving the second assertion and the exact point obstruction. A global $\mathrm{GL}_2(R)$ conjugator can additionally be obstructed by a rank-one module over the quadratic algebra; local cyclicity does not erase this Picard obstruction.

For $R = \mathbb{Z}$, the Gaussian and Eisenstein orders are Euclidean for their norms. Indeed rounding a complex coordinate to the nearest Gaussian lattice point gives squared distance at most $1/2$; for the triangular Eisenstein lattice the corresponding bound is $1/3$. Division with remainder therefore decreases the positive integral norm. Both ideal class groups vanish. A matrix with either irreducible characteristic polynomial makes $\mathbb{Z}^2$ a torsion-free rank-one module over the corresponding order, hence a free one. Thus global conjugators exist for the numerical targets over $\mathbb{Z}$. For r, the identity

$$\det(A)[A, r]_{22} = g^2 + gh + h^2 \quad \left(A = \begin{pmatrix} a & b \\ g & h \end{pmatrix}\right)$$

shows that the $\mathrm{SL}_2(\mathbb{Z})$ image is exactly $Y_{22} > 0$; for s the identity $[A, s] = AA^t/\det A$ shows it is exactly the positive definite target, equivalently $Y_{22} > 0$. The missed integral points are therefore specified completely by $Y_{22} < 0$ in the respective numerical target. No finite list is intended for these infinite sets. □

A.6. Specific formula slips and their corrections [Refuted]

The following formulas, in the stated matrix convention, are false: the trace condition $B + D - C = 1$ in the transcription of C3:266; $s(b) = b^{-t}$ and $bb^t = eI$ for arbitrary $b \in N_s'$ in C3:480; and the claim in C3:401 that ST_s is central in the entire signed $\mathcal{M}$. Their correct forms are, respectively,

$$C + D - B = 1, \qquad sbs^{-1} = (\det b)b^{-t}, \qquad bb^t = e(\det b)I,$$

and centrality of $ST_r \times ST_s$ in $\mathcal{M}^0$ only.

Proof. Direct multiplication gives $\mathrm{tr}(Yr) = C + D - B$. The universal 2×2 adjugate formula gives $sbs^{-1} = (\det b)b^{-t}$. The normalizer condition then gives the displayed equation for bb^t. Taking $b = 2I$ over $\mathbb{Q}$ disproves the formula omitting its determinant. Conjugation by the splitting of $e = -1$ in N_s' inverts ST_s. For example $s \in ST_s(\mathbb{Z})$ is sent to $s^{-1} = -s \neq s$. This is a concrete counterexample to the asserted centrality. The first correction is supported also by C3:151's $C + D - B = 1$; the incorrect order of B, C was checked visually on scan266 itself. The centrality restriction is already corrected in the manuscript at C3:419, (45). The alternatives in C3:328 for the lower-right entry of $R_0^2 S_0$ are tested universally: the correct expression is $-w(w^2 + 2v) + v + w^2$. All these checks and the degree-three identities of C3:327–330 are recorded in the output and readings register. □

A.7. Projective modular pairs, including imperfect fields [Proved]

Consider projective pairs satisfying the order-3, order-2, fppf-local regular unipotence and fibrewise transversality conditions of C3:323, without specifying a unipotent lift. Over a field k of characteristic different from 2 there is one $\mathrm{GL}_2(k)$ orbit, with $\mathrm{SL}_2(k)$ orbits indexed by $k^\times/k^{\times 2}$. In characteristic 2, put $K = k[T]/(T^2 + T + 1)$, a quadratic etale algebra. The $\mathrm{GL}_2(k)$ orbits are indexed by

$$N_{K/k}(K^\times)/k^{\times 2}.$$

Each such orbit has its $\mathrm{SL}_2(k)$ orbits forming a torsor for $k^\times/k^{\times 2}$. Over $\mathbb{Z}, \mathbb{Z}/4, \mathbb{Z}/8, \mathbb{Z}/9$ there is one GL_2 orbit; the numbers of SL_2 orbits are $2, 1, 2, 2$. These projective counts differ from A.4's counts of normalized lifts.

Proof. Transversality implies that both projective elements are non-scalar in every residue field. Otherwise the two unipotent fixed lines would coincide in that fibre. For a representative R of the order-3 element, Cayley–Hamilton gives

$$R^3 = ((\operatorname{tr} R)^2 - \det R)R - (\operatorname{tr} R)\det R\, I.$$

Since R^3 is scalar and the off-diagonal entries and diagonal difference generate the unit ideal, $\det R = (\operatorname{tr} R)^2$. The trace is a unit. Dividing by it gives trace and determinant 1. A cyclic basis conjugates it to r over a field or local ring; over $\mathbb{Z}$ the Eisenstein ideal argument in A.5 gives the same result. The projective involution has a trace-zero representative S, by the same regularity argument applied to its square.

If 2 is invertible, a projectively unipotent matrix V has the unique unipotent normalization $2V/\operatorname{tr} V$. A.3 applies after scaling S compatibly. In characteristic 2 the condition is $\operatorname{tr}(rS) = 0$, since $S^{-1} = S/\det S$. The solutions are precisely $S = zs$ with $z \in K^\times$. Here s induces the nontrivial automorphism $z \mapsto \bar{z}$ of K by conjugation, and $\det S = N(z)$. Scalar changes of S multiply z by $k^\times$; conjugation preserving r multiplies it by $g/\bar{g}$ for $g \in K^\times$. The kernel of the norm consists exactly of such quotients: if $N(w) = 1$ and $w \neq 1$, take $g = 1 + w$, obtaining $g/\bar{g} = w$. The split-algebra case obeys the same formula, and $w = 1$ is immediate. Thus the orbit invariant is exactly the norm modulo squares. These forms are projectively regular unipotent after adjoining the required square root. If their two fixed lines coincided geometrically, r, S would preserve that line; the diagonal entries would then give $\operatorname{tr}(rS) = \lambda \operatorname{tr} r = \lambda \neq 0$. This checks transversality. A stabilizing $g \in K^\times$ must have $g/\bar{g} \in k^\times$. Its norm is 1, so this scalar has square 1 and hence is 1 in a characteristic-2 field. Therefore $g \in k^\times$. The projective stabilizer on field points is trivial, and the determinant argument of A.4 proves the SL_2 assertion.

For $R = \mathbb{Z}/2^n$ with $n \geq 2$, an fppf unipotent normalization λV satisfies $\lambda \operatorname{tr} V = 2$ and $\lambda^2 \det V = 1$. A faithfully flat algebra B preserves $\operatorname{Ann}_B(2) = 2^{n-1}B$. Reduction modulo 2 first shows that $\operatorname{tr} V = 2a$ in R. Then $\lambda a = 1 + 2^{n-1}b$, so a is a unit and $\lambda^2 a^2 = 1$. Hence $\det V = a^2$ already in R. The global scalar a^{-1} supplies a unipotent normalization. Over $\mathbb{Z}$, $(\operatorname{tr} V)^2 = 4 \det V$ with $\det V \in \{\pm 1\}$ likewise gives a global normalization; over $\mathbb{Z}/9$ the argument with invertible 2 applies. Scaling S to make $E = -rS^{-1}$ this normalization makes $\det S = 1$, while its trace stays zero. A.3 now gives the unique GL_2 orbit. Its stabilizer determinants are

$$R^{\times 2}\{1 + \tau : 2\tau = \tau^2 = 0\}.$$

Their indices in $R^\times$ give $2, 1, 2, 2$. The finite program projects all normalized triples and forms their SL_2 orbits, confirming these different projective counts.

The imperfect-field phenomenon is concrete. Over $\mathbb{F}_2(t)$ take $S = \left(\begin{smallmatrix} 1 & t \\ t+1 & 1 \end{smallmatrix}\right)$. Its norm is $t^2 + t + 1$, whose derivative is 1 and which is therefore not a square in $\mathbb{F}_2(t)$. This pair is not conjugate to the standard projective pair, despite satisfying the weak projective conditions. The nonsquare certificate is in the code. □

A.8. Unrestricted uniqueness among all group-scheme isomorphisms [Refuted]

The modular data cannot determine a unique arbitrary group-scheme isomorphism $\mathrm{GL}_2 \to \mathrm{GL}_2$, even over $\mathbb{Z}$ or $\mathbb{Q}$. The correct uniqueness in A.3 is among algebra-induced inner identifications. Indeed

$$\theta(g) = (\det g)^{-1} g$$

is a nonidentity involutive automorphism of GL_2 fixing SL_2 pointwise, hence fixing r, s and both root subgroups.

Proof. Determinants are central scalars, so the formula is multiplicative. In rank two $\det\theta(g) = (\det g)^{-1}$, and hence $\theta^2(g) = g$. On the centre it sends aI to $a^{-1}I$; an inner or algebra-induced automorphism fixes that centre pointwise. For example $2I$ goes to $I/2$ over $\mathbb{Q}$. The program verifies multiplicativity and involutivity universally. Thus a uniqueness claim encompassing all group-scheme isomorphisms needs the inner/algebra condition, independently of the characteristic-2 infinitesimal ambiguity. □

A.9. The general dictionary and its exact remaining comparison [Open]

For the general groups of §49 we use the following dictionary. It is a dictionary of discrete or profinite groups, or of sheaves when those objects are explicitly supplied, and does not silently assert that their automorphism functors are group schemes. For the signed rows we impose $\rho^3 = \sigma^2 = \omega \in Z(C)$ and $\omega^2 = 1$, and retain the inversion involution specified in B.13. The symbol -1 in those rows means this common central element when ω has order two. The signed fibre-product rows additionally use $L_0^{\mathrm{sgn}} \cap Z(C) = 1$, as in B.13; the unsigned and signed cyclic subgroups are distinguished in the table.

Our symbol	Definition; manuscript symbol	C3 scan
C	The group generated by ρ, σ; $\mathcal{C}$ or $\mathcal{G}$	502;530
L_0, L_1	Closed cyclic groups generated by $\sigma^{-1}\rho$, $\rho\sigma^{-1}$ in the unsigned convention	502
B	Automorphisms preserving both conjugacy classes and normalizing L_0; $B_{\rho,\sigma}$	503 (7)
Σ	Numerical commutator pairs with $\eta^{-1}\zeta \in L_1$; $\Sigma_{\rho,\sigma}$	503 (8)
Φ	$u \mapsto (u(\rho)\rho^{-1}, u(\sigma)\sigma^{-1})$; realization is an extra hypothesis	502–503
S_0G, G	$C \rtimes B$ and its quotient by $(\ell^{-1}, \mathrm{Int}\,\ell)$; $S_0G_{\rho,\sigma}, G_{\rho,\sigma}$	505
γ	$B/\mathrm{Int}(L_0)$; $\gamma_{\rho,\sigma}$, previously $\Gamma_{\rho,\sigma}$	506
Z_ρ, Z_σ	Centralizers in G; their kernels over γ are C_ρ, C_σ when $L_0 \cap Z(C) = 1$	509;511
T^0, ST^0	$Z_\rho \times_\gamma Z_\sigma$ and $B \times_\gamma T^0$; $\Gamma^0_{\rho,\sigma}, S_0\Gamma^0_{\rho,\sigma}$	510–511
M^0, SM^0	$T^0 \times_\gamma G$ and $B \times_\gamma M^0$; $M^0_{\rho,\sigma}, S_0M^0_{\rho,\sigma}$	512–514
$\widetilde{K}$	Pullback of a specified global-sign quotient representation; specializes to the definition of $\widetilde{N}_{1,1}$ after the map is fixed	492;524 (2)–(3)
$\varepsilon_2, \varepsilon_3$	Characters of $\widehat{\mathbb{Z}}^\times$ modulo 4 and 3; their additive forms are ν_2, ν_3	531–534
B^{sgn}	Pairs (u, μ) with $u(\sigma^{-1}\rho) = (\sigma^{-1}\rho)^\mu$, $u(\rho) \sim_C \rho^\mu$, $u(\sigma) \sim_C \sigma^\mu$	530 (7)
Z^{sgn}	Tuples $(u, \mu, e_\rho, e_\sigma)$ with the same loop condition and $u(\rho) \sim_C \rho^{e_\rho}$, $u(\sigma) \sim_C \sigma^{e_\sigma}$; the two signs are independent numerical coordinates	549 (6)
$L_0^{\mathrm{sgn}}, e_1, \ell_1$	$\overline{\langle\sigma\rho\rangle}$, $\rho\sigma$, and $(\rho\sigma)^2$; these are the signed loop conventions used in B.13	530–534
D, γ_D, G_D	$D = B^{\mathrm{sgn}}$ or Z^{sgn}; $\gamma_D = D/j(L_0^{\mathrm{sgn}})$, $G_D = (C \rtimes D)/\{(\ell^{-1}, j(\ell))\}$, with $j(\ell) = (\mathrm{Int}\,\ell, 1, 1, 1)$ and the last signs suppressed for B^{sgn}	535–536; 551–553
Σ^{eff}	Numerical signed pairs that are realized by an automorphism; the fourth condition is essential	534–535

Our symbol	Definition; manuscript symbol	C3 scan
N_x, N_x^*, Z_x^*	For $x = \rho, \sigma$, $N_x = \{(g,e) : gxg^{-1} = x^e\}$, $N_x^* = \{(g,e) \in N_x : e = \chi_x(\delta g)\}$, $Z_x^* = C_{G_D}(x) \cap \delta^{-1} \ker \chi_x$, where $\delta : G_D \to \gamma_D$ and χ_x is the recorded sign	540–541; 555–556
$\Gamma_D, S_0\Gamma_D$	$N_\rho^* \times_{\gamma_D} N_\sigma^*$ and $D \times_{\gamma_D} \Gamma_D$; the starred normalizers keep the numerical signs	559 (51), (55)–(56)
$\mathcal{M}_D, S_0\mathcal{M}_D$	$\Gamma_D \times_{\gamma_D} G_D$ and $D \times_{\gamma_D} \mathcal{M}_D$; kernels and the cartesian square are proved in B.13	559 (50), (52)–(54); 560 (57), (59)
D_q, H_q, A_q	For $q : C \twoheadrightarrow C'$, D_q stabilizes $\ker q$; H_q and A_q are the image and kernel of the descended coefficient-group map. Exact formulas are B.15.1–B.15.5	570–572 (88)–(94)
$D_0, L(D_0), G_0$	A restricted coefficient subgroup; $L(D_0) = j^{-1}(D_0)$ and $G_0 = CD_0 \subset G_D$. This keeps all of C as conjugators	571–572 (94)
A, Z^A, L_A, H_A	The different restriction $A = C^\natural \triangleleft C$ is an open normal conjugating subgroup; $L_A = j^{-1}(Z^A)$ and $H_A = \mathrm{im}(A \rtimes Z^A \to G)$. The antidiagonal uses $L_A \cap A$, as proved in B.17	651–664 (1)–(52 bis)
$\mathscr{E}_{1,1,\mathbb{Q}}, \mathscr{E}'_{1,1,\mathbb{Q}}$	The elliptic moduli stack and its rigidification by $[-1]$, written $M_{1,1}, M'_{1,1}$ in the source; their gerbe and coarse j-map are distinguished from the affine group $\mathcal{M}$ in B.16	628 (26)
$\mathcal{E}, \phi, J$	Reconstruction of an extension from (P, T, J, θ, ϕ) satisfying the two compatibilities in B.15.9; compatible gluing additionally requires the diagonal subgroup B.15.11	691–696; fols. 648–650

Proved dictionary, computation, and residual question. The unsigned entries and every map between them are proved in B.11; the global-sign pullback is proved in B.9. The finite construction is executed for S_3, S_4, and B.12 supplies a fully executed failure of $\Sigma^{\mathrm{eff}} = \Sigma$ in $\mathrm{PSL}_2(\mathbb{F}_{13})$. For the signed entries the character definitions follow from $\rho^3 = \sigma^2 = \omega$ and $\omega^2 = 1$. The generator-image conditions fix ω, because the square of either conjugate of $\sigma^{\pm 1}$ is ω. Since μ is odd, replacing $\sigma^{-1}\rho$ by $\sigma\rho = \omega(\sigma^{-1}\rho)$ then preserves the condition that it is raised to its μth power. The two possible loop conventions must nevertheless not be interchanged inside unqualified formulas. A formal signed theorem is given in B.13.

The remaining identifications are distinct questions, rather than one unspecified notation audit. First, numerical signed parameters must be compared with effective automorphism parameters for $C = \widehat{\mathrm{SL}_2(\mathbb{Z})}$. B.8 defines the numerical triples explicitly and proves that each gives a continuous endomorphism; its exact remaining assertion (B.8.5) is surjectivity of every such endomorphism. The finite unsigned and signed realization computations in B.11 and B.14 are proved cases for that question.

Second, B.15 supplies the quotient functoriality of scans 570–572, proves stability of the actual congruence kernel, and gives exact kernel formulas. Its full effective image and compatible lifts remain separately specified. Third, B.16 treats the central twist and normalized conjugators of scans 573–582, retaining the exact named geometric-image and unquotiented-section comparisons of those scans and scans 612–616. Fourth, the continuation of the dictionary in B.18 records the normalized sections of scans 583–595, the normality and index-kernel assertions of scans 635–647, the bisection of scans 648–650, and the extra open-conjugator datum of scans 651–690. B.17 refutes the literal smaller quotient and proves the necessary

intersection repair. B.18 specifies the remaining comparisons first for the named index-six and index-twelve modular subgroups, with proved formal cases and finite calculations. The reconstruction and gluing at scans 691–696 are proved in B.15.

D.7 separately records the first modulus-3 GT lifting instance and the remaining arithmetic comparison. Integral representability and comparison with the schemes of §48 require an additional construction: B.8 proves that the naive constant profinite scheme cannot even realize $\widehat{\mathbb{Z}} \ni 1 \mapsto L_0(1)$ over $\mathbb{Z}$. No untyped fppf quotient or descent theorem is asserted here. These source-indexed residuals are recorded as partly settled in A(a), A(b), B(c), and B(g); none is absorbed into (B.8.5). □

A.1 Low types, orientation, and the three different centres

A.10. The low-type convention table [Known]

Let $S_{g,n}$ be an oriented closed surface of genus g with a set of n distinct marked points removed. Write $\Pi = \pi_1(S_{g,n})$, Mod^+ for orientation-preserving mapping classes allowing permutations of those points, PMod^+ for the subgroup fixing them individually, and $\mathrm{Mod}^\pm$ for the full group allowing orientation reversal. All quotients are by *isotopies through homeomorphisms* preserving the marked set. The points are punctures, not boundary circles fixed pointwise. With $C_2 = \mathbb{Z}/2$, the complete table in the six types specified in Rule 5 is

(g,n)	Π	Mod^+	PMod^+	$\mathrm{Mod}^\pm$
$(0,0)$	1	1	1	C_2
$(0,1)$	1	1	1	C_2
$(0,2)$	$\mathbb{Z}$	C_2	1	$C_2 \times C_2$
$(1,0)$	$\mathbb{Z}^2$	$\mathrm{SL}_2(\mathbb{Z})$	$\mathrm{SL}_2(\mathbb{Z})$	$\mathrm{GL}_2(\mathbb{Z})$
$(0,3)$	F_2	$\mathfrak{S}_3$	1	$\mathfrak{S}_3 \times C_2$
$(1,1)$	F_2	$\mathrm{SL}_2(\mathbb{Z})$	$\mathrm{SL}_2(\mathbb{Z})$	$\mathrm{GL}_2(\mathbb{Z})$

Their centres, keeping the surface group and the two full mapping-class groups separate, are

(g,n)	$Z(\Pi)$	$Z(\mathrm{Mod}^+)$	$Z(\mathrm{Mod}^\pm)$
$(0,0), (0,1)$	1	1	C_2
$(0,2)$	$\mathbb{Z}$	C_2	$C_2 \times C_2$
$(1,0)$	$\mathbb{Z}^2$	$\{\pm I\}$	$\{\pm I\}$
$(0,3)$	1	1	$\{1\} \times C_2$
$(1,1)$	1	$\{\pm I\}$	$\{\pm I\}$

The pure oriented centres are trivial in the sphere rows and $\{\pm I\}$ in the torus rows. The extended pure sphere groups, and hence their centres, are C_2.

Proof. Van Kampen gives F_{n-1} for a sphere with $n > 0$ punctures, $\mathbb{Z}^2$ for the closed torus, and F_2 for the once-punctured torus. These fundamental-group calculations are independent of a choice of marked-point labels. The mapping-class computations are verified in Farb and Margalit (2012, §§2.2.1–2.2.2, Proposition 2.3, Theorem 2.5, the following once-punctured-torus paragraph, and §8.1); the cited PDF is version 5.0, pp. 51–57 and 230–231.

Here is the elementary reduction, including the convention-sensitive parts. On a sphere with three fixed points, an arc joining two points in the complement of the third can be isotoped to its image, with endpoints fixed. After fixing the arc pointwise, cutting along it leaves a disc with one marked point. The Alexander isotopy, with this point at the centre, fixes both the point and the boundary. Thus the pure oriented group is trivial. For fewer marked points, extra points can first be fixed by isotopy in their complement; the same argument applies. Every permutation of at most three points is realized by an orientation-preserving

homeomorphism. A reflection of the sphere fixes three chosen points individually. Consequently the homomorphism recording permutation and orientation is an isomorphism onto $\mathfrak{S}_n \times C_2$ for $0 \le n \le 3$, with $\mathfrak{S}_0 = \mathfrak{S}_1 = 1$. This proves the sphere rows and the assertions about pure groups.

On the torus, action on homology identifies the oriented group with $\mathrm{SL}_2(\mathbb{Z})$ by the cited Theorem 2.5; all integral invertible matrices are realized by linear maps, and orientation is their determinant. Forgetting one marked point is an isomorphism on mapping-class groups. To check injectivity without an erroneous point-pushing kernel, put that point at 0. If f fixes 0 and F_t is an isotopy from f to the identity on the unmarked torus, then $x \mapsto F_t(x) - F_t(0)$ is an isotopy fixing 0. Surjectivity follows from the linear representatives, which already fix 0.

A reduced-word calculation gives $C_{F_2}(a) = \langle a \rangle$ and $C_{F_2}(b) = \langle b \rangle$, so $Z(F_2) = 1$. The other surface-group centres follow from commutativity. The centre of $\mathfrak{S}_3$ is trivial, since commuting with both (12) and (23) forces a permutation to be the identity. A matrix commuting with $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$ is scalar; an invertible scalar integral matrix is $\pm I$. This computes the centres of both torus groups and proves the table. □

Scope of the comparisons. The first four types have $2g - 2 + n \le 0$. Their geometric groups are trivial or abelian, and their profinite completions are respectively $1, 1, \widehat{\mathbb{Z}}, \widehat{\mathbb{Z}}^2$, with those same groups as centres. Hence arguments requiring a centre-free hyperbolic geometric kernel do not apply; no hyperbolic anabelian reconstruction theorem is being asserted for these rows. In particular the forgetful torus map above has no injected $\mathbb{Z}^2$ point-pushing kernel. The types $(0, 3)$ and $(1, 1)$ *are* hyperbolic. Their abstract groups are both $F_2 = \langle a, b \rangle$, but their peripheral classes are respectively $a, b, (ab)^{-1}$ and $[a, b]^{-1}$. Their distinct peripheral data and mapping-class centres must therefore be retained in a comparison.

Even for the cylinder, orientation and loop action differ: if ϵ is orientation sign and s permutation sign, the induced sign on $\Pi = \mathbb{Z}$ is $s\epsilon$. Indeed $z \mapsto 1/z$ exchanges the two punctures and preserves orientation, whereas $z \mapsto \overline{z}$ fixes them and reverses orientation; both act by -1 on Π. This also explains why isotopy cannot be replaced here by arbitrary homotopy.

These are discrete topological groups, not the affine group-scheme automorphism functors in §§48–50. For example the oriented mapping-class group of the sphere is trivial while its complex algebraic automorphism group is $\mathrm{PGL}_2(\mathbb{C})$. Orientation-reversing mapping classes still belong to an orientable surface; they do not supply a non-orientable version of the complex-curve comparison. A real structure adds a specified antiholomorphic involution and its outer action, not just a choice of the abstract extended group. The supplied LM3 A.11 was used as a lead; the calculation and the cited primary exposition were checked independently here.

B The modular schemes, their fibres and their parametrizations

B.1. The solution space and the parametrizations are isomorphisms [Proved]

The scheme $\mathcal{X}$ of A.1 is $\mathbb{G}_a \times \mathbb{G}_m$ over $\mathbb{Z}$. Both maps in C3:368, (2), and C3:371, (10), are isomorphisms of schemes over $\mathbb{Z}$, for the fixed modular matrices. Explicitly,

$$(v, a, b) \longmapsto (\alpha = va^{-1},\ \beta = vb^{-1},\ \mu = \det v,\ e = e(b))$$

identifies $B_0' \times T_r \times N_s'$ with $\mathcal{G}_{r,s}^*$; setting $e = 1$ gives the unsigned version. They are bijective on points over every ring, as well as isomorphisms of fppf sheaves. There are no missed points in these parametrizations over $\mathbb{Z}$, $\mathbb{F}_2$, $\mathbb{F}_3$ or $\mathbb{Z}/4$.

Proof. Put $\eta = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$. The trace condition for s gives $C = B$. The trace condition for $\zeta = \eta L_1(\mu - 1)$ is $\mu D = 1$. The determinant condition then gives

$$\eta = \begin{pmatrix} \mu(1 + B^2) & B \\ B & \mu^{-1} \end{pmatrix}.$$

Thus μ is automatically a unit, even though initially it may have been an affine coordinate. For $v = b_0(\mu, \mu B)$ direct multiplication gives $\eta = [v, s]$ and $\zeta = [v, r]$. This gives a polynomial inverse in the localized coordinate ring and proves the first assertion.

Given (α, β, μ, e) satisfying the signed equation, use $\eta = e[\beta, s]$ to recover the unique v. Set $a = \alpha^{-1}v$ and $b = \beta^{-1}v$. Equality of commutators implies $a \in T_r$, and $[v, s] = e[\beta, s]$ implies $bsb^{-1} = es$. Conversely these identities give the original equation. All formulas are defined over the original ring; no extension is needed. This also proves injectivity, rather than assuming that two centralizers intersect trivially.

The output independently enumerates $\mathcal{X}(\mathbb{F}_2)$, $\mathcal{X}(\mathbb{F}_3)$ and $\mathcal{X}(\mathbb{Z}/4)$: their cardinalities are $2, 6, 8$ and the missed-point sets are empty. The statement does not assert the analogous parametrization for arbitrary pairs (ρ, σ) without the modular hypotheses discussed at C3:370–371. □

B.2. Equations, integral Groebner basis and flatness [Proved]

The final group $\mathcal{M}$ of C3:399, (13), and C3:479, (3), has coordinate algebra

$$\mathbb{Z}[c, d, t, u, U_{11}, U_{12}, U_{21}, U_{22}, e, e', Q_2^{-1}]/(e^2 - 1, (e')^2 - 1, \\ \det U - eQ_2,\ eQ_2 - e'Q_3).$$

It is flat of relative dimension 6 over $\mathbb{Z}$. Its special precursor $S\mathcal{G}^*_{r,s}$ has relative dimension 4; $S\mathcal{M}$ has relative dimension 7. These are different schemes and explain the corrections at scans151 and393.

Proof. A.1 gives $a = cr + dI$, $b = \left(\begin{smallmatrix} u & t \\ -et & eu \end{smallmatrix}\right)$. The displayed equations are precisely the three determinant conditions and the two sign conditions. Since Q_2 is a unit, so is Q_3, and all three matrices are invertible. This proves the asserted functor of points without an unmentioned open condition.

Write $U = D(eQ_2)h$, $h \in \mathrm{SL}_2$. Then, as schemes,

$$\mathcal{M} \simeq \mathrm{SL}_2 \times X, \qquad X = \operatorname{Spec} \frac{\mathbb{Z}[c, d, t, u, e, e', z]}{(Q_3 - ee'Q_2, e^2 - 1, (e')^2 - 1, zQ_2 - 1)}.$$

In lexicographic order

$$h_{11}, h_{12}, h_{21}, h_{22}, c, e, e', z, t, u, d$$

the following five polynomials are a monic Groebner basis over $\mathbb{Z}$:

$$h_{11}h_{22} - h_{12}h_{21} - 1, \quad c^2 + cd + d^2 - ee'(t^2 + u^2), \\ e^2 - 1, \quad (e')^2 - 1, \quad z(t^2 + u^2) - 1.$$

Their leading monomials $h_{11}h_{22}, c^2, e^2, (e')^2, zt^2$ are pairwise coprime. The product criterion reduces every S-polynomial to zero using only monic division, so this is an integral assertion, not a Groebner calculation over $\mathbb{Q}$ misreported as one over $\mathbb{Z}$. The residue classes of the standard monomials are a free $\mathbb{Z}$-basis. In particular there is no integer torsion and the scheme is flat. The same basis remains a Groebner basis in every characteristic. The program computes it independently over $\mathbb{Q}, \mathbb{F}_2, \mathbb{F}_3$ and verifies the integral monicity certificate.

For equidimensionality, $X \to \mu_2^2$ is smooth of relative dimension 3, as proved in B.3; therefore $\mathcal{M}$ has relative dimension 6 at every point. Finally $S\mathcal{G}^* \simeq \mathbb{G}_a \times X$ by B.1 and the determinant conditions, and $S\mathcal{M} \simeq \mathbb{G}_a \times \mathcal{M}$ as schemes. The signed GL_2 precursor $\mathcal{G}^*$ itself has dimension 6, whereas the equations with $\alpha, \beta \in \mathrm{SL}_2$ and no signs have dimension 4. Scan151's marginal correction to 4 concerns this precursor, not the later $\mathcal{M}$. □

B.3. Smooth locus and the fibres at 2 and 3 [Proved]

The smooth locus of $\mathcal{M} \to \operatorname{Spec} \mathbb{Z}$ is precisely $\mathcal{M}_{\mathbb{Z}[1/2]}$. The closed subscheme cut out by the Jacobian ideal for relative dimension 6 is $V(4)$. The fibre at 3 is smooth and has four geometrically irreducible components, each geometrically isomorphic to $\mathrm{SL}_2 \times \mathbb{G}_m^2 \times \mathbb{G}_a$. The fibre at 2 has one geometrically irreducible component and is everywhere nonreduced; its reduction is

$$\mathrm{SL}_{2,\mathbb{F}_2} \times \mathbb{G}_a \times \mathbb{G}_m \times R^1_{\mathbb{F}_4/\mathbb{F}_2}\mathbb{G}_m.$$

Its nilradical is generated by $e-1, e'-1$, its cube is zero, and its generic length over its reduction is 4. The singular locus of that fibre is the entire fibre. The unsigned subgroup $\mathcal{M}^0$ is smooth over all of $\mathbb{Z}$.

Proof. Relative to μ_2^2, X is the open hypersurface $Q_3 = ee'Q_2$, with Q_2 a unit. Its gradient in c, d, t, u is

$$(2c + d, c + 2d, -2ee't, -2ee'u).$$

At a point of residue characteristic different from 2, the last two entries cannot both vanish. In characteristic 2, the first two become d, c; they cannot both vanish since Q_3 is a unit. Thus this hypersurface is smooth over the sign base in every characteristic. It is surjective after an algebraic closure: prescribe a unit norm and solve the two quadratic norm equations. Hence it is faithfully flat over μ_2^2. The geometric descriptions below also prove nonemptiness explicitly.

Away from 2 the sign base is etale. At 2, both sign equations have zero differential, increasing tangent dimension by two. This proves the smooth-locus assertion. More precisely the sign rows in the Jacobian force every maximal minor to be divisible by 4. The remaining gradient entries generate the unit ideal: their ideal contains 2, since Q_2 is invertible; it then contains c, d and hence the unit Q_3. The SL_2 gradient also generates the unit ideal, and the localization equation has invertible z-derivative Q_2. Consequently the minors generate exactly (4).

At 3, $Q_3 = (c-d)^2$. Put $v = c - d \neq 0$. For each sign pair, the remaining equation is $(t/v)^2 + (u/v)^2 = ee'$. This is a smooth norm torsor for $\mathbb{F}_9/\mathbb{F}_3$; both values $1, -1$ occur, for example at $(1, 0)$ and $(1, 1)$. The unused coordinate d is affine. This proves the description and geometric irreducibility of each of the four components.

At 2, put $x = e - 1, y = e' - 1$. Then $x^2 = y^2 = 0$. After reduction $Q_2 = (t + u)^2$. Writing $v = t + u \neq 0$ and dividing c, d by v gives the norm-one equation $a^2 + ab + b^2 = 1$ for $\mathbb{F}_4/\mathbb{F}_2$; the remaining coordinate is affine. This reduction is smooth and geometrically connected. Smoothness over the Artinian sign algebra $\mathbb{F}_2[x, y]/(x^2, y^2)$ proves that its nilradical is exactly (x, y), with nonzero product xy and generic length 4. The same gradient argument with $e = e' = 1$ proves smoothness of $\mathcal{M}^0$ over $\mathbb{Z}$.

In particular the notation "composante neutre" for $\ker(e, e')$ in C3:400 and419 is valid after inverting 2, but is not the identity component of the characteristic-2 fibre: that fibre is connected, and its reduction is a proper closed subgroup. The good behaviour at 3 is stronger than merely smoothness away from 2 and 3. □

B.4. Group law and exact sequences of the modular construction [Proved]

The group law on $\mathcal{M} \subset T_r \times N_s' \times G$ is componentwise multiplication. With $\mathcal{H}$ as in A.1, the following sequences are exact as fppf sheaves:

$$1 \to T_s \to N_s' \xrightarrow{e} \mu_2 \to 1,$$
$$1 \to \mathrm{SL}_2 \to \mathcal{M} \to \mathcal{H} \to 1, \qquad 1 \to \mathbb{G}_a \to S\mathcal{M} \to \mathcal{M} \to 1,$$
$$1 \to \mathcal{M}^0 \to \mathcal{M} \xrightarrow{(e,e')} \mu_2^2 \to 1,$$
$$1 \to ST_r \times ST_s \times \mathrm{SL}_2 \to \mathcal{M} \xrightarrow{(\chi,e,e')} \mathbb{G}_m \times \mu_2^2 \to 1,$$
$$1 \to ST_r \times ST_s \to \mathcal{M}^0 \xrightarrow{U} \mathrm{GL}_2 \to 1,$$
$$1 \to ST_r \times ST_s \to \mathcal{H}^0 \xrightarrow{\chi} \mathbb{G}_m \to 1.$$

The first three have explicit sections and are exact on points over every ring. The other surjections need not be onto on points over a given ring. The larger ambient groups at C3:393 have, after the same change of coordinates,

$$\widetilde{S}\mathcal{M} \simeq B_0' \times T_r \times N_s' \times \mathrm{GL}_2, \qquad \widetilde{\mathcal{M}} \simeq \mathbb{G}_m \times T_r \times N_s' \times \mathrm{GL}_2,$$

and a split exact sequence $1 \to L_0 \to \widetilde{S}\mathcal{M} \to \widetilde{\mathcal{M}} \to 1$. Their relative dimensions are 10 and 9.

Proof. Determinants and signs are characters, so their equalities are preserved under products and inverses. The first section was given in A.1. The section of $\mathcal{M} \to \mathcal{H}$ is $(a, b) \mapsto (a, b, D(\det b))$; conjugation on the normal SL_2 factor is by $D(\chi)$. The section of $S\mathcal{M} \to \mathcal{M}$ appends $D(\chi)$ in the B_0' coordinate. In our order (v, a, b, U) its kernel is $(L_0(z), I, I, I)$; the differing coordinate order in the overwritten formulas at C3:479 must not be ignored.

All kernels follow directly from the equations. To prove fppf surjectivity of the remaining arrows, solve prescribed norm equations by adjoining square roots of the required units and taking scalar elements in the respective quadratic algebras. Such adjoining is finite locally free of rank 2 and faithfully flat, also in characteristic 2. Given (χ, e, e'), require $Q_2 = e\chi$ and $Q_3 = e'\chi$ and take $U = D(\chi)$. This constructs the required local lift. The same construction with $e = e' = 1$ treats the final arrows.

For a pointwise failure over $\mathbb{Z}$, both positive quadratic norms of integral units equal 1, so (χ, e, e') has image only $(1, 1, 1)$ and $(-1, -1, -1)$. Over $\mathbb{F}_3$, Q_3 of a unit is always 1, so $\mathcal{H}^0(\mathbb{F}_3) \to \mathbb{F}_3^\times$ misses -1. These examples disprove pointwise exactness without changing the fppf statements. The table in B.7 gives further complete point images for the individual commutator maps.

For the ambient groups begin with $\mathrm{GL}_2 \rtimes (B_0' \times T_r \times N_s')$, where $v \in B_0'$ acts on GL_2 by conjugation and the other factors act trivially. The change $(g, v, a, b) \mapsto (v, a, b, U = gv)$ converts its product to componentwise multiplication. The diagonal copy of L_0 being killed becomes (v, I, I, I) with $v \in L_0$. Its quotient replaces B_0' by $B_0'/L_0 = \mathbb{G}_m$, with splitting $\mu \mapsto D(\mu)$. This proves the extra sequence, its pointwise exactness and both dimensions. Imposing $\det v = \det b = e' \det a = \det U$ gives the smaller sequence. □

B.5. The distinguished sections and the actual normalizer [Proved]

The subgroups $\mathcal{M}(0)$ and $\mathcal{M}[0]$ are $\mathcal{H}$ and $\mathbb{G}_a \rtimes_\chi \mathcal{H}$ through the embeddings in A.1. Their normalizer comparison is

$$\mathrm{Norm}_{\mathcal{M}}(L_0) = \{(a, b, U) \in \mathcal{M} : U \text{ is upper triangular}\},$$

which properly contains $\mathcal{M}[0]$. The equations $U = b$ and $U = a$ define the section $\mathcal{M}(s)$ and the partial section $\mathcal{M}(r)$ over $e' = 1$. Conjugating the first two appropriate sections by (I, I, r) defines the cycles

$$\{\mathcal{M}(0), \mathcal{M}(1), \mathcal{M}(\infty)\}, \qquad \{\mathcal{M}(1/2), \mathcal{M}(2), \mathcal{M}(-1)\}$$

with $\mathcal{M}(1/2) = \mathcal{M}(s)$ and $\dot{r}(q) = 1/(1-q)$, as in C3:477.

Proof. Multiplying $D(\chi)$ by $L_0(z)$ gives exactly B_0'. Conjugation by $D(\chi)$ sends $L_0(z)$ to $L_0(\chi z)$, which proves the asserted semidirect product. In GL_2, the normalizer of the whole root subgroup L_0 is the full upper triangular Borel: conjugating $I + zE_{12}$ and comparing its coefficient of z shows that the line spanned by E_{12} is preserved; its kernel and image are the first coordinate line, which the conjugator must preserve. The converse is direct multiplication. Since the U projection governs conjugation of the SL_2 factor, the normalizer formula follows. The element $(I, I, -I)$ is already in this normalizer but not in $\mathcal{M}[0]$ over $\mathbb{Z}$. This is the correction explicitly recognized at C3:418, fol.510; it is not treated as a still-maintained assertion.

The two section formulas preserve multiplication and satisfy the determinant constraint precisely on the indicated bases. Finally $r^3 = -I$ acts trivially by conjugation, so the threefold conjugation cycles close. Their homographic labels are the orbits of 0 and $1/2$ under $q \mapsto 1/(1-q)$. The label $-\bar{j}$ names $\mathcal{M}(r)$ in C3:477; the attempted additional choice $\mathcal{M}(-j)$ at scans477–478 is not fixed by the source and is not silently declared canonical here. □

B.6. The central kernels and the square-root cover [Proved]

For $\mathcal{M}^0$ the kernel of $U : \mathcal{M}^0 \to \mathrm{GL}_2$ is $I^0 = ST_r \times ST_s$, and the kernel of its composite with $\mathrm{GL}_2 \to \mathrm{PGL}_2$ is

$$J^0 = \{(a, b, \lambda) : Q_3(a) = Q_2(b) = \lambda^2\} \simeq ST_r \times ST_s \times \mathbb{G}_m.$$

It is the centre of $\mathcal{M}^0$. Forgetting λ gives an fppf central extension

$$1 \to \mu_2 \to J^0 \to \mathcal{H}^0 \to 1,$$

not the unqualified isomorphism to $\mathcal{H}^0$ written in C3:400. Moreover

$$(J^0 \times \mathrm{SL}_2)/\mu_2 \simeq \mathcal{M}^0,$$

where $z \in \mu_2$ is embedded as $((I, I, z), z^{-1}I)$. For the full signed group the centre is

$$\{(a, uI, \lambda I) : a \in T_r,\ u, \lambda \in \mathbb{G}_m, \lambda^2 = u^2,\ (u^2/\det a)^2 = 1\}.$$

Proof. The first equations are immediate. Dividing a, b by λ gives the displayed isomorphism with the product of two norm-one groups and $\mathbb{G}_m$. The map U is fppf surjective by B.4, and the first two coordinates commute; therefore an element is central in $\mathcal{M}^0$ exactly when U is scalar. Adjoining a square root of the common norm proves surjectivity $J^0 \to \mathcal{H}^0$ and identifies its kernel. As a finite check separating the proposed isomorphism, over $\mathbb{F}_3$ these groups have respectively $6 \cdot 4 \cdot 2 = 48$ and 24 points.

The multiplication map sends $((a, b, \lambda), h)$ to $(a, b, \lambda h)$. Its kernel is the indicated diagonal μ_2, and it is locally onto by taking a square root of $\det U$. This proves the quotient assertion as fppf sheaves. For the full centre, projection onto N_s' and GL_2 is fppf surjective, by the norm-root construction. The centre of $N_s' = T_s \rtimes \mu_2$ consists of its scalar matrices: commuting with T_s forces the sign to be 1, and invariance under the universal sign forces the s coefficient to vanish. This argument compares coefficients in $R[e]/(e^2 - 1)$, so it is valid at 2 as a scheme statement. The determinant equations give exactly the formula claimed. □

B.7. Finite point counts and complete small-ring commutator images [Proved]

For the final signed $\mathcal{M}$, the exact counts are

R	$\lvert\mathrm{SL}_2(R)\rvert$	$\lvert\mathcal{H}(R)\rvert$	$\lvert\mathcal{M}(R)\rvert$
$\mathbb{F}_2$	6	6	36
$\mathbb{F}_3$	24	96	2304
$\mathbb{F}_5$	120	384	46080
$\mathbb{F}_7$	336	1152	387072
$\mathbb{F}_{11}$	1320	5760	7603200
$\mathbb{F}_{13}$	2184	6912	15095808
$\mathbb{Z}/4$	48	192	9216
$\mathbb{Z}/8$	384	6144	2359296
$\mathbb{Z}/9$	648	2592	1679616

For $p \geq 5$,

$$|\mathcal{M}(\mathbb{F}_p)| = 4(p-1)\big(p-(\tfrac{-1}{p})\big)\big(p-(\tfrac{-3}{p})\big)p(p^2-1).$$

The individual maps $A \mapsto [A, r]$ and $A \mapsto [A, s]$ have the small-ring images described below; they must be distinguished from the everywhere bijective combined parametrizations of B.1.

Proof. For a finite ring R put $n_i(v) = \#\{(x, y) \in R^2 : Q_i(x, y) = v\}$ and $E(R) = \{e \in R^\times : e^2 = 1\}$. Then the equations give

$$|\mathcal{H}(R)| = \sum_{e,e' \in E(R)} \sum_{v \in R^\times} n_2(v)n_3(ee'v), \qquad |\mathcal{M}(R)| = |\,\mathrm{SL}_2(R)||\mathcal{H}(R)|.$$

The second equality is an equality of point sets from an explicit section, not an assumption about exactness on points. For $p \geq 5$ each nonzero norm has $p-1$ preimages in the split quadratic algebra and $p+1$ in the quadratic field. This follows respectively from $xy = v$ and the cyclic multiplicative group of the field of p^2 elements. The splitting discriminants are -1 and -3, proving the formula. The factor $|\,\mathrm{SL}_2(\mathbb{F}_p)| = p(p^2-1)$ follows by choosing a nonzero first column and then one of the p second columns with determinant 1.

For $p = 3$, $Q_3 = (c-d)^2$, so the nonzero-value counts are $n_3(1) = 6, n_3(2) = 0$ and $n_2(1) = n_2(2) = 4$. For $p = 2$ they are $n_3(1) = 3, n_2(1) = 2$. For $4, 8, 9$ the code enumerates the quadratic values and, independently, every invertible matrix, its determinant, its centralizer or signed-normalizer condition, and every admissible pair. The two methods agree on each count.

For completeness, write a matrix as its four row-major entries. The only points missed by the GL_2 commutator maps on the full numerical targets are

	r	s
$\mathbb{F}_2$	$\varnothing$	$\{(0,1,1,0)\}$
$\mathbb{F}_3$	$\{(2,1,2,0)\}$	$\varnothing$
$\mathbb{Z}/4$	$\varnothing$	$\varnothing$.

They are exactly the nonregular points. For SL_2 the missed sets are the same in the empty cases, with the following two replacements:

$$\begin{aligned} r, \mathbb{F}_3 : \quad & \{(0,2,1,2),(1,1,2,0),(2,0,2,2),\\ & (2,1,0,2),(2,1,2,0)\};\\ s, \mathbb{Z}/4 : \quad & \{(2,1,1,3),(2,3,3,3),(3,0,0,3),\\ & (3,1,1,2),(3,2,2,3),(3,3,3,2)\}. \end{aligned}$$

The $s, \mathbb{F}_2$ singleton still remains. A.5 explains these images by the regularity and norm obstructions; exhaustive enumeration supplies every point, not only a witness. Over $\mathbb{Z}$ the complete missed sets are the inequalities in A.5. All outputs are retained. □

B.8. General pairs and the fundamental realization hypothesis [Open]

For an affine group scheme G of finite presentation, fixed sections ρ, σ, and a fixed morphism $\ell : \mathbb{G}_a \to G$, the functor

$$(\alpha, \beta, \nu) \longmapsto [\alpha, \rho] = [\beta, \sigma]\ell(\nu)$$

is represented by a closed subscheme of $G^2 \times \mathbb{G}_a$. Adding a specified central sign group and prescribed characters is again a fibre-product construction. This representability does not prove the automorphism-realization hypothesis (37) of C3:501, fol.553, or identify every general group of §49 with the modular matrix model.

Proved part and exact obstruction. The two sides are morphisms to the affine separated scheme G; their equalizer is a closed subscheme and represents the functor. The equality is compatible with every base change. In the modular GL_2 case B.1 provides an explicit inverse, and hence an actual transported group law over every base, including $\mathbb{F}_2, \mathbb{F}_3, \mathbb{Z}/4$. Those finite cases were computed above. With no signs the equal-norm group $\mathcal{H}^0$ and the group $\mathcal{M}^0$ are already constructed and smooth over $\mathbb{Z}$; adjoining the two signs is the explicit extension of B.4.

For a topologically generated group $C = \langle \rho, \sigma \rangle$, the assignment of the images of its generators determines at most one continuous homomorphism. Existence requires that every defining relation of C remain true under $\rho \mapsto \alpha\rho\alpha^{-1}$ and $\sigma \mapsto \beta\sigma\beta^{-1}$; invertibility additionally requires an automorphism. The single loop equation alone supplies neither implication. B.12 gives a finite counterexample even when the proposed images generate and the loop exponent is a unit. This is precisely the additional assumption in C3:501 (37)(b).

For the specific completed modular group, existence has a stronger answer. Fix

$$C = \widehat{\mathrm{SL}_2(\mathbb{Z})}, \qquad C^{\pm} = C \rtimes \langle \tau \rangle, \qquad \omega = r^3 = s^2, \quad \omega^2 = 1, \tag{B.8.1}$$

where τ is the involution induced by conjugation by $\mathrm{diag}(-1, 1)$ on the discrete matrix group and then by continuity on its profinite completion. Thus $\tau(s) = s^{-1}$, $\tau(r) = sr^{-1}s^{-1}$, and $\tau(\omega) = \omega$. Let $\nu_2, \nu_3 : \widehat{\mathbb{Z}}^{\times} \to \mathbb{Z}/2$ be defined by $(-1)^{\nu_2(\mu)} = \varepsilon_2(\mu)$ modulo 4 and $(-1)^{\nu_3(\mu)} = \varepsilon_3(\mu)$ modulo 3. Define Σ_C^{sgn} to be the set of triples $\alpha \in C^{\pm}$, $\beta \in C$, $\mu \in \widehat{\mathbb{Z}}^{\times}$ such that the parity of α is $\nu_3(\mu)$ and

$$[\alpha, r] = \omega^{\nu_2(\mu)}[\beta, s](rs^{-1})^{\mu-1}. \tag{B.8.2}$$

The scalar sign is written as a power of the actual central element ω, so (B.8.2) is an equation in C.

Every such triple defines a unique continuous endomorphism $u_{\alpha,\beta,\mu} : C \to C$ by

$$r \longmapsto \alpha r \alpha^{-1}, \qquad s \longmapsto \omega^{\nu_2(\mu)} \beta s \beta^{-1}. \tag{B.8.3}$$

Indeed, the cube of the first image and the square of the second are both ω. The images therefore satisfy the presentation $\langle r, s \mid r^6 = s^4 = 1,\ r^3 = s^2 \rangle$ proved in C.2. They give a homomorphism from the discrete $\mathrm{SL}_2(\mathbb{Z})$ into C. The inverse image of every open normal subgroup of C has finite index, so this homomorphism is continuous for the profinite topology and extends uniquely to its completion. This proves the assertion without assuming automorphism realization.

Put $\ell_0 = s^{-1}r$ and $\ell_1 = rs^{-1}$. Multiplying (B.8.2) gives

$$\begin{aligned} u_{\alpha,\beta,\mu}(\ell_0) &= \omega^{\nu_2(\mu)} s^{-1}[\beta, s]^{-1}[\alpha, r] r \\ &= s^{-1}\ell_1^{\mu-1} r = \ell_0^{\mu}. \end{aligned} \tag{B.8.4}$$

The parity of α and the displayed formula for $\tau(r)$ show that its image of r is C-conjugate to $r^{\varepsilon_3(\mu)} = r^{\mu}$; its image of s is C-conjugate to $s^{\varepsilon_2(\mu)} = s^{\mu}$. This is the signed numerical

condition of C3:534–535. In the convention of B.13, $e_1 = rs$ and the squared loop is e_1^2. Since $rs^{-1} = \omega rs$ and $\mu - 1$ is even,

$$(rs^{-1})^{\mu-1} = (rs)^{\mu-1} = \big((rs)^2\big)^{(\mu-1)/2}.$$

Thus the two loop conventions agree with the stated exponent; no factor of two or central sign is discarded.

The exact group-level assertion remaining here is

$$\text{for every } (\alpha, \beta, \mu) \in \Sigma_C^{\mathrm{sgn}}, \quad u_{\alpha,\beta,\mu} \text{ is surjective.} \tag{B.8.5}$$

Equivalently, the two images in (B.8.3) must topologically generate C. A surjective continuous endomorphism of this finitely generated profinite group is an automorphism. To check the last implication, there are only finitely many open normal subgroups of any given index. Inverse image under a surjective endomorphism permutes this finite set. Every such subgroup N therefore satisfies $u^{-k}N = N$ for some $k > 0$, whence $\ker u \subset N$; intersecting all N gives trivial kernel. We have not proved (B.8.5). B.11 proves the formal unsigned construction, and B.13 proves the formal signed construction on the effective subset. The executed $S_3, S_4, \mathrm{SL}_2(\mathbb{F}_3), \mathrm{SL}_2(\mathbb{Z}/4)$ cases in B.11 and B.14 verify realization in those finite cases. They do not establish (B.8.5) for C. The first separate congruence-to-GT lifting instance in our computation is the modulus-3 tuple of D.7; the arithmetic comparison, its central lift, and its image and kernel are additional questions recorded there.

There is also a precise elementary obstruction to replacing this profinite problem by an unqualified integral fppf statement. Let

$$H^{\mathrm{const}} = \varprojlim_n (\mathbb{Z}/n)_{\mathbb{Z}}$$

be the constant profinite group scheme attached to $\widehat{\mathbb{Z}}$. There is no group-scheme morphism $H^{\mathrm{const}} \to \mathrm{SL}_{2,\mathbb{Z}}$ sending 1 to $L_0(1)$. Its coordinate ring is the filtered union of the finite constant coordinate rings. Since SL_2 is affine of finite presentation, any morphism from H^{const} factors, as a scheme morphism, through $(\mathbb{Z}/n)_{\mathbb{Z}}$ for some n. Its image on $\mathbb{Z}$-points is then finite. A homomorphism sending 1 to $L_0(1)$ would instead send every integer m to the distinct matrix $L_0(m)$, a contradiction. The same argument rules out extending the standard modular matrices from the constant profinite scheme attached to C. Direct multiplication gives $L_0(a)L_0(b) = L_0(a+b)$, which proves the asserted power identity; D.17 records its compatible finite images.

An integral realization must consequently specify an adic or formal construction, or a different schematic envelope, and prove its own representability and comparison maps. Such a construction is not supplied by the abstract profinite fibre products, and is not identified with D.7's finite lifting question. We do not assert an fppf quotient between undefined integral parameter functors. □

B.9. Removing a global sign is not a scheme quotient on adelic points [Proved]

Let K be a profinite group with a specified continuous homomorphism $\phi : K \to \mathrm{GL}_2(\widehat{\mathbb{Z}})/\{\pm I\}$, where the sign subgroup is diagonal and has order 2. Then

$$\widetilde{K} = K \times_{\mathrm{GL}_2(\widehat{\mathbb{Z}})/\{\pm I\}} \mathrm{GL}_2(\widehat{\mathbb{Z}})$$

is a canonical profinite central extension of K by C_2, equipped with a genuine matrix representation. It has a continuous splitting exactly when ϕ has a continuous homomorphic lift. On the other hand the natural map

$$\mathrm{GL}_2(\widehat{\mathbb{Z}})/\{\pm I\} \longrightarrow (\mathrm{GL}_2/\mu_2)(\widehat{\mathbb{Z}})$$

has kernel $\mu_2(\widehat{\mathbb{Z}})/\{\pm 1\}$, which is infinite.

Proof. The fibre product is closed in a product of profinite groups. Projection onto K is onto because the quotient map is onto, and its kernel is the specified diagonal C_2. A splitting and a lift give one another by their two components. For the scheme quotient, exactness as fppf sheaves says that the kernel on points is $\mu_2(\widehat{\mathbb{Z}})$. A unit of $\mathbb{Z}_p$ with square 1 is 1 or -1 since $\mathbb{Z}_p$ is a domain; hence

$$\mu_2(\widehat{\mathbb{Z}}) = \prod_p \{\pm 1\}.$$

Quotienting by the diagonal sign proves the asserted kernel. For example the scalar equal to $-I$ at 5 and I elsewhere is a nontrivial element of it. This gives a concrete obstruction to restating the passage of C3:492 as an unqualified statement about the integral scheme quotient. Identification of the particular $\widetilde{K}$ with the manuscript's $\widetilde{N}_{1,1}$ needs the specified geometric extension class; the universal pullback construction by itself is not that identification. □

B.10. Lead 2 with its strongest explicit modular replacement [Refuted]

The final six-dimensional $\mathcal{M}_{r,s}$ is not the scheme of pairs (α, β) in the preliminary special-linear loop equation, and its T_r, T_s factors are not the norm-one tori. The precise replacements are the schemes, dimensions and split extension in B.1–B.4. The GL_2 centralizers have dimension 2 and the norm-one kernels dimension 1; the final $\mathcal{M}$ is an SL_2 extension of the three-dimensional equal-squared-norm group $\mathcal{H}$.

Proof. The coordinate algebras and inverse maps prove each identification. A dimension-4 precursor cannot be isomorphic to the flat dimension-6 final group. Over an algebraic closure in characteristic different from $2, 3$, T_r, T_s are $\mathbb{G}_m^2$, whereas their norm-one kernels are $\mathbb{G}_m$; dimensions alone disprove their identification. Their degeneration at the ramified primes is explicitly computed in B.3 and in the norm equations, so no torus terminology is extended across those primes without qualification. □

B.11. The general unsigned construction, with its hypotheses [Proved]

Let C be a discrete group, or a topologically finitely generated profinite group, generated by ρ, σ. Put $L_0 = \overline{\langle \sigma^{-1}\rho \rangle}$ (omit the closure in the discrete case), and let $B \subset \mathrm{Aut}(C)$ be the full subgroup preserving both conjugacy classes and normalizing L_0. Write

$$K = L_0 \cap Z(C), \quad \gamma = B/\operatorname{Int}(L_0), \quad G = (C \rtimes B)/\{(\ell^{-1}, \operatorname{Int}\ell) : \ell \in L_0\}.$$

All homomorphisms below are continuous in the profinite case. The map $B \to G$ is injective, the kernel of $C \to G$ is exactly K, and

$$1 \longrightarrow C/K \longrightarrow G \longrightarrow \gamma \longrightarrow 1$$

is exact. Assume now $K = 1$. Put $Z_\rho = C_G(\rho)$, $Z_\sigma = C_G(\sigma)$, $C_\rho = C_C(\rho)$ and $C_\sigma = C_C(\sigma)$. The arrows $Z_\rho, Z_\sigma \to \gamma$ are onto, and the groups

$$T^0 = Z_\rho \times_\gamma Z_\sigma, \quad M^0 = T^0 \times_\gamma G, \quad SM^0 = B \times_\gamma M^0$$

satisfy

$$1 \to C_\rho \times C_\sigma \to T^0 \to \gamma \to 1,$$
$$1 \to C_\rho \times C_\sigma \times C \to M^0 \to \gamma \to 1,$$
$$1 \to L_0 \to SM^0 \to M^0 \to 1,$$
$$1 \to L_0 \times C_\rho \times C_\sigma \times C \to SM^0 \to \gamma \to 1.$$

These are the corrected fibre-product sequences of C3:505–513, §49.II, fols.555–559 (parallel overwritten folios504–512). If the realization map $\Phi : B \to \Sigma$ of C3:503, (9)–(10), is bijective,

the parametrizations (34), (38) are bijective, with the inverse and transported group law described below.

Proof. We use $(c,u)(c',u') = (c\,u(c'), uu')$ in the semidirect product. The map $i(\ell) = (\ell^{-1}, \operatorname{Int}\ell)$ is an injective homomorphism. Its image centralizes C, and conjugation by $u \in B$ sends $i(\ell)$ to $i(u\ell)$, proving normality. In the profinite case it is closed because L_0 is compact. The automorphisms of a finitely generated profinite group form a profinite group: there are only finitely many subgroups of each index, their intersections give a cofinal family of characteristic open subgroups, and compatible automorphisms of their finite quotients are exactly continuous automorphisms of C. The conditions on B are closed, since conjugacy classes are compact and $u(L_0) = L_0$ is a closed condition (use both u and u^{-1}). Thus all quotients have the asserted topology.

An element $(c,1)$ belongs to $i(L_0)$ exactly when $c = \ell^{-1}$ and $\operatorname{Int}\ell = 1$, proving the kernel formula. An element $(1,u)$ can belong to $i(L_0)$ only for $\ell = 1, u = 1$, proving injectivity of B. Projection onto $B/\operatorname{Int}(L_0)$ has kernel precisely the image of C. The construction is universal for two homomorphisms from C, B respecting the given action and identifying $\ell \in L_0$ with $\operatorname{Int}\ell \in B$: this follows from the universal property of a semidirect product and its displayed kernel. For example if $C = C_6$ with generators of orders $3, 2$, then $B = 1, L_0 = C, K = C, G = 1$. Omitting K would be false; the manuscript already corrects the injectivity arrow at scan505.

Given $u \in B$, choose $a \in C$ with $u(\rho) = a\rho a^{-1}$. Then $a^{-1}u \in Z_\rho$ maps to the same element of γ as u. This proves surjectivity; the proof for σ is identical. Their kernels are exactly C_ρ, C_σ. Taking fibre products proves every displayed kernel and surjectivity. The projection $SM^0 \to M^0$ is the pullback of $B \to \gamma$, whose kernel is L_0 because $K = 1$. These arguments prove the corresponding statements for group objects in a sheaf topos when the displayed maps are epimorphisms and the quotients exist there. They do not assert representability of every general quotient.

Here Σ consists of $(\zeta,\eta) \in C^2$ such that $\zeta\rho$ is conjugate to ρ, $\eta\sigma$ is conjugate to σ, and $\eta^{-1}\zeta \in L_1 = \rho L_0 \rho^{-1}$. Assume that $\Phi(u) = (u(\rho)\rho^{-1}, u(\sigma)\sigma^{-1})$ is bijective. From a loop pair $(\alpha,\beta) \in G^2$, recover

$$u = \Phi^{-1}([\alpha,\rho],[\beta,\sigma]), \qquad a = \alpha^{-1}u \in Z_\rho, \quad b = \beta^{-1}u \in Z_\sigma.$$

Conversely $(u,a,b) \mapsto (ua^{-1}, ub^{-1})$ gives that pair. This proves (34). Its restriction to C^2 is exactly the fibre product $B \times_\gamma Z_\rho \times_\gamma Z_\sigma$, proving (38). Transporting componentwise multiplication gives

$$(\alpha',\beta')(\alpha,\beta) = (u'(\alpha)\alpha', u'(\beta)\beta'),$$

where u' acts by conjugation in G and by its given automorphism on C. This proves associativity without assuming that the loop-pair set was already a group.

The symbols ua^{-1}, ub^{-1} at scan510, (38), must therefore be α,β, not the inserted ζ,η, which are their commutators. In scan511, (44), the second SZ_ρ must be SZ_σ. In scan509, (31), the final exponent on ρ must be -1. These are mathematical readings, not claims of final palaeographic settlement.

The finite program constructs all automorphisms by checking generator images against every Cayley-graph edge. For $(C,\rho,\sigma) = (S_3,(123),(12))$ and $(S_4,(123),(34))$ it proves $\Sigma = \Phi(B)$ and obtains

C	$\lvert B\rvert$	$\lvert L_0\rvert$	$\lvert\gamma\rvert$	$\lvert G\rvert$	$\lvert Z_\rho\rvert$	$\lvert Z_\sigma\rvert$	$\lvert M^0\rvert$	$\lvert SM^0\rvert$
S_3	2	2	1	6	3	2	36	72
S_4	8	4	2	48	6	8	576	2304

The algorithms and exact assertions are in `AB_general_finite.py`, with its JSON and text output. □

B.12. The loop equation does not imply realization [Refuted]

In $C = \mathrm{PSL}_2(\mathbb{F}_{13})$ there are two generating pairs (ρ, σ) and (ρ', σ') whose corresponding entries are conjugate, of respective orders $3, 2$ with

$$(\sigma')^{-1}\rho' = (\sigma^{-1}\rho)^2, \qquad |\sigma^{-1}\rho| = 7,$$

yet no homomorphism $C \to C$ sends the first pair to the second. Thus the realization assumption of C3:503, (10)(b), cannot be deleted, even when the loop multiplier is a unit. The general replacement is B.11 with realization verified separately, as it is for the matrix parametrization in B.1.

Proof. All matrices denote projective classes over $\mathbb{F}_{13}$. Take

$$\rho = \begin{pmatrix} 1 & 2 \\ 5 & 11 \end{pmatrix}, \quad \sigma = \begin{pmatrix} 0 & 1 \\ 12 & 0 \end{pmatrix}, \quad \ell = \sigma^{-1}\rho = \begin{pmatrix} 5 & 11 \\ 12 & 11 \end{pmatrix},$$
$$\rho' = \begin{pmatrix} 2 & 5 \\ 9 & 10 \end{pmatrix}, \quad \sigma' = \begin{pmatrix} 1 & 4 \\ 6 & 12 \end{pmatrix}.$$

Conjugators are $\alpha = \left(\begin{smallmatrix} 6 & 6 \\ 10 & 8 \end{smallmatrix}\right)$ and $\beta = \left(\begin{smallmatrix} 6 & 8 \\ 7 & 3 \end{smallmatrix}\right)$. Direct multiplication verifies their determinants, conjugacies, the loop identity and the orders. Breadth-first enumeration from each pair produces all $13(13^2 - 1)/2 = 1092$ projective matrices, proving generation. But

$$[\sigma, \rho] = \begin{pmatrix} 3 & 12 \\ 12 & 5 \end{pmatrix} \quad \text{has order } 7, \qquad [\sigma', \rho'] = \begin{pmatrix} 2 & 0 \\ 11 & 7 \end{pmatrix} \quad \text{has order } 6.$$

The relation $[\sigma, \rho]^7 = 1$ is not preserved, so no homomorphism with these images exists.

With $\zeta = \rho'\rho^{-1}$ and $\eta = \sigma'\sigma^{-1}$, one also checks $\eta^{-1}\zeta = \rho\sigma^{-1} = \varepsilon_1$, exactly scan503's condition. The exponent $2 \bmod 7$ is a unit. For the profinite formulation it lifts to $\widehat{\mathbb{Z}}^\times$ by taking its 7-adic component 2 and all other components 1; the integer 2 is not itself a profinite unit. Every finite equality, order and generation assertion above was executed by the attached program. Its output retains an independent failed relation from the Cayley-graph consistency algorithm as well. □

B.1 The general signed construction and its exact fibres

B.13. Signed normalizers, numerical data, and effective conjugator pairs [Proved]

Let C be a profinite group topologically generated by ρ, σ, with

$$\rho^6 = \sigma^4 = 1, \qquad \omega := \rho^3 = \sigma^2,$$

and suppose that the automorphism τ defined on these generators by

$$\tau(\sigma) = \sigma^{-1}, \qquad \tau(\rho) = \sigma\rho^{-1}\sigma^{-1} \tag{B.13.1}$$

exists. Set $e_0 = \sigma\rho$, $e_1 = \rho\sigma$, $\ell_1 = e_1^2$, and $L_0 = \overline{\langle e_0 \rangle}$. Write $A = \widehat{\mathbb{Z}}^\times$, and let $\epsilon_3 : A \to \{\pm 1\}$ and $\epsilon_2 : A \to \{\pm 1\}$ be reduction modulo 3 and 4. The groups of C3:530, (7), and C3:549, (6), are respectively

$$\begin{aligned} B &= \{(u, \mu) : u(e_0) = e_0^\mu,\ u(\rho) \sim_C \rho^{\epsilon_3(\mu)},\ u(\sigma) \sim_C \sigma^{\epsilon_2(\mu)}\}, \\ Z &= \{(u, \mu, e_r, e_s) : u(e_0) = e_0^\mu,\ u(\rho) \sim_C \rho^{e_r},\ u(\sigma) \sim_C \sigma^{e_s}\}, \end{aligned} \tag{B.13.2}$$

where $u \in \mathrm{Aut}(C)$, $\mu \in A$, and $e_r, e_s \in \{\pm 1\}$. We order the two recorded signs as (ρ, σ); the manuscript often orders them as (σ, ρ). The embedding $B \hookrightarrow Z$ supplies the two displayed characters of μ. It is normal of index 1, 2, or 4.

For either $D = B$ or $D = Z$, retain these two characters, denoted χ_r, χ_s, as well as the multiplier $\chi : D \to A$. Put

$$\begin{gathered} j(\ell) = (\mathrm{Int}\,\ell, 1, 1, 1), \quad K = L_0 \cap Z(C), \quad \gamma_D = D/j(L_0), \\ G_D = (C \rtimes D)/\{(\ell^{-1}, j(\ell)) : \ell \in L_0\}. \end{gathered} \tag{B.13.3}$$

In the B case the last two coordinates of j are suppressed. Then $D \to G_D$ is injective, $\ker(C \to G_D) = K$, and $1 \to C/K \to G_D \xrightarrow{\delta} \gamma_D \to 1$ is exact. Assume $K = 1$ from now on. The three characters descend to γ_D. The element $t = (\tau, -1, -1, -1) \in D$ has order 2, and

$$\begin{gathered} C^{\pm} := C \rtimes \langle t \rangle \hookrightarrow G_D, \qquad t' = \sigma^{-1} t, \\ (t')^2 = 1, \quad t'\rho(t')^{-1} = \rho^{-1}, \quad t'\sigma(t')^{-1} = \sigma^{-1}. \end{gathered} \tag{B.13.4}$$

Here $C^{\pm}$ denotes the whole extension.

For $x = \rho, \sigma$, write $C_x = C_C(x)$, $Z_x = C_{G_D}(x)$, and $\chi_x = \chi_r, \chi_s$, respectively. Define

$$\begin{aligned} E_x &= \{e \in \{\pm 1\} : x^e \sim_C x\}, \\ N_x &= \{(g, e) \in G_D \times \{\pm 1\} : gxg^{-1} = x^e\}, \\ N_x^* &= \{(g, e) \in N_x : e = \chi_x(\delta g)\}, \qquad Z_x^* = Z_x \cap \delta^{-1}(\ker \chi_x). \end{aligned} \tag{B.13.5}$$

These formulas realize the external semidirect products N_x of C3:540–541 and 555 without conflating them with their images in G_D. The precise image and kernel statements are

$$\begin{gathered} \delta(Z_x) = \chi_x^{-1}(E_x), \qquad \mathrm{im}(N_x \to G_D) = N_{G_D}(\langle x \rangle), \\ \ker(N_x \to G_D) = \{(1, e) : x^e = x\}, \qquad N_x^* \longrightarrow G_D \text{ is always injective}, \\ 1 \longrightarrow C_x \longrightarrow N_x^* \xrightarrow{\delta} \gamma_D \longrightarrow 1, \\ 1 \longrightarrow Z_x^* \longrightarrow N_x^* \xrightarrow{e} \{\pm 1\} \longrightarrow 1. \end{gathered} \tag{B.13.6}$$

The second sequence splits by $-1 \mapsto (t', -1)$. Thus $N_x \to G_D$ has a kernel of order 2 exactly when $x^2 = 1$; the starred map remains injective in that case.

The signed groups of C3:559–560, (50)–(59), are the fibre products

$$\begin{aligned} \Gamma_D &= N_\rho^* \times_{\gamma_D} N_\sigma^*, \qquad S_0\Gamma_D = D \times_{\gamma_D} \Gamma_D, \\ \mathcal{M}_D &= \Gamma_D \times_{\gamma_D} G_D, \quad S_0\mathcal{M}_D = D \times_{\gamma_D} \mathcal{M}_D. \end{aligned} \tag{B.13.7}$$

Their kernels over γ_D are, respectively,

$$\begin{aligned} &C_\rho \times C_\sigma, \quad L_0 \times C_\rho \times C_\sigma, \\ &C_\rho \times C_\sigma \times C, \quad L_0 \times C_\rho \times C_\sigma \times C. \end{aligned} \tag{B.13.8}$$

Every arrow to γ_D is onto. Forgetting the D coordinate gives extensions by L_0; forgetting the G_D coordinate gives extensions by C. The square $S_0\mathcal{M}_D = \mathcal{M}_D \times_{\Gamma_D} S_0\Gamma_D$ is cartesian.

Finally, put

$$\Lambda = \{\mu \in A : e_0^\mu = e_0\}.$$

The kernels of forgetting numerical coordinates and retaining only the automorphism u are exactly

$$\begin{aligned} \ker(Z \to \mathrm{Aut}\, C) &= \Lambda \times E_\rho \times E_\sigma, \\ \ker(B \to \mathrm{Aut}\, C) &= \{\mu \in \Lambda : \epsilon_3(\mu) \in E_\rho, \epsilon_2(\mu) \in E_\sigma\}. \end{aligned} \tag{B.13.9}$$

They are central. These kernels must not be confused with the manuscript's distinction between numerical solutions and effective solutions: "effective" there means that the two proposed generator images extend to an automorphism, and the numerical data are still kept. There is an exact conjugator parametrization of that effective subset, described in the proof, with fibres $C_\rho \times C_\sigma$.

Proof. The element ω commutes with both generators and hence with C. Formula (B.13.1) gives $\tau^2 = 1$ on both generators and gives $\tau(e_0) = e_0^{-1}$. It also gives $\tau(e_1) = e_1^{-1}$, using $\sigma^2 = \omega$ and centrality of ω. Composition preserves (B.13.2): for example, if $v(\rho)$ is conjugate to ρ^f and $u(\rho)$ to ρ^e, then $uv(\rho)$ is conjugate to ρ^{ef}. The same calculation for σ, and $u(e_0^\nu) = e_0^{\mu\nu}$, proves closure under composition; applying it to inverses proves the subgroup assertion. The kernel of

$$Z \longrightarrow \{\pm 1\}^2, \qquad (u, \mu, e_r, e_s) \longmapsto (e_r \epsilon_3(\mu)^{-1}, e_s \epsilon_2(\mu)^{-1})$$

is precisely B. This proves normality and the possible indices. The conditions are closed in the product of the profinite groups $\mathrm{Aut}(C)$ and the numerical factors: conjugacy classes are compact, and the power map on L_0 is continuous. The profiniteness of $\mathrm{Aut}(C)$ for topologically finitely generated C was justified in B.11. All groups and arrows in this proof carry these topologies.

Every u in D carries $\ell \in L_0$ to $\ell^{\chi(u)}$. Consequently $j(L_0)$ is normal in D, with kernel K in L_0. In $C \rtimes D$, the antidiagonal in (B.13.3) is an injective image of L_0; it centralizes C, and conjugation by $d \in D$ carries its element indexed by ℓ to the one indexed by $u(\ell)$. It is therefore closed and normal. Intersecting it with C and with D gives the kernel and injectivity assertions in (B.13.3). Projection to $D/j(L_0)$ gives the displayed extension. This also proves the universal property: homomorphisms from C, D respecting the given action factor through G_D exactly when they identify $\ell \in C$ with $j(\ell) \in D$ for every $\ell \in L_0$.

The element t belongs to D by (B.13.1), and $t \notin C$ in G_D because $\chi(t) = -1$, whereas $\chi(C) = 1$. The resulting map $C \rtimes \langle t \rangle \to G_D$ is injective: its intersection with C is injective, and its two cosets have distinct multiplier. Direct multiplication gives $(\sigma^{-1}t)^2 = \sigma^{-1}\tau(\sigma^{-1}) = 1$ and both inversion identities in (B.13.4).

An element $g \in G_D$ above $d \in \gamma_D$ sends x to a C-conjugate of $x^{\chi_x(d)}$. Some C-multiple of g centralizes x if and only if $\chi_x(d) \in E_x$. This proves the first image formula in (B.13.6), including both directions. The only generators of a cyclic group of order dividing 6 or 4 are x, x^{-1}, with possible equality; this proves the normalizer image assertion. The kernel formula follows by putting $g = 1$. The sign map on N_x is split by $(t', -1)$, and its kernel is Z_x. Thus this definition agrees with the external semidirect product $Z_x \rtimes C_2$ even when $t' \in Z_x$.

For a fixed $d \in \gamma_D$, choose g above it and $c \in C$ with $gxg^{-1} = cx^{\chi_x(d)}c^{-1}$. Then $(c^{-1}g, \chi_x(d)) \in N_x^*$ maps to d. The fibre at 1 consists exactly of $(c, 1)$ with $c \in C_x$. If $(1, e) \in N_x^*$, its sign is forced to be $\chi_x(1) = 1$; this proves injectivity for every order of x. Finally the kernel of its sign map is Z_x^*, and $(t', -1)$ is a splitting because $\chi_x(\delta t') = -1$. Taking fibre products now proves every surjectivity, kernel, extension and cartesian assertion of (B.13.7)–(B.13.8). In particular the upper Z_x in C3:556, (41), (43), has to mean Z_x^*; otherwise it need not be a subgroup of N_x^*. The numerical kernel formulas (B.13.9) follow by setting $u = 1$ in (B.13.2). They are central because the numerical factors are abelian and their automorphism coordinate is the identity. No surjectivity of $\chi : \gamma_D \to \widehat{\mathbb{Z}}^\times$ has been used or asserted.

We give the promised exact parametrization. For $D = Z$, let Σ be the set of $(\xi, \eta, \mu, e_r, e_s)$ such that

$$\xi\rho \sim_C \rho^{e_r}, \quad \eta\sigma \sim_C \sigma^{e_s}, \quad \xi = \eta\ell_1^\nu, \qquad \nu = (\mu - 1)/2. \tag{B.13.10}$$

For $D = B$, prescribe $e_r = \epsilon_3(\mu)$, $e_s = \epsilon_2(\mu)$. Division by 2 here is unique in $\widehat{\mathbb{Z}}$ because every profinite unit is odd. Let Σ^{eff} be the subset for which an automorphism u sends (ρ, σ) to

$(\xi\rho, \eta\sigma)$. The automorphism is unique by topological generation. It fixes ω, since its image of σ is conjugate to $\sigma^{\pm 1}$. For $e_0' = \sigma^{-1}\rho = \omega e_0$, the equation $u(e_0) = e_0^{\mu}$ is consequently equivalent to $u(e_0') = (e_0')^{\mu}$. Multiplying this latter equation gives

$$\eta^{-1}\xi = (\rho\sigma^{-1})^{\mu-1} = (\rho\sigma)^{\mu-1} = \ell_1^{\nu}.$$

The middle equality uses the evenness of $\mu - 1$ and $\omega^2 = 1$. Thus the map

$$d \longmapsto \bigl(u(\rho)\rho^{-1}, u(\sigma)\sigma^{-1}, \mu, e_r, e_s\bigr)$$

is a bijection $D \to \Sigma^{\mathrm{eff}}$. This statement keeps the existence condition; it does not identify all of Σ with the effective subset.

Let P_D^{eff} retain, in addition, conjugators $\alpha, \beta \in C^{\pm}$ of orientations e_r, e_s with $u(\rho) = \alpha\rho\alpha^{-1}$ and $u(\sigma) = \beta\sigma\beta^{-1}$. Each fibre over d is a nonempty right torsor under $C_\rho \times C_\sigma$: the class conditions supply conjugators of the required orientation using t', and two such conjugators of the same orientation differ by the corresponding centralizer in C. Write $e_r = (-1)^{v_r}$, $e_s = (-1)^{v_s}$. Using $i_x : N_x^* \hookrightarrow G_D$, the bijection is

$$\begin{aligned} S_0\Gamma_D &\longrightarrow P_D^{\mathrm{eff}}, \\ (d, a, b) &\longmapsto (\alpha, \beta, \mu, e_r, e_s), \\ \alpha = d\, i_\rho(a)^{-1}(t')^{v_r}, &\qquad \beta = d\, i_\sigma(b)^{-1}(t')^{v_s}. \end{aligned} \tag{B.13.11}$$

Equal images in γ_D show that $d\, i_\rho(a)^{-1}$ and $d\, i_\sigma(b)^{-1}$ belong to C. Inverting the prescribed actions on ρ, σ verifies both conjugation identities. Conversely, recover d from the effective images and numerical data, and recover a, b by

$$i_\rho(a) = (t')^{v_r}\alpha^{-1}d, \qquad i_\sigma(b) = (t')^{v_s}\beta^{-1}d. \tag{B.13.12}$$

These elements have the required signs and the same γ_D image as d. Their lifts to the starred groups are unique even if $\rho^2 = 1$ or $\sigma^2 = 1$. This proves both inverseness and the exact fibre assertion. These bijections are homeomorphisms: the source fibre products are compact, the parameter spaces are Hausdorff subspaces of the displayed products, and the formulas are continuous. The parametrization also supplies an associative group law on the effective conjugator parameters by transporting the componentwise law on $S_0\Gamma_D$. It does not impose a group law on numerical solutions that fail realization.

These are statements about profinite groups. For the finite shadows in B.14 they give, after taking constant finite group schemes over $\mathbb{Z}$, exact sequences of fppf sheaves: every quotient map of finite constant groups is already locally split on each constant component. The same formal kernel arguments apply to group objects in a sheaf category with locally stated conjugacy hypotheses and the indicated quotients. This does not prove representability or finite presentation for an arbitrary group-scheme analogue of C; nor does it identify the full noncongruence groups with the matrix schemes of §48. □

B.14. Unrestricted centralizers and a left sign factor do not give the signed formulas [Refuted]

The unqualified inclusions $Z_x \subset N_x^*$ implicit in the upper rows of C3:556, (41), (43), are false in general. The exact replacement is Z_x^* in (B.13.5). Also, with conjugation $g(x) = gxg^{-1}$, the change of conjugator in C3:546, fol. 576, (64), must be

$$\beta' = \beta(t')^{v}, \qquad e_s = (-1)^{v}, \tag{B.14.1}$$

to obtain $\mathrm{Int}(\beta')(\sigma) = \omega^{v}\,\mathrm{Int}(\beta)(\sigma)$. The printed left factor $(t')^{v}\beta$ does not have that property.

The following are exact finite checks of the corrected signed construction. We replace $\widehat{\mathbb{Z}}^\times$ by $U_{12} = (\mathbb{Z}/12)^\times$ throughout the table, take $\rho = (123), \sigma = (12)$ in S_3, and $\rho = (123), \sigma = (34)$ in S_4. In the two matrix groups take

$$\rho = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, \qquad \sigma = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The symbols in each row refer to $D = B_{12}$ or Z_{12}, respectively, and to (B.13.3)–(B.13.8):

C	D	$\lvert D\rvert$	$\lvert\gamma_D\rvert$	$\lvert G_D\rvert$	$\lvert N_\rho^*\rvert$	$\lvert N_\sigma^*\rvert$	$\lvert\mathcal{M}_D\rvert$	$\lvert S_0\mathcal{M}_D\rvert$
S_3	B	8	4	24	12	8	144	288
S_3	Z	32	16	96	48	32	576	1152
S_4	B	16	4	96	12	16	1152	4608
S_4	Z	64	16	384	48	64	4608	18432
$\mathrm{SL}_2(\mathbb{F}_3)$	B	12	4	96	24	16	2304	6912
$\mathrm{SL}_2(\mathbb{F}_3)$	Z	24	8	192	48	32	4608	13824
$\mathrm{SL}_2(\mathbb{Z}/4)$	B	16	4	192	24	32	9216	36864
$\mathrm{SL}_2(\mathbb{Z}/4)$	Z	32	8	384	48	64	18432	73728

In all eight rows $\Sigma = \Sigma^{\mathrm{eff}}$, every fibre in (B.13.6) has the stated order, and (B.13.11) is bijective. These equalities of solution sets are finite-case results.

Proof. First take $C = S_3$ as in the table. Here $\tau = 1$ on C: τ fixes σ, and $\sigma\rho^{-1}\sigma^{-1} = \rho$. Thus $t = (1, -1) \in B$ centralizes all of C, but $\chi_r(\delta t) = \chi_s(\delta t) = -1$. Its positive-sign lift $(t, +1) \in N_x$ is not in N_x^*. This gives an explicit counterexample for both $x = \rho, \sigma$, also in the finite U_{12} shadow. For $\sigma^2 = 1$, the map $N_\sigma \to G_D$ has its additional kernel of order 2; the starred map has no such kernel. Thus neither a blind use of centralizers nor a blind identification of the external normalizer with its image is valid. Formula (B.13.6) is the proved correction.

For (B.14.1), multiplication on the right gives

$$\beta(t')^v\sigma(t')^{-v}\beta^{-1} = \beta\sigma^{(-1)^v}\beta^{-1} = \omega^v\beta\sigma\beta^{-1},$$

using $\sigma^{-1} = \omega\sigma$. The left factor instead conjugates the entire element $\beta\sigma\beta^{-1}$ by t'. The distinction is witnessed over $\mathbb{F}_3$ by $\beta = \rho$ and $t' = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$:

$$(t'\beta)\sigma(t'\beta)^{-1} = \begin{pmatrix} 2 & 2 \\ 2 & 1 \end{pmatrix}, \qquad \omega\beta\sigma\beta^{-1} = (\beta t')\sigma(\beta t')^{-1} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

The attached photograph of scan 546 was inspected: its (64) places the factor τ'^v on the left. The calculation diagnoses a mathematical slip under the stated conjugation convention; it does not claim a final editorial reading of overwritten text. The inverse in C3:544, (54), (56), must likewise be kept as $\beta = u\, i_\sigma(b)^{-1}$ if the equality of δ-images in its (57) is retained. The $b \in Z_\sigma$ printed under C3:545, (60), must be $b \in N_\sigma$, as its sign condition otherwise forces $\epsilon_2(\mu) = 1$.

We describe the executed finite verification so that its scope is reproducible. The standalone program `AB_signed_finite.py` builds the full multiplication tables of the four groups. For every possible pair of generator images of the correct orders, it propagates the proposed map through every edge of the right Cayley graph, rejects an inconsistent assignment, and retains exactly the bijective assignments. It obtains respectively $6, 24, 24, 48$ automorphisms. It constructs B_{12}, Z_{12} from their definitions, checks the antidiagonal normal subgroup, and forms the actual semidirect-product quotient G_D. It enumerates Z_x, N_x, N_x^*, checks every γ_D fibre and every kernel above, and compares all values and inverse values of (B.13.11)–(B.13.12). It also enumerates all numerical solutions of (B.13.10), so the asserted effectivity in these

cases is checked independently of the fibre-product parametrization. Its JSON contains the additional orders of $\Gamma_D, S_0\Gamma_D$, unrestricted normalizers, centralizers, and numerical kernels. The largest displayed product has 73,728 elements by these exact fibre counts.

For clarity, the inverse-conjugacy pairs (E_ρ, E_σ) are

$$(\{\pm1\},\{\pm1\}),\quad (\{\pm1\},\{\pm1\}),\quad (\{1\},\{\pm1\}),\quad (\{\pm1\},\{1\})$$

in the order of the four groups. The numerical kernel orders of $B_{12} \to \mathrm{Aut}(C)$ are $4,2,2,2$; those of $Z_{12} \to \mathrm{Aut}(C)$ are $16,8,4,4$. These calculations verify that forgetting the numerical parameters is not injective even though every numerical solution is effective in these examples.

All powers in the definitions factor through U_{12} because $|e_0| = 2,4,3,4$, respectively. The full profinite objects are $B = B_{12} \times_{U_{12}} \widehat{\mathbb{Z}}^\times$ and $Z = Z_{12} \times_{U_{12}} \widehat{\mathbb{Z}}^\times$, not the finite groups in the table. Reduction $\widehat{\mathbb{Z}}^\times \to U_{12}$ is onto: choose the required units at 2 and 3 and the unit 1 at every other prime. The finite table therefore verifies every signed condition relevant to these quotients without confusing a finite numerical shadow with the full profinite construction. □

B.2 Functoriality, congruence descent, and reconstruction

B.15. The exact descent criterion and the corrected reconstruction formalism [Proved]

We retain the definitions and topology of B.13, including the recorded numerical coordinates. Let $q : C \twoheadrightarrow C'$ be a continuous epimorphism of profinite groups carrying the ordered generators to (ρ', σ') and commuting with the involutions τ. Use either $D = B$ in both groups or $D = Z$ in both groups, and put $N = \ker q$. The maximal subgroup on which quotient automorphisms are defined is

$$D_q = \{d \in D : u_d(N) = N\}. \tag{B.15.1}$$

It is closed, contains $j(L_0)$ and t, and has a canonical continuous homomorphism $q_D : D_q \to D'$, preserving every numerical coordinate. Write $H = q_D(D_q)$, a closed subgroup of D'.

The following conditions are equivalent: $D_q = D$; every effective source numerical solution maps to an effective target solution; and the quotient automorphism $q_D(d)$ exists for every $d \in D$. Replacing normalization of L_0 by conjugacy of L_0 does not change the corresponding kernel-stability criterion. These are the precise equivalent conditions of C3:570, (89)–(92), folio 588. Uniqueness of the resulting map of the enlarged groups includes compatibility with both specified embeddings $C \to G_D$ and $D \to G_D$, as added in C3:571; compatibility with the induced inner actions alone is not being substituted for this condition.

Assume, for the next formulas, that $L_0 \cap Z(C) = L_0' \cap Z(C') = 1$. Form the B.13 groups using D_q and H, respectively, and set

$$A_q = \ker(D_q \to H), \qquad L_N = L_0 \cap N, \qquad K_G = \ker(G_{D_q} \to G_H).$$

The maps of D, γ, G onto their indicated targets are continuous epimorphisms, and their exact kernels are

$$\begin{aligned} K_\gamma &:= \ker(\gamma_{D_q} \to \gamma_H) \simeq A_q / j(L_N), \\ K_G &\simeq (N \rtimes A_q)/\{(\ell^{-1}, j(\ell)) : \ell \in L_N\}, \\ 1 &\longrightarrow N \longrightarrow K_G \longrightarrow K_\gamma \longrightarrow 1. \end{aligned} \tag{B.15.2}$$

For $x = \rho, \sigma$, the induced map of signed normalizers has kernel

$$K_x := \ker(N_x^* \to N_{x'}^*) = C_{K_G}(x). \tag{B.15.3}$$

It need not be onto. If $(g',e)\in N^*_{x'}$ and g is any lift of g', the exact obstruction is

$$(g',e)\text{ lifts}\quad\Longleftrightarrow\quad gxg^{-1}\text{ is }K_G\text{-conjugate to }x^e. \tag{B.15.4}$$

Every nonempty fibre is a right torsor under $C_{K_G}(x)$. The maps of $\Gamma,\mathcal{M},S_0\Gamma,S_0\mathcal{M}$ are the componentwise fibre-product maps. In particular their kernels are

$$\begin{aligned} K_\Gamma &= K_\rho\times_{K_\gamma}K_\sigma, \qquad K_{\mathcal{M}} = K_\Gamma\times_{K_\gamma}K_G,\\ K_{S_0\Gamma} &= A_q\times_{K_\gamma}K_\Gamma, \quad K_{S_0\mathcal{M}} = A_q\times_{K_\gamma}K_{\mathcal{M}}. \end{aligned} \tag{B.15.5}$$

No surjectivity of these last four maps follows merely from surjectivity of q.

The subgroup variant of C3:571–572, (94), has the following precise form. For a closed $D_0\subset D$ containing t, set

$$\begin{aligned} L(D_0) &= \{\ell\in L_0 : j(\ell)\in D_0\}, \qquad \gamma_0 = D_0/j(L(D_0)),\\ G_0 &= CD_0\subset G_D, \qquad G_0\simeq (C\rtimes D_0)/\operatorname{anti}L(D_0),\\ N_x^{0,*} &= N_x^*\cap(G_0\times\{\pm1\}). \end{aligned} \tag{B.15.6}$$

Then $1\to C\to G_0\to\gamma_0\to 1$ is exact, and the remaining groups are the B.13 fibre products over γ_0. The kernel of $D_0\to\gamma_0$ is $j(L(D_0))$, which equals $j(L_0)$ only if the latter is contained in D_0. For descent replace D_0 by $D_0\cap D_q$. The distinguished element required here is $t\in D$; the manuscript's margin reading τ'_Z cannot mean $t'=\sigma^{-1}t$, which need not belong to D. This construction keeps the full conjugating group C. The different restriction of the conjugating group in §49.VI is treated separately in B.17–B.18.

For the particular map

$$q:\widehat{\mathrm{SL}_2(\mathbb{Z})}\twoheadrightarrow\mathrm{SL}_2(\widehat{\mathbb{Z}}),\quad \rho=\begin{pmatrix}0&1\\-1&1\end{pmatrix},\quad \sigma=\begin{pmatrix}0&-1\\1&0\end{pmatrix}, \tag{B.15.7}$$

we have $D_q=D$, for either choice of D. This is the earlier σ-convention of C3:572–573 and B.13. More explicitly, if $q(u(\sigma))_{11}=a$ and the recorded multiplier is μ, then

$$q\circ u=\mathrm{Int}(g(a,\mu))\circ q,\qquad g(a,\mu)=\begin{pmatrix}\mu&a\mu\\0&1\end{pmatrix}. \tag{B.15.8}$$

The restriction $L_0\to L_0'$ is an isomorphism, so here $L_N=1$, $K_\gamma=A_q$, and $K_G=N\rtimes A_q$. Thus congruence-kernel stability is proved; it is not one of the unresolved lifting assertions in B.8 or D.7.

Finally, the reconstruction at the end of §49, C3:691–696, folios 648–650, has a general form. Let P,T be profinite groups, $J\triangleleft T$ closed, $\theta:T\to\mathrm{Aut}(P)$ a continuous action, and $\phi:J\to P$ a continuous homomorphism satisfying

$$\theta(j)=\mathrm{Int}(\phi(j)),\qquad \phi(tjt^{-1})=\theta(t)(\phi(j)). \tag{B.15.9}$$

Then

$$\mathcal{E}=(P\rtimes T)/\{(\phi(j)^{-1},j):j\in J\},\qquad 1\longrightarrow P\longrightarrow\mathcal{E}\longrightarrow T/J\longrightarrow 1 \tag{B.15.10}$$

is defined, and $\ker(T\to\mathcal{E})=\ker\phi$. It reconstructs exactly the data consisting of an extension by P and a map $T\to\mathcal{E}$ onto the quotient, with J the inverse image of P. For a nonempty family of such extensions $\mathcal{E}_i$ with identified kernel P and quotient γ, a compatible transitive system of isomorphisms $\mathcal{E}_i\simeq\mathcal{E}_j$, fixing both, is equivalent to a closed subgroup

$$\mathcal{E}\subset\prod_{i,\gamma}\mathcal{E}_i,\qquad \mathcal{E}\cap P^I=\Delta P,\qquad \mathcal{E}\twoheadrightarrow\gamma. \tag{B.15.11}$$

Such a subgroup is additional data; it need not exist merely because the quotients and the induced actions agree.

Proof. For an automorphism u of C, a quotient automorphism of C/N exists if and only if $u(N) = N$. It is then unique, and its generator images and every numerical coordinate are prescribed. Evaluation on N, both for u and u^{-1}, expresses this equality as closed conditions. Hence D_q is a closed subgroup. Inner automorphisms preserve N, and τ does so by the compatibility assumption. This proves the assertions about $j(L_0)$ and t. Descent is continuous: an open normal subgroup of C' pulls back to an open normal subgroup of C, so all its finite quotient evaluations are continuous. Compactness gives closedness of H.

Suppose a source effective numerical tuple maps to an effective target tuple, represented by u'. Then qu and $u'q$ agree on ρ, σ, hence on the dense subgroup they generate, and therefore everywhere. Their kernels give $u^{-1}(N) = N$. Conversely kernel stability gives the quotient automorphism and hence effectiveness of the target tuple. This proves the three equivalences, without assuming effectiveness of all numerical solutions. If $u(L_0) = cL_0c^{-1}$, replace u by $\mathrm{Int}(c^{-1})u$. Both have the same kernel-stability property since N is normal. This proves the version b') of C3:570.

The map $q : L_0 \to L_0'$ is onto: its image is compact and contains the generator of L_0'. It follows that $q_D(j(L_0)) = j'(L_0')$. The compatible map $C \rtimes D_q \to C' \rtimes H$ carries one antidiagonal onto the other, and therefore induces the continuous epimorphism of the G's. It is the unique map with the two specified compatibilities, since C and D_q generate its source. If a class $d \in \gamma_{D_q}$ maps to the identity, then $q_D(d) = j'(\ell')$. Lifting ℓ' to L_0 and subtracting its j-image gives a representative in A_q. The ambiguity is exactly $j(L_N)$. This proves the first formula in (B.15.2).

Similarly any element of K_G, represented in $C \rtimes D_q$, can be multiplied by a lifted antidiagonal element so that its two coordinates lie in N and A_q. The intersection of $N \rtimes A_q$ with the original antidiagonal is precisely the subgroup indexed by L_N. This proves the second formula and the exact sequence in (B.15.2). The assumption $K = K' = 1$ ensures that both copies of the base group remain embedded; it is not silently discarded.

The signed normalizer map is $(g, e) \mapsto (q_G(g), e)$. In its kernel the sign is $+1$, and its group coordinate centralizes x; this proves (B.15.3). All lifts of g' have the form kg, $k \in K_G$. Such a lift has the prescribed action on x exactly when $k(gxg^{-1})k^{-1} = x^e$, proving (B.15.4). Once one lift is chosen, division on the right identifies all others with $C_{K_G}(x)$. Kernels commute with these explicitly defined fibre products, giving (B.15.5). Their images still require the conjugator-lifting conditions; the proof gives no spurious surjectivity assertion.

For (B.15.6), the inverse image of $G_0 = CD_0$ in $C \rtimes D_0$ has kernel exactly $\{(\ell^{-1}, j(\ell)) : j(\ell) \in D_0\}$. The quotient by C is consequently $D_0/(D_0 \cap j(L_0))$, as asserted. Restricting the conjugacy arguments in B.13 proves the corresponding normalizer and fibre-product formulas; the condition $t \in D_0$ supplies the same sign splittings. The group $D_0 j(L_0)$ gives the same G_0 but a different coefficient group, which explains why these two conventions must not be interchanged without changing the first kernel. The finite check below distinguishes them.

The effective parameter names of C3:565–569, (70)–(87), also have an explicit dictionary. With P_D^{eff} as in B.13, its homeomorphism with $S_0\Gamma_D$ gives

$$\Gamma_D \simeq L_0 \backslash P_D^{\mathrm{eff}}, \qquad S_0\mathcal{M}_D \simeq P_D^{\mathrm{eff}} \times C, \qquad f = dU^{-1}, \quad U = f^{-1}d. \tag{B.15.6a}$$

These are homeomorphisms of parameter spaces, with the transported group laws, not an assertion of a direct-product group law in the second expression. The first follows because multiplication by $(j(\ell), 1, 1)$ sends (α, β) to $(\ell\alpha, \ell\beta)$. In the second, equality of the quotient coordinates puts f in C, and the displayed inverse recovers U. The right action $(\alpha, \beta) \mapsto (\alpha a_0, \beta b_0)$, for $(a_0, b_0) \in C_\rho \times C_\sigma$, corresponds to

$$(d, a, b) \longmapsto (d, (t')^{v_r} a_0^{-1} (t')^{-v_r} a, (t')^{v_s} b_0^{-1} (t')^{-v_s} b). \tag{B.15.6b}$$

This is obtained by inverting (B.13.11). Thus the operator written τ in C3:569, (83), must act here by t'-conjugation; literal t-conjugation need not preserve C_ρ.

We now prove (B.15.8) by identities over $\mathbb{Z}[a,\mu,\mu^{-1}]$. Put

$$U(z)=\begin{pmatrix}1 & z\\ 0 & 1\end{pmatrix},\quad R=q(u(\rho)),\quad S=q(u(\sigma)).$$

The conjugacy conditions give $\operatorname{tr}R=1$, $\operatorname{tr}S=0$, and determinants one. In particular $S^2=-I$. Since $e_0=\sigma\rho=U(-1)$, the multiplier condition gives $SR=U(-\mu)$. Write $S=\left(\begin{smallmatrix}a & b\\ c & -a\end{smallmatrix}\right)$. Then

$$R=-SU(-\mu),\qquad 1=\operatorname{tr}R=c\mu,\qquad b=-\mu(1+a^2). \tag{B.15.12}$$

Consequently

$$S=\begin{pmatrix}a & -\mu(1+a^2)\\ \mu^{-1} & -a\end{pmatrix}=g(a,\mu)\sigma g(a,\mu)^{-1},\quad R=g(a,\mu)\rho g(a,\mu)^{-1}. \tag{B.15.13}$$

These identities hold also in residue characteristics two and three. Conjugation by $g(a,\mu)\in \mathrm{GL}_2(\widehat{\mathbb{Z}})$ is a continuous automorphism of $\mathrm{SL}_2(\widehat{\mathbb{Z}})$. The two homomorphisms in (B.15.8) agree on the topological generators, so agree everywhere. Taking kernels proves $u(N)=N$. Finally the map $\widehat{\mathbb{Z}}\to L_0'$, $z\mapsto U(-z)$, is an isomorphism, and factors through L_0. The surjection $\widehat{\mathbb{Z}}\to L_0$ is thus injective as well. This proves the asserted kernels. Also $U(-z)$ commutes with σ only when $z=0$, so both required intersections of L_0 with the centre are trivial in this example. The surjectivity of (B.15.7) follows from elementary generation modulo every integer and the Chinese remainder theorem, as in C.3.

For the last reconstruction, the map $j\mapsto(\phi(j)^{-1},j)$ is an injective homomorphism. The first identity in (B.15.9) proves the homomorphism assertion and shows that its image centralizes P. The second identity shows that its image is normalized by T. It is therefore closed and normal in $P\rtimes T$. It intersects the first factor trivially, and intersects the second factor in $\ker\phi$; these observations give every assertion in (B.15.10). Conversely, from a map $T\to\mathcal{E}$ onto $\mathcal{E}/P$, conjugation gives θ, the inverse image of P gives J, and restriction gives ϕ. The map $P\rtimes T\to\mathcal{E}$ is onto and has exactly the displayed antidiagonal kernel. The constructions are inverse, including on compatible morphisms. In particular the notation $S_0G=N.\mathcal{G}$ in C3:692, (4), must mean the specified semidirect product, except when its action is trivial.

To prove (B.15.11), each coordinate projection $\mathcal{E}\to\mathcal{E}_i$ is injective: its kernel lies in ΔP and has one coordinate equal to the identity. It is onto: lift the quotient coordinate of an arbitrary $g_i\in\mathcal{E}_i$ to $\mathcal{E}$, then correct its i-th coordinate by an element of ΔP. Compactness makes each continuous bijection an isomorphism. Their composites give the desired transitive system. Conversely, the graph of a transitive system gives such a closed subgroup. This proves the exact correction made in C3:695 to the tentative amalgam idea of C3:694.

All the following checks were executed by `code/AB_functoriality.py`; complete output and elementwise data are in its `.out` and `.json` files. Take $C=S_4$, $\rho=(123)$, $\sigma=(34)$, and let $q:S_4\to S_3$ act on the three partitions $12|34, 13|24, 14|23$. Thus $\rho'=(132)$, $\sigma'=(23)$, $N=\{1,(12)(34),(13)(24),(14)(23)\}$, and $L_N=\{1,(14)(23)\}$. Cayley-graph propagation enumerates every automorphism and verifies kernel stability. We retain $U(12)$, rather than count the infinite numerical group $\widehat{\mathbb{Z}}^\times$, and build every quotient and fibre product. Each entry

below gives source, target, and kernel orders; every image is the full displayed target.

	$D=B$			$D=Z$		
object	$\lvert X\rvert$	$\lvert X'\rvert$	$\lvert\ker\rvert$	$\lvert X\rvert$	$\lvert X'\rvert$	$\lvert\ker\rvert$
D	16	8	2	64	32	2
γ	4	4	1	16	16	1
G	96	24	4	384	96	4
N_ρ^*	12	12	1	48	48	1
N_σ^*	16	8	2	64	32	2
Γ	48	24	2	192	96	2
$\mathcal{M}$	1152	144	8	4608	576	8
$S_0\Gamma$	192	48	4	768	192	4
$S_0\mathcal{M}$	4608	288	16	18432	1152	16

(B.15.14)

Here $A_q = j(L_N)$, $K_\gamma = 1$, and $K_G = N$, exactly as (B.15.2) predicts. The program checks all multiplication pairs for the D, γ, G maps and every image and fibre of all the listed groups. For $D_0 = \langle t\rangle$ it obtains $|D_0| = 2$, $L(D_0) = 1$, $|G_0| = 48$, and verifies that $t' \notin D$ and that conjugation by t' does not preserve L_0. Thus the margin in C3:571 cannot literally require this t' in the coefficient group.

The failure of normalizer surjectivity in (B.15.4) is real. Take $q : \mathrm{SL}_2(\mathbb{F}_3) \to \mathrm{SL}_2(\mathbb{F}_3)/\{\pm I\} \simeq A_4$ with the matrices in (B.15.7). The action on $\mathbf{P}^1(\mathbb{F}_3)$ gives the displayed isomorphism: its kernel is $\{\pm I\}$, and its generator images are even permutations, giving all twelve even permutations. An automorphism acting trivially on this quotient has the form $g \mapsto g\chi(g)$, with $\chi : \mathrm{SL}_2(\mathbb{F}_3) \to C_2$ a homomorphism. The relations $\rho^3 = \sigma^2$ and $(\sigma\rho)^3 = 1$ force $\chi(\rho) = \chi(\sigma) = 1$, so $A_q = 1$. Here $L_N = 1$, whence $K_G = \{\pm I\}$. The element

$$j = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \quad \text{satisfies} \quad j\sigma j^{-1} = -\sigma.$$

Its image centralizes σ' and has trivial quotient coordinate, so belongs to the target $N_{\sigma'}^*$ with sign $+1$. Its only possible lifts are $j, -j$, both of which fail to centralize σ. The program verifies this obstruction and all candidate lifts. With independent $U(12)$ numerics it obtains $|D| = |H| = 24$, $|N_\sigma^*| = |N_{\sigma'}^*| = 32$, and image order 16. Thus a surjective map of the enlarged groups genuinely can induce a nonsurjective map of their starred normalizers.

The symbolic calculation verifies (B.15.12)–(B.15.13), and exhaustive enumeration of determinant-one trace pairs with $SR = U(-\mu)$ gives respectively $2, 6, 8, 20, 32, 54$ solutions over $\mathbb{Z}/n$, for $n = 2, 3, 4, 5, 8, 9$. In each case there are exactly n pairs for each unit μ, parameterized by $a \in \mathbb{Z}/n$. These are trace-pair counts, not an assertion that every independently signed conjugacy condition is automatic over every finite ring.

For the reconstruction and gluing test, take $P = \gamma = C_2$ with trivial action. The groups C_4 and V_4 are two extensions with that kernel and quotient; their different element orders prove that they cannot be isomorphic. The program reconstructs both by (B.15.10). It then enumerates every subgroup of order four in each relevant fibre product of order eight. For two copies of C_4, precisely two subgroups satisfy (B.15.11), the graphs of its two automorphisms. For C_4 and V_4, there are none. This is a complete finite counterexample to reconstruction from only the common quotient and action, and verifies the additional gluing data required in C3:695–696. Passing to constant finite group schemes makes these particular kernel and quotient statements exact in fppf sheaves over $\mathbb{Z}$. □

The congruence comparison left by this theorem. For (B.15.7), write $H_q = q_D(D) \subset D'$ and $A_q = \ker(D \to D')$. The theorem constructs the maps and determines their kernels in terms of these two well-defined groups; it does not determine the full effective image H_q or the

group A_q. Its elements have numerical coordinates $(\mu, e_r, e_s) = (1, 1, 1)$, and satisfy $qu = q$; the coordinates have not been forgotten before taking the kernel. In particular a finite target tuple must lift compatibly to an automorphism of the full completed modular group, not just satisfy its matrix identities. B.8.5, the named comparison of B.16, the restricted-conjugator comparisons of B.18, and the three arithmetic comparison problems in D.7 are separate assertions. The finite map (B.15.14) is a proved nontrivial descent and conjugator-lifting case, not a substitute for any of them.

Here is one completely specified lifting instance not supplied by our present argument. Let $\mu \in \widehat{\mathbb{Z}}^\times$ have 5-adic component 2 and every other component 1, and put

$$d_5' = \big(\mathrm{Int}(\mathrm{diag}(\mu, 1)), \mu, 1, 1\big) \in Z'. \tag{B.15.15}$$

It belongs also to B'. Indeed, both prescribed signs are $+1$. At 5, choose $i \in \mathbb{Z}_5$ with $i^2 = -1$, $i \equiv 2 \pmod 5$; lifting the simple root modulo successive powers of five constructs it. The centralizers contain

$$h_r = (i-1)I + 2\rho, \quad \det h_r = 2, \qquad h_s = I + \sigma, \quad \det h_s = 2.$$

Thus $\mathrm{diag}(2, 1)h_x^{-1} \in \mathrm{SL}_2(\mathbb{Z}_5)$ conjugates x to the required image for $x = \rho, \sigma$. At every other prime take the identity conjugator. The loop condition is immediate from conjugation of $U(-1)$. The code verifies the two resulting $\mathrm{SL}_2(\mathbb{F}_5)$ conjugators, namely $\left(\begin{smallmatrix} 3 & 3 \\ 1 & 3 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1 & 1 \\ 2 & 3 \end{smallmatrix}\right)$. The precise assertion left by this example is

$$\exists u \in \mathrm{Aut}\big(\widehat{\mathrm{SL}_2(\mathbb{Z})}\big) : \quad (u, \mu, 1, 1) \in Z, \qquad qu = \mathrm{Int}(\mathrm{diag}(\mu, 1))q. \tag{B.15.16}$$

Equivalently, $d_5' \in H_q$. The separate conjugacy and norm conditions just proved do not establish this compatible profinite realization. We record (B.15.16) as an exact unresolved instance of our argument, not as a claim that it is open in the literature or the least unit for any order on $\widehat{\mathbb{Z}}^\times$.

B.3 The universal modular comparison and its precise remaining lifts

B.16. Central twists, normalized conjugators, and the universal source comparison [Open]

In this subsection use the normalization of C3:572–582,

$$C = \widehat{\mathrm{SL}_2(\mathbb{Z})}, \quad r = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, \quad s = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \omega = r^3 = s^2 = -I, \quad e = sr = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}. \tag{B.16.1}$$

Thus s is the negative of the final §48 matrix σ. All groups and automorphisms in this subsection are profinite and continuous; they are not integral group schemes. Put

$$\overline{C} = C/\langle\omega\rangle, \quad P = \ker(C \longrightarrow \mathrm{SL}_2(\mathbb{F}_2)), \quad L = \overline{\langle e\rangle}, \quad P_0 = \overline{\langle \omega e^2, \omega(rs)^2\rangle}. \tag{B.16.2}$$

The following assertions are proved below, using the explicitly identified torsion and centralizer theorems of Herfort–Ribes:

$$\begin{gathered} Z(C) = \langle\omega\rangle, \quad C^{\mathrm{ab}} = \mathbb{Z}/12 \quad (r \mapsto 2, s \mapsto 3), \qquad L \simeq \widehat{\mathbb{Z}}, \quad P = P_0 \times \langle\omega\rangle, \\ 1 \longrightarrow \langle\Theta\rangle \longrightarrow \mathrm{Aut}(C) \longrightarrow \mathrm{Aut}(\overline{C}) \longrightarrow 1, \qquad \Theta(x) = x\omega^{\mathrm{sg}(x)}, \end{gathered} \tag{B.16.3}$$

where $\mathrm{sg} : C \to \mathbb{Z}/2$ is the sign after reduction to $\mathrm{SL}_2(\mathbb{F}_2) = S_3$. In particular $\Theta(r) = r$ and $\Theta(s) = \omega s$. It fixes L^2 pointwise, but does not carry L to a C-conjugate of L.

Let $\mathcal{A}$ consist of the automorphisms of C preserving P and the conjugacy class of L, and let $\overline{\mathcal{A}}$ be the analogous subgroup of $\mathrm{Aut}(\overline{C})$ for P/ω and the image of L. Then

$$\mathcal{A} \xrightarrow{\sim} \overline{\mathcal{A}}. \tag{B.16.4}$$

Let Z be the group of automorphisms preserving L and the two conjugacy classes of cyclic subgroups $\langle r\rangle, \langle s\rangle$. The multiplier μ and the two signs are unique and satisfy

$$u(e) = e^{\mu}, \quad u(r) \sim_C r^{\epsilon_3(\mu)}, \quad u(s) \sim_C s^{\epsilon_2(\mu)}. \tag{B.16.5}$$

Consequently the independent-sign and linked-sign groups of B.13 coincide in this particular universal case. Define

$$\gamma = Z/\operatorname{Int}(L), \qquad G = (C \rtimes Z)/\{(\ell^{-1}, \operatorname{Int}\ell) : \ell \in L\}. \tag{B.16.6}$$

The adjoint action induces a topological isomorphism $G/\langle\omega\rangle \simeq \mathcal{A}$.

Both P and P_0 are normal in G. The action on $C/P = S_3$ defines $\kappa : G \to \mathrm{Aut}(S_3) = S_3$; put $G^! = \ker\kappa$ and $Z^! = Z \cap G^!$. There are exact sequences

$$\begin{aligned} 1 &\longrightarrow L^2 \longrightarrow Z^! \longrightarrow \gamma \longrightarrow 1, \\ 1 &\longrightarrow P \longrightarrow G^! \longrightarrow \gamma \longrightarrow 1. \end{aligned} \tag{B.16.7}$$

In particular $Z^!$ has index two in Z; the distinction corrected in the margin of C3:582 and again at C3:613 is necessary.

Write τ for conjugation by $\mathrm{diag}(-1,1)$ and $\tau' = s^{-1}\tau$ in G, as in B.13. For every $u \in Z^!$ there are unique

$$\alpha \in P_0\tau'^{\nu_3(\mu)}, \qquad \beta \in P_0, \qquad (-1)^{\nu_i(\mu)} = \epsilon_i(\mu), \tag{B.16.8}$$

such that

$$u(r) = \alpha r \alpha^{-1}, \qquad u(s) = \omega^{\nu_2(\mu)}\beta s \beta^{-1}. \tag{B.16.9}$$

This proves the normalization of C3:579(27),582(38)–(40) for the precisely defined subgroup $Z^!$.

Identify N_r^*, N_s^* with the subgroups of G in B.13. The maps of C3:612(118)–(123) have the following rigorous formal version:

$$\overline{a}(u) = \tau'^{\nu_3(\mu)}\alpha^{-1}u \pmod{\omega}, \qquad b(u) = \beta^{-1}u, \qquad \overline{a} : Z^! \to N_r^*/\omega, \quad b : Z^! \to N_s^*. \tag{B.16.10}$$

They are continuous homomorphisms over γ and satisfy

$$\overline{a}(e^{2t}) = 1, \qquad b(e^{2t}) = \omega^{t \bmod 2} \quad (t \in \widehat{\mathbb{Z}}). \tag{B.16.11}$$

Here e^{2t} in the domain means its embedded inner automorphism. They induce sections

$$a_\gamma : \gamma \to N_r^*/\omega, \qquad b_\gamma : \gamma \to N_s^*/\omega, \tag{B.16.12}$$

and therefore

$$N_r^*/\omega \simeq C_3 \rtimes_{\epsilon_3} \gamma, \qquad N_s^*/\omega \simeq C_2 \times \gamma. \tag{B.16.13}$$

These are the quotient sections in C3:615(130)–(131). The stronger unquotiented σ-section of (132)–(140), its chosen geometric interpretation, and the source comparisons specified below are not claimed by this statement.

Proved part. We first specify the sole external group-theoretic input. Herfort and Ribes, *On Free Pro-$\mathcal{C}$-Products of Pro-$\mathcal{C}$-Groups*, Theorems 1 and 2, pp. 31–34 Herfort and Ribes (1984), apply to the class of all finite groups. For a free profinite product they put finite-order elements in conjugates of the factors and give $C_{A\amalg B}(a) = C_A(a)$ for $1 \neq a \in A$. Their hypotheses require closure under subgroups, quotients and extensions, all of which the class of all finite groups satisfies. The directly inspected primary article is an announcement with a proof outline; it explicitly refers to their subsequent full paper. We use these two established theorems, not a claim that a discrete reduced-word proof works profinitely.

By C.2 and the universal property of completion, $\overline{C} = C_3 \amalg C_2$. Indeed the completion of the discrete quotient by its finite central subgroup is C/ω: both groups have exactly the same continuous maps to finite groups. The involution survives modulo three, so the kernel has order two. The centralizer theorem, first for the order-three factor and then for the order-two factor, gives

$$C_C(r) = \langle r \rangle, \qquad C_C(s) = \langle s \rangle, \qquad Z(C) = \langle \omega \rangle. \tag{B.16.14}$$

For example an element centralizing r projects into $\langle \overline{r} \rangle$, whose inverse image is exactly $\langle r \rangle$; the reverse inclusion is immediate.

Abelianizing the presentation in C.2 gives $\langle r, s \mid 6r = 4s = 0, 3r = 2s \rangle$. Its order is twelve, and the displayed map to $\mathbb{Z}/12$ is onto, so it is an isomorphism. The profinite abelianization has the same universal property for finite abelian targets, hence is that same finite group. If v fixes ω and induces the identity on $\overline{C}$, write $v(x) = x\omega^{\chi(x)}$. Centrality makes $\chi : C \to \mathbb{Z}/2$ a continuous homomorphism; fixing ω requires $\chi(\omega) = 0$. Conversely such a character defines an involutory automorphism. There are exactly two characters, the trivial one and sg, and both vanish on ω, whose abelianization is 6. This proves the kernel in (B.16.3), and also proves the same assertion for endomorphisms inducing the identity on the central quotient.

For surjectivity in (B.16.3), let $v \in \mathrm{Aut}(\overline{C})$. The torsion theorem gives conjugators for $v(\overline{r}) \sim \overline{r}^{\pm 1}$ and $v(\overline{s}) \sim \overline{s}$. Lift those conjugators to C. The resulting proposed images of r, s have common cube and square ω, so C.2 and completion extend them to an endomorphism $\widetilde{v}$ of C. Make the same construction for v^{-1}. Both compositions induce the identity on $\overline{C}$ and fix ω; the preceding kernel calculation makes them automorphisms. Thus $\widetilde{v}$ is an automorphism too.

The element e has order n in every $\mathrm{SL}_2(\mathbb{Z}/n)$. Consequently $t \mapsto e^t$ embeds $\widehat{\mathbb{Z}}$ into C. Modulo three, e has order three and ωe has order six. Hence $\Theta(L)$ and L cannot be conjugate, whereas $\Theta(e^2) = e^2$. An automorphism of $\overline{C}$ preserving the indicated peripheral subgroup class sends $\overline{e}$ to a conjugate of $\overline{e}^{\mu}$ for a unique unit multiplier after its conjugator is chosen. Either lift to C sends e to that lift or its negative; composing with Θ switches these two choices. Exactly one lift preserves the class of L. Preservation of P is equivalent to preservation of P/ω. This proves (B.16.4).

The two signs in (B.16.5) are unique, since r, r^{-1} have distinct images $2, 10$ in C^{ab}, and s, s^{-1} have distinct images $3, 9$. In that abelianization the loop equation is

$$2e_r + 3e_s = 5\mu \pmod{12}.$$

Reducing modulo three gives $e_r \equiv \mu \pmod 3$; reducing modulo four gives $e_s \equiv \mu \pmod 4$. This proves the two linked signs. It also proves that the numerical coordinates of B.13 introduce no kernel in this particular universal case.

E.15 proves that the discrete Π_0 is normal of index twelve, free on $\lambda_0 = \omega e^2, \lambda_1 = \omega(rs)^2$, and that $\Pi = \Pi_0 \times \langle \omega \rangle$. A subgroup of finite index carries its full profinite topology: the core in the ambient group of any finite-index subgroup is still finite-index. Completion therefore gives $P = P_0 \times \langle \omega \rangle$, with P_0 free profinite on λ_0, λ_1. Equivalently, P_0 is the closed normal subgroup of C generated by λ_0; conjugation by r supplies λ_1. Every $u \in Z$ sends λ_0 to λ_0^{μ} and fixes ω. It therefore preserves both P_0 and P.

For completeness, $N_C(L) = L\langle\omega\rangle$. An element of this normalizer has reduction modulo two in $C_{S_3}(\overline{e}) = \langle\overline{e}\rangle$. Multiplying by e if necessary puts it in P. If an element of P carries e^2 to e^{2v}, its conjugation on $P_0^{\mathrm{ab}} = \widehat{\mathbb{Z}}^2$ forces $v = 1$. The centralizer theorem applied to $P_0 = \langle\lambda_0\rangle \amalg \langle\lambda_1\rangle$ then puts it in $\langle\lambda_0\rangle\langle\omega\rangle$. This is $L^2\langle\omega\rangle$, proving the claim. In particular $L \cap Z(C) = 1$.

The torsion theorem also says that every automorphism of C carries r to a conjugate of $r^{\pm1}$ and s to a conjugate of $s^{\pm1}$. For r, the possible extra central sign after lifting from $\overline{C}$ is ruled out by its cube being ω. Thus $\mathcal{A} = \mathrm{Int}(C)Z$. The action of $C \rtimes Z$ on C has kernel consisting of $(c, \mathrm{Int}(c^{-1}))$ with $c \in N_C(L)$. The preceding normalizer computation and the antidiagonal quotient in (B.16.6) leave exactly $\langle\omega\rangle$ as kernel. All groups are compact Hausdorff, so the resulting continuous bijection $G/\omega \to \mathcal{A}$ is an isomorphism.

Since Z and C preserve P_0, P, so does G. The action of $u \in Z$ on S_3 fixes the transposition $\overline{e}$, because μ is odd. Since $\mathrm{Aut}(S_3) = \mathrm{Int}(S_3)$, its image under κ lies in $\langle\overline{e}\rangle$. The element $\mathrm{Int}(e)$ maps onto that order-two subgroup. Thus $Z^!$ has index two, and multiplying by a suitable element of L makes any coset in γ lie in $Z^!$. Its kernel is L^2. Similarly $\kappa(C) = S_3$ corrects any element of G into $G^!$, whose intersection with C is P. This proves (B.16.7).

We now prove normalization without supposing it in advance. For $u \in Z^!$, write $u(s) = cs^{\epsilon_2}c^{-1}$. Modulo P the conjugator c centralizes $\overline{s}$; multiply it by a suitable power of s to put it in P. Removing its central ω-factor puts it in P_0, giving β. For r the same argument works when $\epsilon_3 = 1$, using $C_{S_3}(\overline{r}) = \langle\overline{r}\rangle$. When $\epsilon_3 = -1$, use $\tau(r) = sr^{-1}s^{-1}$. If $u(r) = cr^{-1}c^{-1}$, then cs^{-1} conjugates $\tau(r)$ to $u(r)$ and has reduction in $\langle\overline{r}\rangle$. Multiplication by a suitable power of $\tau(r)$, and then removal of the central sign, produces $\alpha \in P_0\tau$. Two such β differ by $C_C(s) \cap P_0 = 1$; two such α of the prescribed parity differ by $C_C(r) \cap P_0 = 1$, by (B.16.14) and the injections of the order-four and order-six factors into C/P_0 proved in E.15. The coordinate maps are continuous: their graphs are closed, and a continuous bijection from a compact space to a Hausdorff space has continuous inverse.

The element $b(u)$ in (B.16.10) is characterized uniquely by

$$b(u) \in N_s^*, \qquad b(u) \equiv u \pmod{P_0}. \tag{B.16.15}$$

Indeed $N_s^* \cap P_0 = C_C(s) \cap P_0 = 1$. Since P_0 is normal in G, (B.16.15) proves multiplicativity. Likewise $a(u) = \tau'^{\nu_3}\alpha^{-1}u$ belongs to N_r^*, lies over the same element of γ, and satisfies

$$a(u) \equiv s^{-\nu_3}u \pmod{P}. \tag{B.16.16}$$

Here $N_r^* \cap P = \langle\omega\rangle$, so the class of $a(u)$ modulo ω is uniquely characterized by this congruence. Elements of $Z^!$ commute with C/P in G/P. For $u', u \in Z^!$ their right sides in (B.16.16) multiply to $s^{-(\nu_3'+\nu_3)}u'u$; replacing the sum by its value modulo two changes a power of $s^2 = \omega \in P$ only. This proves that $\overline{a}$ is a homomorphism.

For $u = e^{2t} = \omega^{t \bmod 2}\lambda_0^t$ the two characterizations give (B.16.11). Hence the quotient maps (B.16.12) exist and are sections over γ. Their kernels in $N_r^*/\omega, N_s^*/\omega$ are respectively $\langle r\rangle/\omega = C_3$ and $\langle s\rangle/\omega = C_2$. Conjugation in the first case is the sign ϵ_3; every automorphism of C_2 is the identity. This proves (B.16.13). In particular the reference to ϵ_2 acting on the order-two quotient in C3:615 must be read as a trivial action; inversion becomes visible on C_4. □

Executed finite model and its exact limitation. The stand-alone program `AB_modular_comparison.py` enumerates every matrix and every automorphism from the images of r, s, checking every edge of the complete Cayley graph. For $H_4 = \mathrm{SL}_2(\mathbb{Z}/4)$ it gives

$$|H_4| = 48, \quad H_4/\omega \simeq S_4, \quad |\,\mathrm{Aut}(H_4)| = 48, \quad |\ker(\mathrm{Aut}(H_4) \to \mathrm{Aut}(S_4))| = 2, \tag{B.16.17}$$

and the image has order twenty-four. The isomorphism to S_4 is certified by conjugation on the four cyclic order-three subgroups: the permutation image has order twenty-four and

kernel $\{1, \omega\}$. This also gives $|\operatorname{Aut}(S_4)| = 24$: an automorphism acts on those four intrinsic subgroups, and an automorphism acting trivially there must commute with the faithful inner permutation action, which forces it to fix every group element. The central-character argument above bounds the remaining kernel by two, while the sign twist realizes both.

In this finite group all forty-eight automorphisms preserve the conjugacy class of $\langle e \rangle$. In fact

$$\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} e \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}^{-1} = -e^{-1} \pmod 4. \tag{B.16.18}$$

The finite counterpart of P_0 is the normal Klein four group

$$\left\{ I, \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, \begin{pmatrix} 3 & 2 \\ 0 & 3 \end{pmatrix}, \begin{pmatrix} 3 & 0 \\ 2 & 3 \end{pmatrix} \right\}.$$

Using the numerical group $(\mathbb{Z}/12)^\times$, the program finds sixteen linked automorphism–multiplier pairs. Eight act trivially modulo two; each has one normalized α of the prescribed parity and two choices of β. For the identity with multiplier one the latter are already I and $(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix})$. Thus finite quotient centralizers do not prove profinite normalization; the centralizer theorem is essential to our reduction. The additional modulo-three computation confirms the orders three and six used above. It also verifies that the proposed central twist does not descend to an automorphism of $\mathrm{SL}_2(\mathbb{F}_3)$; its central-quotient automorphism kernel is trivial.

Exact source dictionary and unresolved comparisons. To prevent a collision with the final matrix scheme $\mathcal{M}$ and the fibre-product group $\mathcal{M}_D$ of B.13, write $\mathscr{N}$ for the source $\widetilde{N}_{1,1}$ and $\mathscr{P}$ for the source $\widetilde{\mathcal{M}}_{0,3}$. The source begins with $1 \to \langle \omega \rangle \to \mathscr{N} \to \mathscr{P} \to 1$ at C3:573(1). The formal groups G, Z, γ have been constructed above; identifying them with these named geometric groups is an additional comparison. The remaining assertions are the following.

(R1) **C3:573–576(3)–(5),(13),582,622(4 bis).** For the source-defined action $\iota : \mathscr{P} \to \operatorname{Aut}(\overline{C})$, prove

$$\iota : \mathscr{P} \xrightarrow{\sim} \overline{\mathcal{A}}, \qquad \mathscr{N} \xrightarrow{\sim} G \quad \text{over } C \text{ and the central } \omega. \tag{B.16.R1}$$

Our (B.16.4) and (B.16.6) give the unique peripheral lift and the formal central extension once this geometric comparison is supplied. They do not identify an earlier geometric π_1 construction merely by giving its abstract automorphism candidate the same name. Nor is γ thereby identified with the absolute Galois group. The further GT and Galois questions are D.7.

(R2) **C3:615–618(132)–(143),623(8)–(12).** The first additional unquotiented lifting question is exactly whether the proved section b_γ in (B.16.12) admits a continuous homomorphic lift

$$\widetilde{b}_\gamma : \gamma \longrightarrow N_s^*, \qquad \widetilde{b}_\gamma \bmod \omega = b_\gamma. \tag{B.16.R2}$$

Equivalently, the explicitly defined central extension

$$E_s = \gamma \times_{N_s^*/\omega} N_s^*, \qquad 1 \to \langle \omega \rangle \to E_s \to \gamma \to 1 \tag{B.16.R2a}$$

must split. The homomorphism b we proved on $Z^!$ does not itself descend to γ, since $b(e^2) = \omega$. The manuscript obtains a lift from a chosen source section $\mathscr{N}(0) \simeq \gamma$ of $Z^! \to \gamma$. That section and its agreement with the earlier geometric construction have not been constructed here.

The additional source objects consequently have the following precise conditional dictionary, after choosing that section and making (R1):

source object and scan	defining subgroup or intersection
$\mathscr{N}^!$ 616(NB), 623(8)	$\ker\kappa = G^!$
$\mathscr{N}_0^!$ 616(135), 623(8)	$\mathscr{N}(0)P_0$
$\mathscr{N}(1/2)$ 617(139), 623(9)	$N_s^* \cap \mathscr{N}_0^!$
$\mathscr{N}'(-\bar{j})$ 618(142), 624(17)	$N_r^* \cap \mathscr{N}_0^!$
γ' 618	$\ker(\epsilon_3 : \gamma \to \{\pm 1\})$

(B.16.R2b)

The remaining assertions are that the third row maps isomorphically onto γ, that the fourth row maps isomorphically onto γ', and that these are the particular geometric sections at $1/2$ and $-\bar{j}$ in C3:627(25). These are specific section comparisons, not consequences of the matrix-scheme section calculations in B.5.

(R3) **C3:619–621(144)–(149),625–626(21)–(24).** The other proved quotient section defines the exact central extension

$$E_r = \gamma \times_{N_r^*/\omega} N_r^*, \qquad 1 \to \langle\omega\rangle \to E_r \to \gamma \to 1. \tag{B.16.R3}$$

The remaining comparison identifies this particular extension with the source $\widetilde{\gamma}_{r,s}$ and $\mathcal{N}_r^?$ and identifies its class with the pullback of the finite cocycle on C3:596–610. D.8 proves that finite cocycle and D.12 computes the stated arithmetic cup product; neither alone identifies the present profinite extension. The source's claims about its nontriviality and about its restriction to named geometric sections require this comparison.

The diagram C3:628(26), folio 617 verso (parallel page 627), introduces different objects again. Write $\mathscr{E}_{1,1,\mathbb{Q}}$ for the stack of elliptic curves with origin and $\mathscr{E}'_{1,1,\mathbb{Q}}$ for its rigidification by the central involution $[-1]$. These are the source's $M_{1,1}, M'_{1,1}$; neither is the affine group scheme $\mathcal{M}$ of Part B. Concretely, rigidification stackifies the same objects with isomorphisms divided by this central C_2. Its fibres are therefore locally nonempty groupoids with central automorphism group $C_2 = \mu_2$ over $\mathbb{Q}$, which is the gerbe meaning required in that diagram. The j-invariant factors through this map to the coarse j-line. A specified object of the rigidification is additional data beyond its j-coordinate. The question in scan 628 about existence of a curve of invariant zero over $\mathbb{Q}$ has the explicit answer

$$E : y^2 = x^3 + 1, \qquad c_4 = 0, \quad c_6 = -864, \quad \Delta = -432 \neq 0, \quad j = c_4^3/\Delta = 0. \tag{B.16.19}$$

The point at infinity supplies the origin and the nonzero discriminant proves smoothness. The invariant identity $c_4^3 - c_6^2 = 1728\Delta$ and the displayed values are checked in `D_motives.py`. Over an algebraic closure the automorphism $(x, y) \mapsto (\zeta_3 x, y)$ survives rigidification, so that object is not identified with a scheme point merely by dividing out $[-1]$. The source itself distinguishes a lift of a rigidified object from a lift of its coarse invariant in C3:629, folio 618 (parallel page 628). Both photographs were inspected. The exact geometric question in (R3) is the lift of the particular object or section specified by the source, not the existence of an elliptic curve with that coarse invariant. No integral extension of this stack comparison at residue characteristic two is asserted here.

The further lift laws and filtered groups on C3:583–595 and 635–650 are separately recorded in B.18.

The "Open" label applies to (R1)–(R3), as unfinished assertions of this report. It does not label the established Herfort–Ribes theorems or our normalized conjugator calculation as open.

B.4 Late normalized sections and the extra open-subgroup datum

The constructions of §49.V–VI continue beyond C3 scan 650: scan 651 starts folio 628 and scans 689–690 are folio 647 and its verso. In this subsection groups are discrete or profinite, with continuous maps in the latter case. We do not identify their automorphism functors with integral group schemes. The finite examples may also be read as constant finite group schemes. Their constant C_2 is not replaced by μ_2 in characteristic two.

B.17. The smaller antidiagonal quotient needs an intersection [Refuted]

The definition in §49.VI, fols. 631 verso–632 verso, C3:658 (18), 659 (28), and 660 (29)–(32), cannot be used literally with all the data it permits. Its loop subgroup $L_0^\natural$ need not lie in the given open normal $C^\natural$. In that case its antidiagonal does not lie in $C^\natural \rtimes Z$, so the displayed smaller quotient is undefined. This occurs already in the modular finite quotient $C = \mathrm{SL}_2(\mathbb{Z}/4)$, with $C^\natural$ the image of Π_0.

Here is the exact repair. Let C be a group, or a profinite group, and let D be a group of automorphisms with retained numerical coordinates. In the profinite case assume D is profinite, its action is continuous, and the map j below is continuous. Let $A \triangleleft C$ and $L \subset C$ be D-stable subgroups, closed in the profinite case. Suppose

$$j : L \longrightarrow D, \qquad j(l) = (\mathrm{Int}(l), 1, 1, 1)$$

is defined, and D's conjugation on $j(L)$ agrees with its given action on L. Write

$$\Delta L = \{(l^{-1}, j(l)) : l \in L\}, \quad G = (C \rtimes D)/\Delta L, \quad H_A = \mathrm{im}(A \rtimes D \longrightarrow G), \quad K = L \cap Z(C).$$

Then the image construction is well-defined and

$$H_A \simeq (A \rtimes D)/\Delta(L \cap A), \qquad 1 \longrightarrow A/(A \cap K) \longrightarrow H_A \longrightarrow D/j(L \cap A) \longrightarrow 1. \quad \text{(B.17.1)}$$

If $K = 1$, the kernel of $H_A \to D/j(L)$ is the image of AL, and it equals the image of A if and only if $L \subset A$. Thus replacing the quotient by its image does not also preserve the manuscript's claimed smaller exact sequence without this extra condition.

Suppose more specifically that $D \subset \mathrm{Aut}(C) \times T \times \{\pm 1\}^2$, where T is the multiplier group. In the profinite case take T profinite and this embedding continuous and closed. Put

$$P_A = \{(f, m) \in A \times T : (\mathrm{Int}(f), m, 1, 1) \in D\}.$$

For the homomorphism forgetting the multiplier but retaining the automorphism and both signs, the exact kernel is

$$\ker(H_A \longrightarrow \mathrm{Aut}(C) \times \{\pm 1\}^2) \simeq P_A/\{(l, 1) : l \in L \cap A\}. \quad \text{(B.17.2)}$$

For the manuscript's specified normalizer-with-multiplier $P_A = S\widetilde{N}_0^\natural$, this gives the correctly typed form of C3:664 (52 bis). Its denominator can be all of $L_0^\natural$ only when that subgroup is contained in $C^\natural$.

Proof. We use the multiplication $(c, d)(c', d') = (c\, d(c'), dd')$. The map $l \mapsto (l^{-1}, j(l))$ is a homomorphism: its first coordinate on a product is $(ll')^{-1}$. Its image centralizes the copy of C, and conjugation by D carries its value at l to its value at $d(l)$. Hence ΔL is normal. Its intersection with $A \rtimes D$ is exactly $\Delta(L \cap A)$, proving the first isomorphism. This intersection is normal in the smaller semidirect product. Projection onto $D/j(L \cap A)$ gives the exact sequence; the kernel of $A \to H_A$ is $A \cap \ker j = A \cap K$. When $K = 1$, a representative (a, d) maps trivially to $D/j(L)$ precisely when $d = j(l)$; its image in G is then the image of al. This proves the assertion about AL. All subgroups and quotients are closed profinite groups when

the data are profinite, since they are continuous images or closed subgroups of compact groups. No quotient-topology identification is omitted.

For the second assertion use the injective homomorphism

$$P_A \longrightarrow A \rtimes D, \qquad (f, m) \longmapsto (f^{-1}, (\mathrm{Int}(f), m, 1, 1)).$$

Multiplying two displayed pairs gives the value at (ff', mm'). Its image is precisely the inverse image of the asserted automorphism-and-sign kernel. Its intersection with ΔL consists of $(l, 1)$ with $l \in L \cap A$, proving (B.17.2).

For the finite modular witness take, modulo 4,

$$\rho = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, \quad \sigma = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad e_0 = \sigma\rho, \quad l = e_0^2 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix},$$

and

$$A = \left\{ I, \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, \begin{pmatrix} 3 & 0 \\ 2 & 3 \end{pmatrix}, \begin{pmatrix} 3 & 2 \\ 0 & 3 \end{pmatrix} \right\}.$$

This is the normal Klein four subgroup used in E.15–E.16. Here $l \notin A$, but $h = -l \in A$, and $\mathrm{Int}(l) = \mathrm{Int}(h)$. Consequently the tuple $(\mathrm{Int}(l), 1, 1, 1)$ satisfies C3:653 (4): both conjugators can be h, the loop is fixed, A is stable, and the induced action on C/A is already trivial. It therefore satisfies both the stronger and the weakened versions of condition d). The involution $\tau = \mathrm{diag}(-1, 1)$ normalizes A, as required by the orientation extension.

The same obstruction occurs in $C = \widehat{\mathrm{SL}_2(\mathbb{Z})}$ itself. Take $A = \widehat{\Pi}_0$, $l = e_0^2$, and $h = -e_0^2$. By E.15, A is normal of index 12 and h is one of its specified generators; the modulo-four calculation proves $l \notin A$. Conjugation by τ inverts the two generators of Π_0, so it preserves its closure. Again $\mathrm{Int}(l) = \mathrm{Int}(h)$, and both conjugators may be h. Thus $l \in L_A \setminus A$ for the actual named modular subgroup, not only for an abstract finite model.

The program `B_late_sections.py` enumerates all 48 matrices, verifies generation by ρ, σ, normality and these identities, and checks the actual positive-sign loop preimage in the four-element $\langle e_0 \rangle$. It is $L = \{1, l\}$. In checking the weakened condition d), the image of Π is the eight-element kernel of reduction modulo 2. Moreover $L \cap A = L \cap Z(C) = 1$, so the example does not exploit a failure of the earlier injectivity condition. For the numerical subgroup $D = \langle (\mathrm{Int}(l), 1, 1, 1) \rangle$, the corrected image is AL, of order 8, whereas $|A| = 4$ and $D/j(L) = 1$. Its automorphism kernel is $\{I, -I\}$. These are executed checks of both the quotient obstruction and the exact repair. □

B.18. The late normalized sections and their remaining profinite comparisons [Open]

The following dictionary records the additional constructions of §49.V–VI. It does not infer the existence of the actual profinite normalizer sections from the finite cocycle calculation in D.8. B.16 supplies normalized conjugators and quotient sections in its precisely defined G_Z; the further unquotiented sections and source identifications below require additional arguments. We distinguish the smaller fibre product of scan 648, written $\mathcal{B}$ below, from the later open-subgroup construction depending on $A = C^{\natural}$. The superscript $\natural \mid \sharp$ is a working reading, not a palaeographical conclusion.

Object or map	Definition or assertion to be compared	Source: C3 scan; folio
a_0, b	For normalized conjugators, $a_0 = \alpha^{-1}u$, $b = \beta^{-1}u$. The b-coordinate is multiplicative; the image of a_0 has the twisted law (47), not an ordinary product law.	583–585 (41)–(48); 595–596
$A(u), S_0\Gamma^!$	The fibre over u consists of $a = f'^{-1}u, b = f^{-1}u$, with $f' = \alpha\rho^j(\tau')^{\nu_3}$, $f = \beta\sigma^i$, $j \bmod 6, i \bmod 4$. The normalized section chooses $i = j = 0$; now $a = (\tau')^{\nu_3}\alpha^{-1}u$.	586–587 (49)–(53); 596v–597
$S_0^{\Pi}\Gamma^!$	The congruence-defined subgroup in (62) is asserted to extend $Z^!$ by the first central C_2. The normalized section has cocycle (63); the formula for α'' requires the final commutator correction.	590–595 (59)–(64); 598v–601
$M^0(-\bar{j}), M'''(-\bar{j})$	$M^0(-\bar{j}) = \ker(\epsilon_2 \vert M'(-\bar{j}))$; $M'''(-\bar{j}) = \langle\tau'\rangle\, M^0(-\bar{j})$. The subgroup lattice, normality, and its stated indices require the actual normalized sections.	635–642 (32)–(44); 621–623v
$\Theta, \ker\Theta$	The map $N_\rho^* \to \mathrm{GL}_2(\mathbb{Z}/4)/V_4$ is asserted to have kernel $M^0(-\bar{j})$ and quotient of order 24. Computing the target is not a proof of this kernel.	643–647 (45)–(46); 624–626
$E_\rho, S_\sigma, \mathcal{B}$	$E_\rho = N_\rho^? \to \gamma$ has kernel μ_ρ; $S_\sigma = Z^!_{\sigma 0} \to \gamma$ is an isomorphism; $\mathcal{B} = E_\rho \times_\gamma S_\sigma \times_\gamma G$. These two normalizer maps are substantive inputs.	647–648 (50)–(55); 626–626v
$\Gamma^\flat$, bisection	$\Gamma^\flat = E_\rho \times_\gamma S_\sigma \subset \Gamma$; its kernel over γ is μ_ρ. The resulting canonical map lands in $\mathcal{M}_{\rho,\sigma}/\mu_\rho$.	649–650 (56)–(60); 627–627v
$A, C_\tau^\natural, Z^A$	An open normal, τ-stable $A \subset C$ is extra data; conjugators in (4) belong to $A \rtimes \langle\tau\rangle$, with their stated orientation signs. The surviving condition d) requires stability of A and identity on $(C/A)/\operatorname{Im}\Pi$.	651–653 (1)–(4); 628–629
L_A, γ_A, H_A	$L_A = \{l \in L_0 : j(l) \in Z^A\}$, $\gamma_A = Z^A/j(L_A)$. The image construction H_A and its smaller kernel use B.17, not an undefined antidiagonal quotient.	658–660 (18)–(32); 631v–632v
$S\widetilde{N}_0^A$	Pairs (f, m) with $f \in A$, $\mathrm{Int}(f)(e_0) = e_0^m$, and $(\mathrm{Int}(f), m, 1, 1) \in Z^A$. Forgetting the multiplier gives the kernel in (B.17.2).	663–664 (45)–(52 bis); 634–634v
M^A, G^A	In (107), the three images of ρ, σ, e_0 are A-conjugate to their specified τ-twists or powers, and u stabilizes A. The proposed identifications (108), (115)–(116) require matching these conditions with the repaired construction.	682–686 (107)–(118); 643v–645v
$\mathcal{M}^{A,?}$	Intersection of the later $\mathcal{M}^A$ with $\mathcal{M}^?$. Its image in M^A is determined by the images of $N_\rho^{A,?}$ and $Z_{\sigma 0}^{A,!}$ in Γ_0^A; the claimed surjectivity is not checked in the manuscript.	688–690 (122)–(128); 646v–647v

Proved reductions, executed finite cases, and exact residuals. The normalized multiplication laws. Write $t = \tau_\gamma$, $t^2 = 1$, and suppose $\delta(u) = t^{\nu(u)}\delta(a_0(u))$, with $\nu(u'u) = \nu(u') + \nu(u)$. Multiplication alone gives

$$\delta(a_0(u'u)) = t^{\nu(u)}\delta(a_0(u'))t^{-\nu(u)}\delta(a_0(u)). \tag{B.18.1}$$

Thus the untwisted law is equivalent to the commutation condition in scan 585 (48). In the finite group S_3, take $t = (12)$, ν the permutation sign in $\mathbb{F}_2$, and $a_0(g) = t^{\nu(g)}g$. The program checks (B.18.1) for all 36 ordered pairs and exhibits failures of ordinary multiplicativity.

For scan 595 use $t = \tau'$, $a = t^{\nu_3}\alpha^{-1}u$, and the analogous primed values. If $a'' = \varepsilon a' a$, with ε central of order dividing two, direct substitution gives the exact identity

$$\alpha'' = \varepsilon\, u'(\alpha)\alpha'\, t^{\nu'_3}[a', t^{\nu_3}]t^{-\nu'_3}. \tag{B.18.2}$$

Indeed the left side is $\varepsilon u' u a^{-1} a'^{-1} t^{\nu_3+\nu'_3}$; multiplying the right side and cancelling the middle factors gives that same word. The program verifies the word identity for all $u, u', a, a' \in S_3$ and both sign bits, and finds a failure when the commutator is omitted. This is a test of the stated algebraic identity, not a realization claim for arbitrary such tuples in the universal profinite group.

The source-order cocycle in scan 595 (63), with the left factor primed, is

$$c((u', v'), (u, v)) = u(u' + v') \in \mathbb{F}_2, \quad (\epsilon_3, \epsilon_2) = ((-1)^u, (-1)^v). \tag{B.18.3}$$

The associated group on $\mathbb{F}_2^3$ has law $(s, u, v)(t, x, y) = (s + t + x(u + v), u + x, v + y)$. Its bilinearity proves associativity. The coordinate change $(s, u, v) \mapsto (s + uv, u, v)$ identifies it with D.8's convention. Hence that classification applies, but its cocycle must not be copied with the factor order reversed. The program verifies all 512 associativity triples, the coordinate change, and all 16 sign pairs. It confirms the criterion $\varepsilon = 1$ exactly when $\epsilon_3(u) = 1$ or $\epsilon_2(u')\epsilon_3(u') = 1$, thereby correcting the incompatible prose at the end of scan 594.

The early bisection, conditional on its typed inputs. Let N_r, N_s, G map onto γ, with kernels C_r, C_s, C. Suppose subgroups $E \subset N_r$ and $S \subset N_s$, closed in the profinite case, satisfy

$$1 \to \mu_r \to E \to \gamma \to 1, \qquad S \xrightarrow{\sim} \gamma, \tag{B.18.4}$$

where $\mu_r \subset C_r$ is central in N_r. Define $\mathcal{M} = N_r \times_\gamma N_s \times_\gamma G$, $\mathcal{B} = E \times_\gamma S \times_\gamma G$, $\Gamma = N_r \times_\gamma N_s$, and $\Gamma^\flat = E \times_\gamma S$. Pullback of (B.18.4) proves

$$1 \to \mu_r \to \mathcal{B} \to G \to 1, \quad \mathcal{B}/\mu_r \simeq G \hookrightarrow \mathcal{M}/\mu_r, \quad \Gamma^\flat \cap (C_r \times C_s) = \mu_r \times 1. \tag{B.18.5}$$

Every element of Γ differs from an element of $\Gamma^\flat$ by an element of its kernel $C_r \times C_s$, so $\Gamma = \Gamma^\flat(C_r \times C_s)$. After quotienting by μ_r, this is a split extension with kernel $(C_r/\mu_r) \times C_s$ and complement $\Gamma^\flat/\mu_r \simeq \gamma$; the action is the conjugation action of that complement. This proves exactly the formal bisection assertions of scans 648–650 under (B.18.4), without manufacturing the two required profinite normalizer maps.

If $\mu_s \subset C_s$, $\mu_G \subset C$ are the other central factors, then in the displayed coordinate order (r, s, G)

$$\mathcal{B} \cap (\mu_r \times \mu_s \times \mu_G) = \mu_r \times 1 \times \mu_G. \tag{B.18.6}$$

Thus the $1 \times \mu \times \mu$ printed in the checked transcription of scan 648 is incompatible with that order and with its definition of μ_ρ. The second inclusion of scan 647 (50) must have target N_ρ^*, and the lower right object of scan 649 (57) must be γ, for the displayed kernels. The final arrow of scan 650 has the canonical target $\mathcal{M}/\mu_r$, as in scan 649 (56); a lift to $\mathcal{M}$ needs a splitting not supplied by the fibre-product argument.

For an executed finite case take $\gamma = C_2^2$, the source-order D_8 extension E above, $N_r = E \times C_3$, $N_s = \gamma \times C_4$, $S = \gamma \times \{0\}$, and $G = \gamma \times C_2$. Then $\mathcal{M} \simeq E \times C_3 \times C_4 \times C_2$ has order 192, $\mathcal{B}$ has order 16, and its kernel over the order-eight group G has order 2. The program verifies (B.18.5)–(B.18.6), the order-96 group Γ, its order-eight bisection and their order-two intersection. Both lifts of the element $((1, 0), 0) \in G$ have order 4. Hence $\mathcal{B} \to G$ has no section: omitting the quotient in the purported canonical lift would fail even in this

finite fibre-product model. These groups are a model of the formal hypotheses, not asserted to be normalizers in the same universal modular group.

Finally, for any subgroup restrictions $E_A \subset E$, $S_A \subset S$, and $G_A \subset G$, the image of the third projection is exactly

$$\operatorname{im}(E_A \times_\gamma S_A \times_\gamma G_A \longrightarrow G_A) = q^{-1}\big(\operatorname{im} E_A \cap \operatorname{im} S_A\big). \tag{B.18.7}$$

Both inclusions follow by choosing the two lifts of the same image in γ. This proves the reduction in scan 690: surjectivity of the two normalizer maps would suffice, but must itself be established.

Exact residuals. For the universal profinite modular input, B.16 constructs the normalized conjugators and the maps $Z^! \to N_\rho^*/\langle\omega\rangle$ and $Z^! \to N_\sigma^*$, together with their quotient sections over γ, inside the formal G_Z. It remains to match the congruence-defined source subgroup $S_0^{\Pi}\Gamma^!$ of scans 583–595 and identify its central extension with the computed sign cocycle. Prove the normality and index assertions of scans 635–642 and the precise kernel

$$\ker(N_\rho^* \xrightarrow{\Theta} \mathrm{GL}_2(\mathbb{Z}/4)/V_4) = M^0(-\bar{j}). \tag{B.18.8}$$

This modulo-four target is already the first finite quotient needed by that construction. E.19 and its program compute the target of order 24; they do not compute the kernel of this profinite-source map. In particular, stability alone does not prove the congruence $\tau'(x) \equiv x \pmod{\Pi_0}$ in scan 641 (42); its action on the quotient must be checked. For the typed inputs (B.18.4), B.16 isolates the remaining unquotiented lift $\gamma \to N_\sigma^*$ and the identification of the central pullback in N_ρ^* with the source's $N_\rho^?$. Thus its proved quotient-section maps from scans 612–616 are used, while the additional assertions are retained as explicit residuals.

For the later extra datum take first

$$C = \widehat{\mathrm{SL}_2(\mathbb{Z})}, \qquad A = \widehat{\Pi} = \ker(C \to \mathrm{SL}_2(\mathbb{F}_2)),$$

the first named proper subgroup in scan 656, of index 6, and then $A = \widehat{\Pi}_0$, of index 12. Use τ, not $\tau' = \sigma^{-1}\tau$, in the allowed conjugator subgroup. In particular $\tau(\rho) = \sigma\rho^{-1}\sigma^{-1}$, so changing the condition to A-conjugacy to ρ^{-1} is not justified when $\sigma \notin A$. With the surviving weakened version of condition d) fixed, determine the actual L_A, the corrected image H_A, and the map to the M^A of scan 682 (107); prove or correct scan 686 (115)–(116). Then determine the images of $N_\rho^{A,?}$ and $Z_{\sigma 0}^{A,!}$ in Γ_0^A, and therefore the image in (B.18.7), as requested in scans 689–690. These are explicit remaining comparisons for the two named modular subgroups, not a claim that the corresponding questions are open in the literature. The $A = \widehat{\Pi}_0$ construction already has the modulo-four obstruction proved in B.17; the original smaller quotient cannot be silently used in its place.

The finite calculations and formal reductions prove cases of these exact statements. They do not prove that the normalizer subgroups, their images, and their quotient maps descend to the integral schemes of §48. That additional representability and fppf comparison remains within the explicit scope limitation of A.9 and B.8. □

C The presentation on scan 336

We work on the big fppf site of the indicated base. A constant group means the associated locally constant sheaf, not a diagonalizable group. In this section alone, s denotes the earlier lift s_- of the conventions table, because it is the lift used on scan 336. We write $x(t) = \left(\begin{smallmatrix} 1 & t \\ 0 & 1 \end{smallmatrix}\right)$, $s = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, and $z = -I$. In the normalization of §48.VI, scan 336, displayed folio 335, the matrices are

$$\sigma = s, \qquad \rho = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, \qquad \varepsilon_0(t) = x(-t).$$

Thus $\rho^{-1}\sigma\varepsilon_0(1) = \rho^3 = \sigma^2 = z$. The manuscript really says $\mathbb{Z}/2$, $\mathbb{Z}/3$, $\mathbb{Z}/4$, and $\mathbb{Z}/6$: the photograph was inspected. Replacing these by μ_n changes the question.

C.1. The literal sheaf presentation [Proved]

Put $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, $\overline{\Gamma} = \mathrm{PSL}_2(\mathbb{Z})$ and let $\mathbb{Z} \to \Gamma$ send 1 to $x(1)$. The two sheaves specified by scan 336 are

$$A_{\mathrm{SL}} = \underline{\Gamma} *_{\underline{\mathbb{Z}}} \mathbb{G}_a, \qquad A_{\mathrm{P}} = \underline{\overline{\Gamma}} *_{\underline{\mathbb{Z}}} \mathbb{G}_a. \tag{C.1}$$

Here the asterisk is a pushout in sheaves of groups, and the other map from $\underline{\mathbb{Z}}$ sends n to $n \cdot 1$. They have canonical comparison homomorphisms $A_{\mathrm{SL}} \to \mathrm{SL}_2$ and $A_{\mathrm{P}} \to \mathrm{PGL}_2$. The same generators and relation that the manuscript assigns to GL_2 give $A_{\mathrm{SL}} \to \mathrm{SL}_2 \hookrightarrow \mathrm{GL}_2$.

Proof. The first two finite factors in the special-linear version give $C_4 *_{C_2} C_6$, with the common element $\sigma^2 = \rho^3$ of order two. The additional relation (63) identifies $\varepsilon_0(1)$ with $\sigma^{-1}\rho z$. Eliminating this generator gives precisely the pushout (C.1); changing t to $-t$ gives our x. In the projective version one starts with $C_2 * C_3$ and eliminates $\varepsilon_0(1)$ using (62). The identification of these discrete groups with Γ and $\overline{\Gamma}$ is proved next. All displayed matrices have determinant one. Consequently the asserted GL_2 presentation cannot include matrices with arbitrary determinant. This conclusion uses the actual matrices, independently of any doubtful word surrounding the formulas. □

C.2. The discrete integral amalgam and its reduction [Known]

The comparison maps

$$C_2 * C_3 \xrightarrow{\sim} \mathrm{PSL}_2(\mathbb{Z}), \qquad C_4 *_{C_2} C_6 \xrightarrow{\sim} \mathrm{SL}_2(\mathbb{Z})$$

are isomorphisms of discrete groups. This is the conclusion of Conrad, Theorem C.1 and Theorem 1.1, in the present normalization Conrad (nd); the following proof gives the reduction and the central lift explicitly.

Proof. For $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ of determinant one over $\mathbb{Z}$, left multiplication by $x(n)$ adds n times the second row to the first and left multiplication by s interchanges the rows with a sign. The Euclidean algorithm reduces (a, c) to $(\pm 1, 0)$. The remaining matrix is a power of $x(1)$, possibly multiplied by $s^2 = -I$. Hence $s, x(1)$ generate.

For injectivity in the projective group, cyclically reduce an alleged nontrivial kernel word in $C_2 * C_3$. A word in a single factor has nonidentity image. Otherwise cyclically rotate it to the form $(s\rho^{i_1}) \cdots (s\rho^{i_m})$ with $i_j = 1$ or 2. Direct multiplication gives $s\rho = x(-1)$ and $s\rho^2 = y(-1)$. Conjugation by $\mathrm{diag}(-1, 1)$ makes these blocks $x(1), y(1)$. Starting with I, multiplication by either positive block strictly increases the sum of the nonnegative entries, because both rows and columns of an invertible such matrix have positive sum. A nonempty product therefore has entry sum greater than two and cannot be I; nonnegative entries also exclude $-I$. This is Conrad's positive-matrix argument, with the change of normalization exhibited. It proves injectivity without a geometric fundamental-domain assumption.

In $C_4 *_{C_2} C_6$ the common involution is central, because it is central in each factor. Quotienting by it gives $C_2 * C_3$. The preceding injectivity shows that the kernel of the special-linear comparison is contained in this order-two subgroup; its nonidentity element maps to $-I \neq I$. The kernel is therefore trivial. This also proves that the parabolic subgroup used in (C.1) is an embedded copy of $\mathbb{Z}$. □

C.3. Images and cokernels in the fppf topology [Proved]

Over every base, $A_{\mathrm{SL}} \to \mathrm{SL}_2$ is an epimorphism even for the Zariski topology, and $A_{\mathrm{P}} \to \mathrm{PGL}_2$ is an fppf epimorphism. The image of the literal general-linear map is SL_2 and its

sheaf cokernel is $\mathbb{G}_m$, by determinant. For a local ring R, the subgroup of $\mathrm{SL}_2(R)$ generated by s and the $x(t)$ is all of $\mathrm{SL}_2(R)$. For a local ring, or for $R = \mathbb{Z}, \mathbb{Z}[1/p]$, the subgroup they generate in $\mathrm{PGL}_2(R)$ is the kernel of

$$\det \bmod (R^\times)^2 : \mathrm{PGL}_2(R) \longrightarrow R^\times/(R^\times)^2. \tag{C.2}$$

This last assertion concerns the subgroup generated by the displayed global points, not an interchange of evaluation with a sheaf pushout.

Proof. Put $y(v) = sx(-v)s^{-1}$ and

$$w(u) = x(u)sx(u^{-1})s^{-1}x(u), \qquad h(u) = w(u)s = \mathrm{diag}(u, u^{-1}) \quad (u \in R^\times).$$

These identities hold over $\mathbb{Z}[u, u^{-1}]$. If a is a unit, then a determinant-one matrix factors as

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = y(c/a)h(a)x(b/a). \tag{C.3}$$

If c is a unit, apply the same formula to $s^{-1}g$. The two open sets $D(a)$ and $D(c)$ cover the base since $ad - bc = 1$. This proves the Zariski epimorphism and generation over local rings. Over $\mathbb{Z}$ use C.2; over $\mathbb{Z}[1/p]$ clear denominators and perform the integral Euclidean algorithm. The resulting gcd is a power of p and hence a unit, so (C.3) finishes the reduction.

A projective-linear point is fppf locally represented by a matrix in GL_2. Adjoining a square root of its unit determinant is a finite free faithfully flat extension, including in characteristic two. Scalar multiplication then makes the determinant one. This proves the projective fppf epimorphism. If $\mathrm{Pic}(R) = 0$, a projective-linear point has a matrix representative; it has a determinant-one representative exactly when that determinant is a square. All rings specified in the statement have trivial Picard group. Finally determinant gives the split exact sequence $1 \to \mathrm{SL}_2 \to \mathrm{GL}_2 \to \mathbb{G}_m \to 1$, with section $d \mapsto \mathrm{diag}(d, 1)$. □

C.4. The proposed presentation fails over every nonempty base [Refuted]

Neither comparison in C.1 is an isomorphism over any nonempty base scheme. Their kernels are noncentral after a field extension of every geometric fibre. In particular there is no nonempty open subset of $\operatorname{Spec}\mathbb{Z}$, including $\operatorname{Spec}\mathbb{Z}[1/6]$, over which the literal presentation is correct. The kernel is therefore not the central K_2 kernel of a Steinberg presentation. Lead 1, in the form proposed in the prompt, is false.

Proof. We give a word surviving fppf sheafification; this is stronger than merely finding a relation absent from a formal list. In characteristic zero put $H = \mathbb{Z}$ and let Γ_0 be Γ or $\overline{\Gamma}$. In characteristic p, put

$$\Gamma_p = \Gamma_0/\langle\langle x(1)^p \rangle\rangle, \qquad H = C_p.$$

There is a homomorphism from Γ_p to the appropriate matrix group over $\mathbb{F}_p$. Its restriction to H is injective, and $s \notin H$, as seen from the lower-left entry. On a connected algebra in that characteristic, the presheaf underlying (C.1) is the ordinary amalgam $\Gamma_p *_H (R, +)$; in characteristic zero it is $\Gamma_0 *_{\mathbb{Z}} (R, +)$. This follows by first imposing exactly the kernel of $\mathbb{Z} \to (R, +)$ and then taking the ordinary pushout.

Let k be any field and work in $k(T, U, V)$, with independent variables. The matrix word

$$K = h(T)x(U)h(T)^{-1}x(-T^2U)$$

is the identity, since $h(T) = \mathrm{diag}(T, T^{-1})$. Conjugating it by $x(-T)$ gives the following alternating word L:

$$\begin{aligned} L = {} & sx(T^{-1})s^{-1}x(T)sx(U)s^{-1}x(-T) \\ & sx(-T^{-1})s^{-1}x(-T^2U). \end{aligned} \tag{C.4}$$

Every $s^{\pm 1}$ lies outside H, and every displayed additive parameter lies outside H. Thus (C.4) is a cyclically reduced word of length twelve in the amalgam, and is nonidentity. The commutator $[L, x(V)]$ has length twenty-four after cancellation of the adjacent additive parameters: its central additive parameter is V, and all other additive syllables are still nonconstant. Hence it too is nonidentity, proving noncentrality.

For completeness, the normal-form fact used here is elementary. Choose representatives for right cosets of the common subgroup in each factor. Move common-subgroup elements to the right successively; induction on the number of alternating factors gives a unique reduced representative. A reduced alternating word of positive length cannot represent the identity. This proof also proves injectivity of the two factors. It applies to our two factors because the maps from H are injective, as checked above.

No faithfully flat extension can invalidate this argument. A nonzero algebra over $k(T, U, V)$ contains that field injectively. Its displayed parameters therefore remain outside the constant subgroup H on every nonempty connected piece. The same reduced word remains nonidentity on every fppf cover. Thus it is nonidentity in the sheaf pushout, and so is its commutator. Every nonempty base has a field point; base change preserves the sheaf presentation. This proves the assertion for every base.

There are smaller rational witnesses. In the special-linear case $[x(1/2), z]$ is a reduced nonidentity word but the identity matrix. In the projective case $w(2/3)^2$ is a nonidentity reduced amalgam word but has matrix $-I$. The symbolic program verifies both matrix identities. Its integral calculation of (C.4) verifies the same identity simultaneously in characteristics $2, 3, 5$. □

The distinction from the K-theoretic literature is substantive. Brylinski–Deligne define central extensions on the big Zariski site and classify those extensions, not the much larger source (C.1) (Brylinski and Deligne, 2001, Introduction and Theorem 7.2). Moreover their introduction explicitly distinguishes type C_n, including SL_2, in the description of universal central extensions. Our noncentral kernel already prevents application of that classification to the literal manuscript presentation.

C.5. An explicit normal generating family for the entire kernel [Proved]

The kernel of $A_{\mathrm{SL}} \to \mathrm{SL}_2$ is the normal subsheaf generated by the following four families of words. All symbols below denote words in s and $x(t)$, so this is an explicit presentation. For u invertible put

$$N_0(v, u, t) = y(v)h(u)x(t), \qquad N_1(v, u, t) = sN_0(v, u, t), \qquad D = 1 + u^2vt.$$

For $i = 0, 1$ impose

$$N_i(v, u, t)s = N_i(v + u^{-2}t^{-1}, ut, -t^{-1}) \qquad (t \text{ invertible}), \tag{C.5}$$

$$N_i(v, u, t)s = N_{1-i}(-u^2t/D, (-1)^iD/u, -u^2v/D) \qquad (D \text{ invertible}). \tag{C.6}$$

These relations are valid over $\mathbb{Z}$, with no primes inverted. They identify the entire kernel as a normal subsheaf, rather than only exhibiting some of its elements.

Proof. Multiplication gives

$$N_0(v,u,t)=\begin{pmatrix} u & ut \\ uv & uvt+u^{-1}\end{pmatrix}, \qquad N_1(v,u,t)=\begin{pmatrix} -uv & -uvt-u^{-1} \\ u & ut\end{pmatrix}.$$

Substitution proves (C.5)–(C.6). The attached program independently reduces each of the four matrix differences to zero over the corresponding universal Laurent ring. The open sets $D(t)$ and $D(D)$ cover, because $D-u^2vt=1$.

Let Q be the sheaf quotient by these relators. A local word in $s, x(t)$ reduces, after a Zariski cover, to one of the forms N_i. Indeed the identity is $N_0(0,1,0)$: $h(1)=1$ is already an identity in the discrete factor Γ. Right multiplication by $x(a)$ changes t to $t+a$. Right multiplication by s is handled by (C.5)–(C.6) on their covering open sets. Inverses cause no problem: $x(a)^{-1}=x(-a)$ and $s^{-1}=s^3$. Induction on word length proves the asserted local reduction.

If a section of Q maps to the identity matrix, reduce it locally. It cannot have form N_1, whose lower-left entry is a unit. In form N_0 its entries force $u=1$, $v=t=0$, so it is the identity in Q. Thus $Q\to \mathrm{SL}_2$ is a monomorphism of sheaves; surjectivity is C.3. The quotient is exactly SL_2 and the stated normal subsheaf is exactly the kernel. Notice that the proof is valid over rings with nilpotents as well as fields. □

C.6. The corresponding presentations for projective and general linear groups [Proved]

For A_{P}, add the images of (C.5)–(C.6) and also

$$h(\zeta)=1 \quad \text{for every } \zeta\in\mu_2. \tag{C.7}$$

Their normal closure is exactly $\ker(A_{\mathrm{P}}\to \mathrm{PGL}_2)$. Over $\mathbb{Z}[1/2]$, (C.7) follows locally from $s^2=1$ after (C.5)–(C.6); over characteristic two it cannot be omitted.

A presentation of GL_2 is obtained from the corrected special-linear presentation by adjoining a copy $m(\mathbb{G}_m)$ and imposing

$$m(d)x(t)m(d)^{-1}=x(dt), \qquad m(d)sm(d)^{-1}=h(d)s. \tag{C.8}$$

Here $m(d)$ maps to $\mathrm{diag}(d,1)$.

Proof. The source A_{P} is the quotient of A_{SL} by the normal subsheaf generated by the constant element $z=s^2$. After the Gaussian relations, quotienting by all scalar sections ζI, $\zeta^2=1$, gives the fppf quotient $\mathrm{SL}_2/\mu_2=\mathrm{PGL}_2$, by the same square-root argument as C.3. Since $h(\zeta)=\zeta I$ in SL_2, (C.7) imposes exactly this quotient. When two is invertible, the roots of X^2-1 are fppf locally $1,-1$, so the constant relation suffices locally. In $R=\mathbb{F}_2[e]/(e^2)$, however, $\zeta=1+e$ satisfies $\zeta^2=1$ and $\zeta\neq 1$. The scalar $h(\zeta)=\zeta I$ is nonidentity in $\mathrm{SL}_2(R)$ and identity in $\mathrm{PGL}_2(R)$. This finite counterexample is checked in the program. In characteristic two the constant $-I$ is already I; the extra scalar cannot be removed by that constant relation.

For (C.8), conjugation by $\mathrm{diag}(d,1)$ has exactly the two stated effects. Since $x(t)$ and s generate as a sheaf, these relations impose its complete conjugation action on SL_2. The presented sheaf is therefore the semidirect product $\mathrm{SL}_2\rtimes\mathbb{G}_m$ for that action. Every invertible matrix g has the unique expression $(g\,\mathrm{diag}(\det(g)^{-1},1))\,\mathrm{diag}(\det(g),1)$, which proves its isomorphism with GL_2. □

C.7. Executed point presentations and their kernels [Proved]

Let $P_{\mathrm{SL}}(R)$ and $P_{\mathrm{P}}(R)$ mean the ordinary abstract presentations obtained by evaluating the generators and relations before sheafification. These are distinguished from $A_{\mathrm{SL}}(R), A_{\mathrm{P}}(R)$. For $R = \mathbb{Z}/n$,

$$\begin{aligned} P_{\mathrm{P}}(R) &= \langle s, t \mid s^2 = (st)^3 = t^n = 1 \rangle, \\ P_{\mathrm{SL}}(R) &= \langle s, t \mid s^4 = 1, (st)^3 = s^2,\ t^n = 1 \rangle. \end{aligned} \tag{C.9}$$

For $n = 2, 3, 4, 5$ their respective orders are $(6, 12, 24, 60)$ and $(6, 24, 48, 120)$. In each case the comparison is an isomorphism onto the subgroup generated by the corresponding matrices. For $R = \mathbb{F}_4$, both point presentations are $S_3 *_{C_2} (C_2 \times C_2)$, and the kernel of the comparison to $\mathrm{SL}_2(\mathbb{F}_4)$ is a free group of rank six. For $R = \mathbb{F}_7$ and $\mathbb{Z}/9$, the kernels of both point comparisons onto their generated matrix images are surface groups of genera three and ten, respectively.

Proof. For a cyclic additive ring, every $x(a)$ is a power of $x(1)$. Eliminating those generators gives (C.9), using the equivalent standard discrete generators $s, x(1)$ from C.2. The central element s^2 in the special-linear presentation has order at most two; its quotient is the projective presentation.

The program performs Todd–Coxeter enumeration for the trivial subgroup for $n = 2, 3, 4, 5$ in both presentations. It saves every coset table, checks that each generator column is a permutation with the indicated inverse, and checks every relator at every coset. The eight completed enumerations have the orders stated. For an independent geometric description of the projective upper bounds, they are the rotational triangle groups of the spherical triangles $(2, 3, n)$, with orders $6, 12, 24, 60$; the enumeration does not rely on this description. The complete matrix enumerations give precisely the same orders for the images. A surjective map between these finite groups of the same order is injective.

In characteristic two the discrete quotient Γ_2 has order six, as the enumeration for $n = 2$ proves, and is S_3; the prime additive subgroup has order two. The pushout with $(\mathbb{F}_4, +) = C_2^2$ is therefore the asserted amalgam. Its two factors inject into $\mathrm{SL}_2(\mathbb{F}_4)$, which has order sixty. The kernel acts freely on the amalgam tree. The quotient graph has $60/6 + 60/4 = 25$ vertices and $60/2 = 30$ edges; it is connected. Its fundamental group is free of rank $30 - 25 + 1 = 6$. The tree construction is explicit: vertices are cosets of the two factors and edges are cosets of C_2; normal forms prove connectedness and absence of circuits. A group acting freely on a tree identifies with the fundamental group of its quotient graph by lifting paths. This supplies the required reduction to the elementary graph calculation.

For $n = 7, 9$, let Q be the projective matrix image, of order $168, 324$ respectively. The three elements s, st, t have exact orders $2, 3, n$. The regular permutation action of Q constructs an oriented surface as follows. Take $|Q|$ edges, with cyclic orders at black and white vertices prescribed by the permutations of orders two and three. Glue in the faces prescribed by the permutation of order n. This produces a connected orientable surface because the generators act transitively; its numbers of vertices, edges and faces are $|Q|/2 + |Q|/3$, $|Q|$, and $|Q|/n$. The exact orders ensure that no vertex or face has residual branching. The path-lifting construction, or van Kampen applied to the glued polygons, identifies its fundamental group with the kernel of the triangle presentation in (C.9): the only local loops killed upstairs are precisely $s^2, (st)^3, t^n$. Its Euler characteristic is

$$|Q|(1/2 + 1/3 + 1/n - 1) = -4, -18,$$

so its genus is three or ten. In particular both presentations are infinite, a fact not inferred from failure of enumeration. For the special-linear comparison, s^2 maps to $-I \neq I$. Projection identifies its kernel with the projective kernel: any lift of a projective-kernel element maps to I or $-I$, and multiplication by s^2 selects the unique lift mapping to I. □

The complete matrix counts are as follows. "Image" means the subgroup generated by the global points of the specified factors. For every row the special-linear image is all of $\mathrm{SL}_2(R)$.

R	$\lvert\mathrm{SL}_2\rvert$	$\lvert\mathrm{GL}_2\rvert$	$\lvert\mathrm{PGL}_2\rvert$	|projective image|	projective index
$\mathbb{F}_2$	6	6	6	6	1
$\mathbb{F}_3$	24	48	24	12	2
$\mathbb{F}_4$	60	180	60	60	1
$\mathbb{F}_5$	120	480	120	60	2
$\mathbb{F}_7$	336	2016	336	168	2
$\mathbb{Z}/4$	48	96	48	24	2
$\mathbb{Z}/9$	648	3888	648	324	2

All entries are obtained by enumerating all four matrix coordinates, testing determinants, and closing the stated generators under multiplication. Projective equivalence is tested by all unit scalar multiples. The predicted index in (C.2) is verified independently.

C.8. Sheafification changes points over finite rings [Proved]

For $q = p^a$, let Γ_p denote the appropriate discrete quotient used in C.4. There is an exact description

$$A(\mathbb{F}_q) = \left(\Gamma_p *_{C_p} (\overline{\mathbb{F}}_p, +)\right)^{\mathrm{Frob}_q}. \tag{C.10}$$

Frobenius fixes the discrete factor and acts by $u \mapsto u^q$ on the additive factor. In particular both sheaves have $A(\mathbb{F}_2) = S_3$ and $A(\mathbb{F}_4) = S_3 *_{C_2} (C_2 \times C_2)$. In contrast, $A_{\mathrm{SL}}(\mathbb{F}_3)$ is infinite although $P_{\mathrm{SL}}(\mathbb{F}_3)$ has order twenty-four. Both $A_{\mathrm{SL}}(\mathbb{Z}/4)$ and $A_{\mathrm{P}}(\mathbb{Z}/4)$ are infinite, although their point presentations have orders forty-eight and twenty-four. There is also an explicit additional infinite-order descent kernel element in $A_{\mathrm{SL}}(\mathbb{Z}/9)$.

Proof. Every nonempty fppf cover of an algebraically closed field has a rational point: a nonempty scheme of finite presentation over that field has a closed point with that residue field. Thus sheafification does not change the value of the presentation presheaf at $\overline{\mathbb{F}}_p$. This gives the amalgam on the right of (C.10). Any word uses finitely many algebraic parameters, so it is defined over a finite extension. Equality of reduced words is also detected over a finite extension. Finite Galois descent now identifies sections over $\mathbb{F}_q$ with the Frobenius-invariant words. This proves (C.10), including both injectivity and surjectivity.

For $p = 2$, $\Gamma_2 = S_3$ and its subgroup C_2 is malnormal: $C_2 \cap gC_2g^{-1} = 1$ when $g \notin C_2$. The normal-form comparison of a word and its Frobenius transform allows only insertions of elements of C_2 between adjacent syllables. At each discrete syllable outside C_2, malnormality forces both adjacent inserted elements to be the identity. Thus every additive parameter in a reduced invariant word is itself Frobenius-fixed. The remaining cases, a word in just one factor, give the same conclusion. This proves the two asserted characteristic-two values.

For the contrasting characteristic-three example, take $\alpha \in \mathbb{F}_{27}$ with $\alpha^3 - \alpha = 1$. The cubic $X^3 - X - 1$ has no root in $\mathbb{F}_3$ and is irreducible. In the discrete factor $\Gamma_3 = \mathrm{SL}_2(\mathbb{F}_3)$, $z = -I$ is a nonidentity central involution outside C_3. The word

$$b = x(\alpha)zx(-\alpha)$$

is Frobenius-fixed, since $\alpha^3 = \alpha + 1$ and $x(1)$ commutes with z in the discrete factor. Hence it descends to $A_{\mathrm{SL}}(\mathbb{F}_3)$. The word bz is cyclically reduced of length four, so $(bz)^m$ has length $4m$ for every $m > 0$. It has infinite order and maps to the identity matrix. The program verifies $\alpha^3 = \alpha + 1$ by exact arithmetic in $\mathbb{F}_3[X]/(X^3 - X - 1)$. Thus the finite coset tables in C.7 must not be described as the values of the sheaf pushout.

For the four-element base ring put $R = \mathbb{Z}/4$ and $S = R[b]/(b^2 + b + 1)$. This is a finite etale Galois cover with involution $F(b) = -b - 1$. The element $a = 2b$ satisfies $F(a) = a + 2$,

$2a = 0$, and $a \notin R$. In each discrete point factor from C.7 put $g = y(2)$. Its matrix shows $g \notin \langle x(1) \rangle$ and $gx(2) = x(2)g$. These are identities in the discrete point factor because C.7 proves its isomorphism with its matrix image. Consequently $x(a)gx(-a)$ is F-invariant and descends. The word

$$x(a)gx(-a)g^{-1}$$

is cyclically reduced of length four and has infinite order. Its matrix is the identity: $2a = 0$ gives $x(a)y(2) = y(2)x(a)$. The nontriviality persists on fppf covers. Indeed every nonzero flat algebra over the local Artinian ring S is faithfully flat and hence contains S injectively; the parameter a cannot become an element of the embedded constant subgroup $\mathbb{Z}/4$ on any nonempty component. The reduced-word argument therefore applies just as in C.4.

Finally take $R = \mathbb{Z}/9$ and $S = R[b]/(b^3 - b - 1)$. The derivative is a unit and the reduction of the polynomial is irreducible, so this is a finite etale Galois cover of degree three. Its Frobenius generator is $F(b) = 1 + 4b + 3b^2$; direct calculation verifies the polynomial relation and $F^3(b) = b$. Now $a = 3b$ satisfies $F(a) = a + 3$. With $z = s^2$ in the special-linear discrete factor, $x(a)zx(-a)z$ descends, maps to the identity matrix, and is cyclically reduced of length four. Its powers remain reduced over every fppf cover for the same local-Artinian argument. All ring identities in these two examples are executed exactly in the supplied program; no appeal to a numerical approximation or to unspecified class field theory is involved. □

C.9. The two requested localized integral amalgams [Known]

For each prime p, in particular $p = 2, 3$, put $D_p = \mathrm{diag}(1, p)$ and $\Gamma_0(p) = \{g \in \mathrm{SL}_2(\mathbb{Z}) : c(g) \equiv 0 \pmod{p}\}$. There is an isomorphism of discrete groups

$$\mathrm{SL}_2(\mathbb{Z}[1/p]) = \mathrm{SL}_2(\mathbb{Z}) *_{\Gamma_0(p)} D_p \mathrm{SL}_2(\mathbb{Z}) D_p^{-1}. \tag{C.11}$$

The edge subgroup has index $p + 1$ in both vertex groups. This is the arithmetic application of the tree-amalgam theorem of Serre (Serre, 1969, §1.2); we prove the required arithmetic reduction here. It is not an assertion that (C.1) becomes an isomorphism after localizing the base.

Proof. Use the tree whose vertices are homothety classes of $\mathbb{Z}_p$-lattices in $\mathbb{Q}_p^2$, with adjacent lattices of index p. The elementary-divisor theorem shows that this graph is connected and that there is a unique nonbacktracking path between any two vertices: write their relative elementary divisors and change the difference one step at a time. It is therefore a tree. The determinant-one group preserves the parity of that difference and has no edge inversions.

The stabilizer in $G = \mathrm{SL}_2(\mathbb{Z}[1/p])$ of the standard lattice class is $H_0 = \mathrm{SL}_2(\mathbb{Z})$: determinant one forces a stabilizing homothety to have valuation zero, and $\mathbb{Z}[1/p] \cap \mathbb{Z}_p = \mathbb{Z}$. The adjacent vertex $D_p\mathbb{Z}_p^2$ has stabilizer $H_1 = D_p H_0 D_p^{-1}$. Their intersection is exactly $\Gamma_0(p)$, by multiplying out the matrices.

The neighbors of the standard vertex are the $p + 1$ lines in $\mathbb{F}_p^2$. Reduction of H_0 is onto $\mathrm{SL}_2(\mathbb{F}_p)$, since the latter is generated by elementary matrices and these lift integrally. This group acts transitively on those lines. The conjugate statement holds at the other vertex. By following paths in the tree, the group generated by H_0, H_1 is transitive on edges. Given $g \in G$, some such product has the same effect on the base edge; their quotient fixes the edge and lies in $H_0 \cap H_1$. Thus these two subgroups generate G and a single edge is a fundamental domain. Finally a reduced alternating word in the two stabilizers moves the base edge along a nonbacktracking path, so cannot be the identity. This proves the amalgam and the index assertion. It is the direct proof of the comparison required to apply Serre's theorem, rather than a citation without reduction. □

C.10. The correction over the 2-adic and 3-adic bases [Proved]

Over $\operatorname{Spec}\mathbb{Z}_2$ and $\operatorname{Spec}\mathbb{Z}_3$, the corrected special-linear presentation is exactly C.5, the projective presentation is exactly C.6, and the general-linear presentation is (C.8) together with C.5. The literal kernels remain noncentral. No correction consisting only of replacing a non-etale μ_4 or μ_6 can address the manuscript's question: those are not its factors.

Proof. All defining identities and their local-normal-form proofs are over $\mathbb{Z}$, and so remain valid under these base changes. C.4 applies after passage to the generic field, and also gives independent witnesses in each characteristic. The constant cyclic schemes in scan 336 are etale in every characteristic; a cyclic map is not necessarily an embedding. For example $C_4 \to \mathrm{SL}_{2,\mathbb{F}_2}$ sending a generator to s has kernel C_2, since $s^2 = I$ there. The same central collapse occurs for the C_6 map. The actual infinitesimal projective correction at two is (C.7), exhibited over the four-element ring in C.6. At three, μ_2 is etale and that extra infinitesimal phenomenon is absent; the noncentral Gaussian relations are still necessary. □

C.11. The remaining evaluation problem in Part C(d) [Open]

The sheaf kernels and corrected presentations have been determined in C.3–C.6 and C.10. We do not provide a finite or otherwise simplified group presentation of the Frobenius fixed subgroup in (C.10) for $A_{\mathrm{SL}}(\mathbb{F}_3)$, nor a corresponding complete descent computation of $A_{\mathrm{SL}}(\mathbb{Z}/4)$, $A_{\mathrm{P}}(\mathbb{Z}/4)$ and the two values over $\mathbb{Z}/9$. Thus Part C(d) is partly settled: the raw point presentations (C.9) have been computed, while the complete descent calculations for these actual sheaf values remain.

Precise partial result, computation, and obstruction. The strongest explicit description obtained for the smallest unresolved finite-field value is

$$A_{\mathrm{SL}}(\mathbb{F}_3) = \left(\mathrm{SL}_2(\mathbb{F}_3) *_{C_3} (\overline{\mathbb{F}}_3, +)\right)^{u \mapsto u^3}.$$

C.8 proves this formula and constructs an infinite cyclic subgroup of its matrix kernel. C.7 computes and completely certifies the smaller order-twenty-four point presentation; the Artin–Schreier calculation proves why this computation does not settle the sheaf value. C.8 does settle both characteristic-two field values requested in the prompt.

The missing step is an explicit presentation of the fixed subgroup displayed above, and the analogous descent of normal forms over the two nonreduced finite rings. Equivalently, one must classify the compatible changes of reduced amalgam syllables under finite etale descent, including the nontrivial C_3-torsor and the two unramified ring examples exhibited in C.8; imposing only relations among global additive parameters is insufficient. The first missing case in increasing finite-field size is the indicated special-linear value over $\mathbb{F}_3$. No claim of undecidability or of a previously established open problem is intended: this is an exact boundary of the present calculation. □

The theorem answering the substantive sheaf question is therefore C.3–C.6: the generators do generate in the stated topology, their one displayed relation does not present the group over any nonempty base, their full kernel is the explicit noncentral normal relator sheaf, and the Gaussian relations supply a presentation valid integrally. The remaining point evaluation question is kept separately in C.11.

D Points, comparison maps, and central extensions

We use the final signed modular scheme of Part B, not an earlier twisted-conjugacy parameter space. Thus

$$\rho = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, \qquad \sigma = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad A_R = R[r]/(r^2 - r + 1), \quad B_R = R[i]/(i^2 + 1).$$

The norm forms are $N_A(d + cr) = c^2 + cd + d^2$ and $N_B(u + ti) = u^2 + t^2$. The signs e, e' below are respectively $\varepsilon_2, \varepsilon_3$, and $\chi = \det U$. This dictionary is the one in §48, folios 542–543, scans 479–481, especially (3) and (12bis). It matters that $\mu_2(\widehat{\mathbb{Z}}) = \prod_p \{\pm 1\}$ whereas the constant group $C_2 = \{\pm 1\}$ has only two elements. We distinguish these groups throughout.

D.1 Integral, local, and adelic points

D.1. An explicit group law for the requested points [Proved]

Let $R = \mathbb{Z}, \mathbb{Z}_p$, or $\mathbb{Q}_p$. Define

$$H_R = \{(a, z, e, e') : a \in A_R^\times,\ z \in B_R^\times,\ e, e' \in \{\pm 1\},\ N_A(a) = ee' N_B(z)\}.$$

Put $\kappa_e(z) = z$ for $e = 1$ and $\kappa_e(z) = \bar{z}$ for $e = -1$. Then the group operation is

$$(a, z, e, e')(a_1, z_1, e_1, e_1') = (aa_1, z\kappa_e(z_1), ee_1, e'e_1'). \tag{D.1}$$

There is an isomorphism of groups, and of topological groups in the local cases,

$$M(R) \simeq \mathrm{SL}_2(R) \rtimes H_R, \qquad h(S) = D_{\chi(h)} S D_{\chi(h)}^{-1}, \quad D_\mu = \mathrm{diag}(\mu, 1), \quad \chi(h) = eN_B(z). \tag{D.2}$$

For $\mathbb{Z}$ this specializes to

$$M(\mathbb{Z}) \simeq C_6 \times (\mathrm{SL}_2(\mathbb{Z}) \rtimes D_8), \tag{D.3}$$

where $D_8 = C_4 \rtimes C_2$ has order eight and acts on $\mathrm{SL}_2(\mathbb{Z})$ through its quotient C_2, by conjugation with $\mathrm{diag}(-1, 1)$.

Proof. The final equations give $a = d + c\rho$ and

$$b = \begin{pmatrix} u & t \\ -et & eu \end{pmatrix} = (u + et\sigma)\,\mathrm{diag}(1, e).$$

Thus $z = u + eti$ has norm $u^2 + t^2$. Conjugation by $\mathrm{diag}(1, e)$ acts on B_R as κ_e; this proves (D.1). The remaining equation is $\det U = eN_B(z) = e'N_A(a)$, exactly the equation defining H_R. For fixed h, write $U = SD_{\chi(h)}$, uniquely with $S \in \mathrm{SL}_2(R)$. Multiplication of the matrices proves (D.2); both maps use polynomial operations and inversion of units, so are continuous with continuous inverses.

Over $\mathbb{Z}$, the two positive definite norms are positive units and therefore equal to one. The solutions for a are the six powers of r; those for z are $1, i, -1, -i$. The signs satisfy $e' = e$. Conjugation inverts i, so $H_\mathbb{Z} = C_6 \times (C_4 \rtimes C_2)$, and $\chi = e$. This proves the asserted integral group, including its action. □

D.2. All local determinant and sign restrictions [Proved]

Set $A_R^1 = \ker N_A$ and $B_R^1 = \ker N_B$. For $R = \mathbb{Z}_p, \mathbb{Q}_p$ the description in D.1 has the exact sequence of topological groups

$$1 \longrightarrow A_R^1 \times B_R^1 \longrightarrow H_R \xrightarrow{(\chi, e, e')} D_R \longrightarrow 1, \quad D_R = \{(\mu, e, e') : e\mu \in N_B(B_R^\times),\ e'\mu \in N_A(A_R^\times)\}. \tag{D.4}$$

Here D_R has the multiplication inherited from $R^\times \times C_2^2$. For integral local points the norm images are

$$\begin{array}{c|cc} & N_A(A^\times_{\mathbb{Z}_p}) & N_B(B^\times_{\mathbb{Z}_p}) \\ \hline p=2 & \mathbb{Z}_2^\times & 1+4\mathbb{Z}_2 \\ p=3 & 1+3\mathbb{Z}_3 & \mathbb{Z}_3^\times \\ p\geq 5 & \mathbb{Z}_p^\times & \mathbb{Z}_p^\times. \end{array} \tag{D.5}$$

Consequently μ is any p-adic unit; at 2 one must have $e = \chi_4(\mu)$, at 3 one must have $e' = \left(\frac{\mu \bmod 3}{3}\right)$, and there are no other sign restrictions. Here $\chi_4(1) = 1, \chi_4(3) = -1$. The norm-one kernels themselves have the following noncanonical topological decompositions:

$$\begin{array}{c|cc} & A^1_{\mathbb{Z}_p} & B^1_{\mathbb{Z}_p} \\ \hline p=2,3 & C_6 \times \mathbb{Z}_p & C_4 \times \mathbb{Z}_p \\ p\geq 5 & C_{p-(\frac{-3}{p})} \times \mathbb{Z}_p & C_{p-(\frac{-1}{p})} \times \mathbb{Z}_p. \end{array} \tag{D.5a}$$

For $\mathbb{Q}_p$, write $\mu = p^r v$ with $v \in \mathbb{Z}_p^\times$. The condition on r is

$$r \in 2\mathbb{Z} \quad \text{unless } p \equiv 1 \pmod{12}; \qquad r \in \mathbb{Z} \quad \text{if } p \equiv 1 \pmod{12}. \tag{D.6}$$

The restrictions at $2, 3$ are the same as above with v in place of μ, and at other primes the two signs are arbitrary. Equations (D.1)–(D.6), including the explicitly specified quadratic norm-one groups, compute $M(\mathbb{Z}_p)$ and $M(\mathbb{Q}_p)$ as groups, not just their determinant images.

Proof. The map in (D.4) has the displayed kernel. Its image is exactly the displayed set: choose independently elements with the required two norms. This also proves surjectivity without assuming a section.

A split quadratic algebra has surjective norm on units. In an unramified quadratic extension, the residue-field norm is surjective. Norms on successive principal-unit quotients are the residue trace, which is surjective; lifting successively and using completeness proves surjectivity on all units. For A the unramified or split cases are $p \neq 3$, and for B they are $p \neq 2$. At 3, the nonzero values of $c^2 + cd + d^2$ reduce to 1. Conversely every unit $w \equiv 1$ modulo 3 is $1 + d + d^2$: Hensel's lemma applies at $d = 0$ to $d^2 + d + 1 - w$, whose derivative is 1 modulo 3. At 2, a unit sum of two squares is 1 modulo 4. If $w \equiv 1 \pmod 8$, it is a square in $\mathbb{Z}_2^\times$; if $w \equiv 5 \pmod 8$, $w - 4$ is such a square and $w = (\sqrt{w-4})^2 + 2^2$. This proves (D.5).

In an unramified quadratic field the valuation of a norm is even; all even valuations occur. In the two ramified fields the uniformizers $1 + i$ and $\sqrt{-3}$ have norms 2 and 3. Thus their norm images have arbitrary valuations and the unit restrictions just proved. For $p \geq 5$, the two algebras split respectively when $p \equiv 1 \pmod 3$ and $p \equiv 1 \pmod 4$. Both split precisely when $p \equiv 1 \pmod{12}$. At 2, A is unramified; at 3, B is unramified. This proves (D.6) and all the sign assertions. For (D.5a), logarithm on a sufficiently deep principal-unit subgroup identifies the norm-one subgroup with the kernel of the trace, a free $\mathbb{Z}_p$-module of rank one. The compact abelian group therefore has an open copy of $\mathbb{Z}_p$. The structure theorem for finitely generated abelian pro-p groups splits its free rank-one part from its torsion. The torsion consists of the roots of unity of norm one. At the split odd primes there are $p - 1$; in an unramified quadratic field there are $p + 1$. At 2, the unramified field $\mathbb{Q}_2(\sqrt{-3})$ has norm-one roots μ_6, and the Gaussian field has μ_4. At 3, the ramified Eisenstein field has μ_6 and the unramified Gaussian field has norm-one roots μ_4. This proves the table. Over $\mathbb{Q}_p$, a field norm-one element is a unit, so these kernels do not change; in the split case the kernel is instead $\mathbb{Q}_p^\times = \mathbb{Z} \times \mathbb{Z}_p^\times$. The exhaustive norm computations modulo $4, 8, 16, 9, 27, 25, 49$ in `code/D_gt.py`, recorded in `D_gt.out`, give independent finite checks, including both ramified primes. □

D.3. Integral and finite adelic points [Proved]

For the affine finite-type group scheme $M/\mathbb{Z}$,

$$M(\widehat{\mathbb{Z}}) = \prod_p M(\mathbb{Z}_p), \qquad M(\mathbb{A}_f) = \prod_p{}' M(\mathbb{Q}_p), \tag{D.7}$$

as topological groups; the restricted product is relative to the subgroups $M(\mathbb{Z}_p)$. The group laws and local factors are those of D.1–D.2. This answers the adelic question of §49, folio 576, scan 546, in the final modular normalization.

Proof. An R-point is a tuple of finitely many coordinates satisfying finitely many polynomial identities, with the specified inverse of the norm. A point over a product of rings is therefore exactly a product of points. The finite adeles are the union, over finite sets S of primes, of $\prod_{p\in S}\mathbb{Q}_p \times \prod_{p\notin S}\mathbb{Z}_p$. All coordinates of a point, including its inverse norm, belong to one such ring, because there are finitely many of them. Conversely a restricted product of local points defines these coordinates in $\mathbb{A}_f$. These inverse maps respect the group law and the defining product and restricted-product topologies. □

D.2 What the congruence comparison proves, and what it does not

We write $G_{\mathbb{Q}}$, $\widehat{\mathrm{GT}}$, and $\widehat{\mathrm{GT}}_0$ separately. The last group means the continuous automorphisms described by pairs $(\lambda, f) \in \widehat{\mathbb{Z}}^\times \times [\widehat{F}_2, \widehat{F}_2]$ satisfying Drinfeld's relations (I),(II); $\widehat{\mathrm{GT}}$ also satisfies the pentagon. We do not identify either with $G_{\mathbb{Q}}$. The expressions in the manuscript initially yield maps modulo central signs (§48, folio 543, scan 481, (24)–(25)). A representative in an unquotiented matrix group is extra data.

D.4. The typed congruence map and an obstruction to openness [Proved]

Let $\mathscr{S}_K$ be a group of genuinely lifted, normalized loop solutions attached to $K = G_{\mathbb{Q}}, \widehat{\mathrm{GT}}$, or $\widehat{\mathrm{GT}}_0$, with its chosen comparison to the profinite modular group and the triple coordinates of scan 479. Suppose its geometric words have a genuine continuous matrix realization extending the congruence homomorphism

$$q : \widehat{\mathrm{SL}_2(\mathbb{Z})} \longrightarrow \mathrm{SL}_2(\widehat{\mathbb{Z}}) \tag{D.8}$$

and its multiplier is evaluated by the specified character to $\widehat{\mathbb{Z}}^\times$. Evaluating the three coordinates then defines a continuous homomorphism $\Phi_K : \mathscr{S}_K \to M(\widehat{\mathbb{Z}})$. This conditional construction is not an assertion that those lift and realization data have been constructed on each bare K. If the unlifted comparison is only $\bar{\Phi}_K : K \to M(\widehat{\mathbb{Z}})/\Sigma$, its canonical replacement is the pullback group

$$K^\sharp = K \times_{M(\widehat{\mathbb{Z}})/\Sigma} M(\widehat{\mathbb{Z}}), \tag{D.9}$$

not a homomorphism from K to $M(\widehat{\mathbb{Z}})$. A homomorphism of the latter kind exists precisely when the pullback extension has a continuous section.

For every lift that preserves the two original global signs, the image lies in

$$M(\widehat{\mathbb{Z}})_{\mathrm{diag}} = \{m : (e_p(m), e'_p(m)) \text{ is independent of } p\}. \tag{D.10}$$

This is a closed subgroup of infinite index with empty interior. Thus the proposed openness in the full signed $M(\widehat{\mathbb{Z}})$ is false whenever that comparison is defined with the manuscript's global signs. The same obstruction applies to $\widehat{\mathrm{GT}}$ and $\widehat{\mathrm{GT}}_0$. It also applies after quotienting by the manuscript's finite central scalar-sign group Σ, since the two characters are trivial on Σ. It does not decide openness in an appropriately restricted target.

Proof. The congruence homomorphism is induced by the compatible reduction maps on the dense discrete group. All identities defining a lifted normalized triple are word and determinant identities, hence are preserved by reduction. The matrix triple then satisfies the equations of M. Products of triples reduce to products, and reduction is continuous. This proves the asserted map on the specified source, without asserting that the required lift choices exist for every unlifted source.

The fibre product (D.9) has its usual group law and projection; a section is exactly a compatible selection of a representative for each element of K. This proves the lifting assertion. The signs of a genuine profinite solution are elements of the single constant group C_2. Applying them in every congruence coordinate embeds them diagonally. This proves (D.10). For every $p \geq 5$, D.2 shows that $M(\mathbb{Z}_p) \to C_2^2$ is surjective. An open subgroup of a product contains the full factors outside a finite set, whereas (D.10) permits no independent change of a sign at one of those primes. It has empty interior; its quotients obtained by recording signs at arbitrarily many primes have unbounded order, so its index is infinite.

For an explicit point added by the scheme, let $s \in \mathbb{Z}_5$ be the square root of -1 congruent to 2 modulo 5. Take at 5

$$a = sI, \qquad b = I, \qquad U = I, \qquad e = 1,\ e' = -1,$$

and take the identity at every other prime. Its norm equation is $1 = (-1)(-1)$, so D.3 gives a point of $M(\widehat{\mathbb{Z}})$. Its second sign is not diagonal, so it comes from no global-sign-preserving solution. The existence of s follows from Hensel's lemma applied to $s^2 + 1$ at 2 modulo 5. Multiplying a, b, U by scalar signs changes neither their determinants nor the conjugation sign e, so both sign characters descend modulo Σ. The nonopenness and the extra point consequently persist in that quotient. □

D.5. Executed finite tests of relations (I) and (II) [Proved]

In $\mathrm{SL}_2(\mathbb{Z}/n)$ put

$$x = \begin{pmatrix} -1 & 2 \\ 0 & -1 \end{pmatrix}, \quad y = \begin{pmatrix} -1 & 0 \\ -2 & -1 \end{pmatrix}, \quad z = (xy)^{-1}, \qquad K_n = \langle x, y \rangle.$$

We use the convention

$$f(y,x)f(x,y) = 1, \qquad f(z,x)z^m f(y,z)y^m f(x,y)x^m = 1, \qquad m = (\lambda - 1)/2. \tag{D.11}$$

These are the displayed relations (I),(II) in Schneps (2012, p. 19). Let L_n be generated by (x, y, z, y) and (y, x, x, z) in K_n^4. Allow precisely the tuples in $[L_n, L_n]$ as simultaneous evaluations $(f(x,y), f(y,x), f(z,x), f(y,z))$, and require that $x \mapsto x^\lambda,\ y \mapsto f(x,y)^{-1} y^\lambda f(x,y)$ descend to an automorphism of K_n. The exact enumeration is

n	$\lvert K_n \rvert$	$\lvert L_n \rvert$	$\lvert [L_n, L_n] \rvert$	λ modulus	λ residues	solutions per residue
2	1	1	1	2	1	1
3	24	24	8	12	1, 5, 7, 11	3
4	4	4	1	4	1, 3	1
6	24	24	8	12	1, 5, 7, 11	3
12	96	96	8	12	1, 5, 7, 11	3.

(D.12)

Every enumerated solution passes the stated automorphism test. These are finite equation solutions, not a determination of the images of any of the three infinite groups.

Proof. The simultaneous substitution homomorphism $F_2 \to K_n^4$ has image L_n, and maps F_2' onto $[L_n, L_n]$. Thus using this derived subgroup enforces that all four evaluations arise from the same commutator word, rather than choosing four unrelated matrices. In a group generated by two elements the derived subgroup is generated by the conjugates of their commutator. The program first forms L_n, then this normal closure, and checks the two matrix products in (D.11) for every resulting tuple. Powers depend only on λ modulo twice the least common multiple of the orders of x, y, z. Finally a breadth-first traversal of the Cayley graph assigns the proposed image to each vertex. Conflicting assignments reject a pair; a well-defined permutation of the vertices proves that the endomorphism is an automorphism. These finite procedures are exhaustive, and `code/D_gt.py` was executed: `D_gt.out` records every count and every surviving $f(x, y)$, not merely the totals.

For example at $n = 3, \lambda = 1$, the three surviving evaluations are I, $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, and $\left(\begin{smallmatrix} 1 & 2 \\ 2 & 2 \end{smallmatrix}\right)$. Compatibility with all finite quotients, and the pentagon in the case of $\widehat{\mathrm{GT}}$, have not been imposed. Passing those tests is not a consequence of this enumeration. □

D.6. The congruence kernel is already visible through a simple quotient [Proved]

The kernel of (D.8) is nontrivial. More precisely, $\widehat{\mathrm{SL}_2(\mathbb{Z})}$ has a quotient $\mathrm{SL}_2(\mathbb{F}_8)$ which cannot factor through any $\mathrm{SL}_2(\mathbb{Z}/n)$. This conclusion concerns elements of the modular group; it does not construct an element of $\widehat{\mathrm{GT}}_0$, or a solution of all profinite loop equations, in that kernel.

Proof. Use $\mathbb{F}_8 = \mathbb{F}_2[t]/(t^3 + t + 1)$. The matrices

$$u = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad v = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$

have orders $2, 3$. The execution in `D_gt.out` verifies that they generate all 504 elements and that the conjugacy-class sizes are

$$1,\ 56,\ 56,\ 56,\ 56,\ 63,\ 72,\ 72,\ 72.$$

No proper nonempty union of the nonidentity classes, together with the identity, has order dividing 504; the program checks every such union. Thus the group is simple. The modular presentation $\mathrm{PSL}_2(\mathbb{Z}) = C_2 * C_3$ gives the required surjection.

By the Chinese remainder theorem, any congruence quotient is a product of $\mathrm{SL}_2(\mathbb{Z}/p^a)$. A surjection from that product to the nonabelian simple group $S = \mathrm{SL}_2(\mathbb{F}_8)$ must be surjective on one factor: images of factors are commuting normal subgroups, so at most one is nontrivial. The kernel of reduction modulo p in that factor is a p-group, by the filtration $1 + p^j X \mapsto X \bmod p$. Its normal image in S is trivial. The surjection would therefore factor through $\mathrm{SL}_2(\mathbb{F}_p)$. This group is generated by its upper and lower unipotent matrices, each of order p. If $p \notin \{2, 3, 7\}$, all their images in a group of order $504 = 2^3 3^2 7$ are trivial. For $p = 2, 3, 7$, the source orders are $6, 24, 336$, all less than 504. No such surjection exists.

If (D.8) were injective, compactness would make it a topological isomorphism onto $\mathrm{SL}_2(\widehat{\mathbb{Z}})$. Every finite continuous quotient of the latter factors through some $\mathrm{SL}_2(\mathbb{Z}/n)$, contradicting the exhibited quotient. □

D.7. The exact remaining comparison problem [Open]

The following parts of D(b) remain undecided here. For each of $G_{\mathbb{Q}}$, $\widehat{\mathrm{GT}}$, and $\widehat{\mathrm{GT}}_0$, specify the actual normalized comparison (with its central lift, or in its correct quotient), and determine the kernel and image. In particular we have not proved that the $\widehat{\mathrm{GT}}_0$-image strictly contains

the $G_{\mathbb{Q}}$-image. Nor have we produced a nonidentity solution of all profinite loop equations annihilated by every congruence comparison.

There are concrete starting instances. The first nontrivial finite lifting question in (D.12) is whether the tuple at $n = 3, \lambda = 1, f(x, y) = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ extends to a pair in $\widehat{\mathrm{GT}}_0$ with $\lambda = 1$, and then whether it extends to $\widehat{\mathrm{GT}}$ or comes from $G_{\mathbb{Q}}$. For the requested invisible solution the first specific noncongruence test supplied by our attack is $\mathrm{SL}_2(\mathbb{F}_8)$: find a loop solution whose congruence coordinates are all trivial but whose coordinate in this quotient is not, or prove none exists.

The partial results are the obstruction and extra point of D.4, the exhaustive five-modulus computation D.5, and the actual noncongruence quotient D.6. The unproved step is extending finite equation solutions compatibly through all finite quotients. Nontriviality of the kernel in D.6 alone supplies no such extension.

D.3 The finite extension and the geometric group-scheme calculation

In this subsection $V = C_2^2$, with additive coordinates $(u, v) \in \mathbb{F}_2^2$ corresponding to $(\varepsilon_3, \varepsilon_2) = ((-1)^u, (-1)^v)$. The finite group in §49, folios 602–608, scans 597–610, is denoted E. This is an ordinary finite group of order eight. Over a base on which 2 is invertible we may identify its constant kernel C_2 with μ_2.

D.8. Classification of the eight finite extension classes [Proved]

Over an algebraically closed field k of characteristic different from 2, the central extensions of the constant group scheme V by μ_2, with both end identifications fixed, form

$$H^2_{\mathrm{fppf}}(BV, \mu_2) = H^2(V, C_2) = \mathbb{F}_2\{u^2, uv, v^2\}. \tag{D.13}$$

They are uniquely determined by

$$X^2 = z^a, \qquad Y^2 = z^b, \qquad [X, Y] = z^c, \qquad a, b, c \in \mathbb{F}_2, \quad z^2 = 1, \tag{D.14}$$

where X, Y lift the ordered basis of V. The underlying groups are C_2^3 once, $C_4 \times C_2$ three times, D_8 three times, and the quaternion group Q_8 once.

The manuscript's square rule in scan 602, (90), determines

$$E = \mathbb{F}_2^3, \qquad (s, u, v)(t, x, y) = (s + t + u(x + y), u + x, v + y). \tag{D.15}$$

Thus its class is $u(u + v)$, its group is D_8, and it is nonsplit. The canonical complements over $u = 0$ and $u = v$ are not normal in E. Their intersection is trivial.

Proof. The kernel is central, so the squares of lifts and their commutator belong to $\{1, z\}$. Changing either lift by z does not change these three quantities. Conversely, putting every element in the unique normal form $z^s X^u Y^v$ gives a multiplication with central exponent $s + t + aux + bvy + cvx$; it is associative because the added function is bilinear. It gives an extension for every triple, and any extension with those three quantities has that normal form and multiplication. Hence all eight triples occur and are inequivalent as extensions. Addition of cocycles adds the triples, proving the three-dimensional cohomology. Since the finite group schemes in question are étale and k is algebraically closed, they are constant; the same argument classifies their fppf central extensions. The description by cohomology of the classifying stack is the usual normalized descent description: a torsor on V with an associative multiplication is a central extension.

If $c = 0$, (D.14) is abelian; it is C_2^3 when $a = b = 0$ and $C_4 \times C_2$ otherwise. If $c = 1$, every noncentral element has order four exactly when $a = b = 1$, giving Q_8. In the other three cases there is a noncentral involution and a cyclic subgroup of order four which it inverts, giving D_8.

In (D.15), the square of (s,u,v) is $(u(u+v),0,0)$. This is precisely the manuscript's condition that order divide two when $\varepsilon_3 = 1$ or $\varepsilon_2\varepsilon_3 = 1$. The lift over $(u,v) = (1,0)$ has order four, so no section of $E \to V$ exists. The two canonical subgroups are $\{(0,0,v)\}$ and $\{(0,u,u)\}$. The commutator of lifts of $(u,v),(x,y)$ is $(uy+xv,0,0)$. Conjugating their nonidentity elements by a suitable lift therefore multiplies them by z, which is outside the chosen complement. Both are nonnormal.

The complete enumeration in `code/D_finite.py` checks all 2^9 normalized two-cochains: 16 are cocycles, 2 are coboundaries, and the eight signatures in (D.14) all occur. It also checks the order distribution $1,2,2,2,2,2,4,4$ and the two failures of normality. □

D.9. The full geometric central-extension group of M [Proved]

For the final signed scheme over $\mathbb{C}$, central extensions by μ_2 form

$$\mathrm{Ext}_{\mathrm{cen}}(M_{\mathbb{C}},\mu_2) = H^2_{\mathrm{fppf}}(BM_{\mathbb{C}},\mu_2) \simeq \mathbb{F}_2^5. \tag{D.16}$$

The same answer holds for $M(0)_{\mathbb{C}}$ and $M[0]_{\mathbb{C}}$; the latter is the $Z^!$ to which scan 610 pulls back the finite extension. Three generators are the inflations of (D.13). After choosing the two embeddings of $A_{\mathbb{C}}$, a fourth is the pullback of the squaring map on $\mathbb{G}_m$ along the first eigenvalue $a_1 : M_{\mathbb{C}} \to \mathbb{G}_m$; a fifth is its pullback along $\chi = \det U$. In particular inflation of the manuscript's class $u(u+v)$ is nonzero. Its two inflated complements are nonnormal; their intersection, the identity-sign subgroup with the chosen trivial lift, is normal.

Proof. Choose eigenvalue coordinates (a_1,a_2) on $A_{\mathbb{C}}^{\times}$ and (z_1,z_2) on $B_{\mathbb{C}}^{\times}$. In the identity component the equations are

$$a_1a_2 = z_1z_2 = \det U = \delta.$$

Thus

$$M^0_{\mathbb{C}} \simeq \mathbb{G}_m^2 \times \mathrm{GL}_2, \qquad (a_1,z_1,U) \longmapsto ((a_1,\delta/a_1),(z_1,\delta/z_1),U). \tag{D.17}$$

This is an isomorphism of algebraic groups. Put $a_* = (2\rho - I)/\sqrt{-3}$. It has square I and determinant -1. The elements

$$g_{e'} = (a_*,I,I), \qquad g_e = (a_*,\mathrm{diag}(1,-1),\mathrm{diag}(-1,1)) \tag{D.18}$$

are commuting involutions with the indicated respective negative signs. They give a section $V \to M_{\mathbb{C}}$. In particular inflation from V is injective.

The coordinate ring in (D.17) is a localization of a polynomial ring, hence is factorial and has trivial Picard group. Its units modulo squares are generated by a_1, z_1, δ; constant units are squares in $\mathbb{C}$. The Kummer sequence therefore gives

$$H^1_{\mathrm{fppf}}(M^0_{\mathbb{C}},\mu_2) = \mathbb{F}_2\{a_1,z_1,\delta\}. \tag{D.19}$$

Every such torsor is the pullback of squaring along a character, so has a central group-extension structure. That structure, with its identity fixed, is unique: the difference of two multiplications would be a morphism from the connected scheme $M^0_{\mathbb{C}} \times M^0_{\mathbb{C}}$ to μ_2, hence the constant identity. Equivalently the same argument may be made after trivializing the torsors fppf-locally and then descending. Thus (D.19) also computes central extensions of $M^0_{\mathbb{C}}$.

Conjugation by $g_{e'}$ is trivial. Conjugation by g_e fixes a_1,δ, sends z_1 to δ/z_1, and acts on U by an inner automorphism. On the exponent vector (α,β,γ) modulo 2 it acts by

$$(\alpha,\beta,\gamma) \longmapsto (\alpha,\beta,\beta+\gamma). \tag{D.20}$$

The invariant subspace has dimension two, with basis a_1,δ. Any extension of $M_{\mathbb{C}}$ restricts to an invariant extension of $M^0_{\mathbb{C}}$; both invariant generators extend globally by the two character constructions in the statement.

If an extension restricts trivially to $M^0_{\mathbb{C}}$, its splitting there is unique, since $\mathrm{Hom}(M^0_{\mathbb{C}}, \mu_2) = 0$. Conjugation consequently preserves that splitting. Quotienting by its image gives an extension of V, and the original extension is its pullback. The kernel of restriction is therefore exactly the three-dimensional inflated subspace. Surjectivity onto the two-dimensional invariant subspace proves (D.16). Over $\mathbb{C}$ the base has vanishing positive-degree cohomology with these finite coefficients, so the central-extension description is all of $H^2(BM_{\mathbb{C}}, \mu_2)$.

For $M(0)$, the identity component has coordinate ring $\mathbb{C}[a_1^{\pm 1}, z_1^{\pm 1}, \delta^{\pm 1}]$. For $M[0]$, adjoining the upper-right entry of $U = \left(\begin{smallmatrix} \delta & b \\ 0 & 1 \end{smallmatrix}\right)$ adds one polynomial variable. Both rings are factorial, with the same three unit classes. The section (D.18) belongs to $M(0)$, and the action on those unit classes is again (D.20). Every step of the restriction and inflation argument therefore applies to both subgroups.

The section (D.18) detects the nonsplitting and nonnormality proved in D.8. The intersection of the complements is the kernel of the sign map with its unique prescribed lift, and the cocycle (D.15) is zero whenever one argument has both signs zero. This intersection is therefore normal. `D_components.py` checks (D.18) modulo 7, where $\sqrt{-3} = 2$, and enumerates all eight characters in (D.20), finding the four invariant ones. □

D.10. A genuine obstruction to an integral root-of-unity model [Proved]

There is no central extension of finite flat group schemes

$$1 \longrightarrow \mu_2 \longrightarrow E_2 \longrightarrow \mu_2^2 \longrightarrow 1 \tag{D.21}$$

over $\mathbb{Z}_2$, or over $\mathbb{Z}$, whose generic fibre is the dihedral extension (D.15). In fact every central extension of the displayed kind is commutative. This does not prevent the constant group D_8 from being defined over $\mathbb{Z}$; its integral kernel and quotient are constant groups, not the indicated root-of-unity group schemes.

Proof. The commutator of a central extension of a commutative group is an alternating bicharacter $\mu_2^2 \times \mu_2^2 \to \mu_2$. By diagonalizable-group duality,

$$\underline{\mathrm{Hom}}(\mu_2^2, \mu_2) = (C_2^2)_{\mathrm{const}}. \tag{D.22}$$

For completeness, a homomorphism to μ_2 is a group-like element of the Hopf algebra of μ_2^2. The coefficient identities for being group-like say that its coefficients are pairwise orthogonal idempotents summing to one; on each connected piece precisely one monomial is selected. This is the constant character group in (D.22). The bicharacter is consequently a morphism $\mu_2^2 \to (C_2^2)_{\mathrm{const}}$.

The source is connected over $\mathbb{Z}_2$: its finite coordinate algebra has reduction $\mathbb{F}_2[s,t]/(s^2, t^2)$, a local ring, and hence has no nontrivial idempotent. It is also connected over $\mathbb{Z}$. Indeed an idempotent would remain nontrivial over $\mathbb{Z}_2$ unless it or its complement became zero there; injectivity of the base change of the free coordinate module then forces the same vanishing over $\mathbb{Z}$. Thus the morphism to the constant group is constant, and its value at the identity is the identity. The commutator is trivial. A dihedral generic fibre is noncommutative, giving the required contradiction. □

D.11. Arithmetic descent and the remaining integral H^2 [Open]

D.8 classifies the relevant finite geometric extensions; D.9 computes the complete central-extension group over $\mathbb{C}$; D.10 excludes the literal root-of-unity extension at 2. We have not computed the central-extension groups of $M_{\mathbb{Z}_2}$ or $M[0]_{\mathbb{Z}_2}$, nor its restriction map to the generic

fibre, nor the full arithmetic descent of all five geometric classes to $\mathbb{Q}$. One exact first remaining instance is the kernel and image of

$$\mathrm{Ext}_{\mathrm{cen}}(M[0]_{\mathbb{Z}_2}, \mu_2) \longrightarrow \mathrm{Ext}_{\mathrm{cen}}(M[0]_{\mathbb{Q}_2}, \mu_2). \tag{D.23}$$

The computations of all finite cocycles and all invariant identity-component characters are attached in `D_finite.out` and `D_components.out`. Their limitation is precise: they classify étale geometric extensions and characters; they do not classify flat μ_2-torsors on the nonreduced special fibre of M. That descent problem is not resolved by declaring the constant dihedral group to be a root-of-unity group scheme.

D.4 Two distinct Brauer obstructions

D.12. The cup product on scan 611 is zero [Refuted]

The finite extension (D.15), pulled back along the cyclotomic characters

$$u = \chi_{-3}, \qquad v = \chi_{-1} \quad \text{in } H^1(G_{\mathbb{Q}}, \mu_2),$$

has class

$$c = u \smile (u + v) = \chi_{-3} \smile \chi_3 = (-3, 3) = 0 \quad \text{in } \mathrm{Br}(\mathbb{Q}). \tag{D.24}$$

It has order one, and remains zero over $\mathbb{Q}(i)$. Thus the claimed nonvanishing in §49, folio 609, scan 611, is false for the cup product actually written there. The two quadratic characters belong to H^1, not the displayed H^2. An explicit full dihedral lift is the restriction map

$$G_{\mathbb{Q}} \twoheadrightarrow \mathrm{Gal}(\mathbb{Q}(\sqrt[4]{3}, i)/\mathbb{Q}) \simeq D_8; \tag{D.25}$$

its only ramified rational primes are 2 and 3.

Proof. The normalized cocycle in (D.15) is $(u, v), (x, y) \mapsto u(x + y)$, so its pullback is exactly the cup product in (D.24). The Kummer classes of the two quadratic fields named on scan 611 are -3 and 3; the second is the product of the classes -3 and -1. Under the Kummer-to-Brauer map their cup product is the quaternion symbol. Explicitly, let

$$I = \begin{pmatrix} 0 & -3 \\ 1 & 0 \end{pmatrix}, \qquad J = \begin{pmatrix} 0 & 3 \\ 1 & 0 \end{pmatrix}.$$

Then $I^2 = -3$, $J^2 = 3$, and $IJ = -JI$. The matrices $1, I, J, IJ$ have determinant -36 as four vectors in $M_2(\mathbb{Q})$, so these assignments give an algebra isomorphism $(-3, 3) \simeq M_2(\mathbb{Q})$. This is also the norm identity $N_{\mathbb{Q}(\sqrt{-3})/\mathbb{Q}}(\sqrt{-3}) = 3$. The splitting is checked exactly by `D_finite.py`, and the Hilbert symbols are checked by `D_number_fields.py`.

Put $\alpha = \sqrt[4]{3} > 0$. The polynomial $X^4 - 3$ is Eisenstein at 3, so $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 4$; this field is real, and adjoining i doubles its degree. The automorphisms $r(\alpha) = i\alpha, r(i) = i$ and $s(\alpha) = \alpha, s(i) = -i$ satisfy $r^4 = s^2 = 1$, $srs = r^{-1}$, and generate its group. The fixed field of the centre is $\mathbb{Q}(i, \sqrt{3})$. On that biquadratic field, r has signs $(u, v) = (1, 0)$, whereas s has signs $(1, 1)$; these are exactly the order-four and involution fibres specified in (D.15). This identifies (D.25) as a full lift. The polynomial discriminant is $-2^8 3^3$, so the splitting field is unramified outside $2, 3$. Both primes ramify: the quartic is Eisenstein at 3, and the field contains the ramified quadratic extension $\mathbb{Q}(i)$ at 2. □

D.13. The nonzero symbol and the meaning of C_4 on scan 631 [Proved]

The different quaternion algebra

$$(-1,-3) = (-3,-1) \tag{D.26}$$

is ramified exactly at 3 and the real place. Its Brauer class has order two and generates C_2, not C_4. The C_4 in §49, folio 619, scan 631, is the inverse image of a transposition in the finite group of order twelve, not the order of a Brauer class. Pulling $0 \to C_2 \to C_4 \to C_2 \to 0$ back along χ_{-3} gives precisely the nonzero class (D.26). Consequently Lead 4 conflates two different extensions.

Proof. For a normalized lift of C_2 to C_4, the central two-cocycle is $(u,x) \mapsto ux$. Its pullback is $\chi_a^2 = (a,a) = (a,-1)$; the last equality follows from $(a,-a) = 0$, proved by the norm $N(\sqrt{a}) = -a$. Taking $a = -3$ gives (D.26). This differs from (D.24) by the term χ_{-3}^2.

At infinity both entries of (D.26) are negative, so the quaternion algebra is nonsplit. At 3, the odd-prime Hilbert-symbol formula gives $(-1/3) = -1$. At 2, the formula for two odd units is $(a,b)_2 = (-1)^{(a-1)(b-1)/4}$, which is 1 here. At every other finite prime both entries are units and the odd-prime formula gives 1. These computations appear in `D_number_fields.out`. The class is killed by two because it is a cup product with μ_2 coefficients, and is nonzero by its real localization; hence its order is exactly two. The exact local norm interpretation used here is Milne (2020, Chapter III, Theorem 4.4(d)). The scan itself identifies C_4 as a finite inverse-image group and notes that its chosen σ has order four. Nothing there asserts that a Brauer class has order four. □

D.5 The class-field exercise: constant and arithmetic forms

D.14. The constant lift with imaginary quadratic sign [Refuted]

The group in §49, folio 606, scan 605, and folio 618, scan 629, is

$$\widetilde{G}_3 = \langle r, s \mid r^6 = s^4 = 1,\ r^3 = s^2,\ srs^{-1} = r^{-1} \rangle \simeq \mathrm{Dic}_3 \simeq S_3 \times_{C_2} C_4. \tag{D.27}$$

It has order twelve, with a unique element of order two. For any continuous homomorphism $q : G_{\mathbb{Q}} \to S_3$ whose sign is χ_{-3}, there is no homomorphism $G_{\mathbb{Q}} \to \widetilde{G}_3$, with $\widetilde{G}_3$ a *constant* finite group, lifting q. In particular making q surjective does not remove the obstruction in scan 634.

Proof. Every word in the presentation is r^j or $r^j s$ with $0 \le j < 6$; all twelve occur in $C_3 \rtimes C_4$, where the generator of C_4 inverts C_3. The quotient by its central involution is S_3. Pairing this quotient with the abelianization $\mathrm{Dic}_3 \to C_4$ gives the fibre-product description in (D.27): the pair map is injective and both groups have order twelve. Every element over an odd permutation has order four; the only involution is the central kernel element.

This presentation agrees with the indicated modular quotient. The normal subgroup of $\mathrm{SL}_2(\mathbb{Z}/4)$ generated by $-\ell_0, -\ell_1$ is the four-element group

$$\left\{ I,\ \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix},\ \begin{pmatrix} 3 & 0 \\ 2 & 3 \end{pmatrix},\ \begin{pmatrix} 3 & 2 \\ 0 & 3 \end{pmatrix} \right\}. \tag{D.28}$$

The quotient has the presentation (D.27). `D_finite.py` enumerates its twelve cosets, verifies the order distribution $1^1, 2^1, 3^2, 4^6, 6^2$, and checks that all six elements above odd permutations have order four.

Complex conjugation $c_\infty \in G_{\mathbb{Q}}$ has order two, but $\chi_{-3}(c_\infty) = -1$. A lift would send it to an element above an odd permutation, which would have order four. A homomorphism cannot increase the order of this element, giving a contradiction. This proof applies even when q is not surjective. □

D.15. The exact class-field criterion for constant full lifts [Proved]

Let k be a field of characteristic different from two and let $q : G_k \to S_3$ be a continuous homomorphism with nontrivial sign χ_a. It lifts to the constant group Dic_3 if and only if

$$\chi_a : G_k \to C_2 \text{ lifts to } C_4 \quad \Longleftrightarrow \quad (a,-1) = 0 \quad \Longleftrightarrow \quad a \in N_{k(i)/k}(k(i)^\times). \tag{D.29}$$

Here if $i \in k$, the norm condition means the split quadratic algebra and is automatic. If q is surjective, every lift is surjective. When a lift exists its choices form a torsor under $H^1(G_k, C_2)$.

For $k = \mathbb{Q}$, write the square class of a as a nontrivial squarefree integer d. The criterion is

$$d > 0, \qquad \text{every odd prime divisor of } d \text{ is } 1 \pmod 4. \tag{D.30}$$

Equivalently d is a sum of two rational squares. This is a full criterion for the constant finite-group exercise; the sign $d = -3$ in the manuscript fails it.

Proof. By (D.27), a lift is exactly a pair (q, η) with $\eta : G_k \to C_4$ reducing to χ_a. The obstruction cocycle of $C_4 \to C_2$ is $\chi_a \smile \chi_a$, calculated in D.13 as $(a,-1)$. A quaternion symbol $(-1,a)$ is split exactly when a is a norm from $k(i)$: to see this directly, write the quaternion algebra as $k(i) \oplus k(i)j$ with $j^2 = a$ and $jz = \bar{z}j$. If $a = N(b)$, replacing j by $b^{-1}j$ makes its square one, and its action together with multiplication by $k(i)$ identifies the algebra with $\mathrm{End}_k(k(i))$. Conversely a nonzero zero divisor $z + wj$, which exists in a split algebra, has reduced norm $N(z) - aN(w) = 0$; necessarily $w \neq 0$, giving $a = N(z/w)$. This proves (D.29). The ratio of two lifts takes values in the central kernel and is a character, proving the torsor assertion. If q is onto, a proper image of a lift would have order six and split the extension of S_3, impossible because of its involutions. Its image is therefore all twelve elements.

Here is the class-field-theoretic reduction over $\mathbb{Q}$. By Milne (2020, Chapter I, Theorem 4.16), $G_{\mathbb{Q}}^{\mathrm{ab}} \simeq \widehat{\mathbb{Z}}^\times$ through the cyclotomic action. Thus the problem is precisely whether the quadratic Dirichlet character χ_d has a continuous quartic character whose square is χ_d. Such a square is trivial on every element of order two in $\prod_p \mathbb{Z}_p^\times$. At an odd ramified prime the local quadratic character is the Legendre character; on -1 it is $(-1)^{(p-1)/2}$. Hence primes $3 \bmod 4$ cannot occur. Complex conjugation gives $d > 0$.

Conversely, for each $p \equiv 1 \pmod 4$ dividing d, choose a generator of $\mathbb{F}_p^\times$ and send it to i; this gives a quartic character whose square is the Legendre character. If $2 \mid d$, the remaining local character is χ_8. On $\mathbb{Z}_2^\times = \{\pm 1\} \times \overline{\langle 5 \rangle}$, the character sending -1 to 1 and 5 to i factors modulo 16 and has square χ_8. Multiplying these finitely many local characters gives the required continuous quartic character. This proves (D.30) by class field theory, rather than merely asserting a lifting theorem. The equivalence with a sum of squares is the norm expression $N_{\mathbb{Q}(i)/\mathbb{Q}}(x + iy) = x^2 + y^2$ in (D.29). □

D.16. Explicit full lifts with complete local ramification [Proved]

There are two useful corrections to the exercise on scan 634.

Constant group over $\mathbb{Q}$. Let L_5 be the splitting field of $X^3 - 12X - 14$. Then

$$K_5 = L_5\,\mathbb{Q}(\zeta_5), \qquad \mathrm{Gal}(K_5/\mathbb{Q}) = \mathrm{Dic}_3, \tag{D.31}$$

has degree twelve and sign field $\mathbb{Q}(\sqrt{5})$. Its ramified primes, with ramification index e, residue degree f, and number g of primes, are

p	e	f	g
2	3	4	1
3	3	4	1
5	4	1	3.

(D.32)

The imaginary sign with its arithmetic determinant action. Retain the order-two arithmetic automorphism τ of Dic_3 which fixes r and sends s to s^{-1}. Then

$$G_{\mathrm{ar}} = \mathrm{Dic}_3 \rtimes \langle \tau \rangle \simeq \mathrm{GL}_2(\mathbb{Z}/4)/V_4 \simeq S_3 \times_{C_2} D_8. \tag{D.33}$$

where V_4 is the subgroup (D.28). Let L be the splitting field of $X^3 - 2$, and set

$$K = L\,\mathbb{Q}(\sqrt[4]{3}, i) = \mathbb{Q}(\sqrt[3]{2}, \sqrt[4]{3}, i). \tag{D.34}$$

Then $\mathrm{Gal}(K/\mathbb{Q}) = G_{\mathrm{ar}}$ has order twenty-four, its determinant character is χ_{-1}, and its S_3-sign is χ_{-3}. Over $\mathbb{Q}(i)$ it is a genuine full Dic_3-extension. The only ramified rational primes are $2, 3$; at both of them $K/\mathbb{Q}$ has $(e, f, g) = (12, 2, 1)$. For $K/\mathbb{Q}(i)$, the corresponding pairs (e, f) are $(6, 2)$ at the prime over 2 and $(12, 1)$ at the prime over 3.

More generally, every surjection $G_{\mathbb{Q}} \to S_3$ with sign χ_{-3} has a full lift to G_{ar} with determinant χ_{-1}, by the same compositum with $\mathbb{Q}(\sqrt[4]{3}, i)$. Over $\mathbb{Q}$ this gives a Dic_3-valued *twisted cocycle*, not the constant-group homomorphism excluded by D.14.

Proof. The cubic $X^3 - 12X - 14$ has no rational root, and its discriminant is $1620 = 18^2 \cdot 5$. Its splitting group is therefore S_3 and its quadratic subfield is $\mathbb{Q}(\sqrt{5})$. It is Eisenstein at 2; after replacing X by $X - 1$ it becomes $X^3 - 3X^2 - 9X - 3$, Eisenstein at 3. The corresponding local integer rings are generated by these uniformizers. Only $2, 3$ can divide the index, since their squares are the only prime squares dividing 1620; the two Eisenstein index tests show that neither does. The field discriminant is 1620. The cyclic quartic field $\mathbb{Q}(\zeta_5)$ has the same quadratic subfield. Its intersection with an S_3 extension is at most quadratic, since every abelian quotient of S_3 factors through its sign. Thus the intersection is exactly $\mathbb{Q}(\sqrt{5})$. The compositum has the fibre-product Galois group in (D.27), proving (D.31).

At $2, 3$ the cubic fields are totally ramified of degree three. In each case $\mathbb{Q}_p(\sqrt{5})$ is the unramified quadratic extension, so the local splitting fields have $(e, f) = (3, 2)$. The order of 2, respectively 3, modulo 5 is four, so adjoining ζ_5 gives the unramified extension of degree four. The composita have $(e, f) = (3, 4)$. At 5 the cubic reduces to $(X - 2)^2(X - 1)$. Its simple root lifts, and its remaining quadratic splitting field is $\mathbb{Q}_5(\sqrt{5})$. It is contained in the totally ramified quartic field $\mathbb{Q}_5(\zeta_5)$, so $(e, f, g) = (4, 1, 3)$. The discriminants show there is no other ramification. The exact irreducibility, index, factorization, Hilbert-symbol, and quartic-character checks were executed in `D_number_fields.py`; its output also reports an independent round-two computation of the field discriminant.

For the second construction, the subgroup (D.28) is normal also in $\mathrm{GL}_2(\mathbb{Z}/4)$. Conjugation by $\mathrm{diag}(-1, 1)$ fixes the class of ρ and inverts that of σ. `D_arithmetic.py` verifies normality in all 96 matrices, the 24 cosets, and these two conjugation identities. Factoring G_{ar} by its normal C_3 gives D_8, and factoring by the appropriate normal subgroup of order four gives S_3. The resulting injection into their fibre product is an isomorphism by orders; both quotients have the same parity map to C_2. This proves (D.33). To specify that parity without ambiguity, in $F = \mathbb{Q}(\alpha, i)$, $\alpha = \sqrt[4]{3}$, write $R(\alpha) = i\alpha$, $R(i) = i$, and let B be complex conjugation. The common character is χ_{-3}; it is negative on both R and B. The determinant character is χ_{-1}, negative on B and positive on R. In the fibre product we can take $s = (\text{a transposition}, R)$ and $\tau = (1, RB)$. Thus τ has trivial common parity, inverts s, and fixes the generator $r = (\text{a three-cycle}, R^2)$.

The splitting field L has group S_3 and unique quadratic field $\mathbb{Q}(\sqrt{-3})$, by the discriminant -108. The dihedral field $F = \mathbb{Q}(\sqrt[4]{3}, i)$ also contains that quadratic field. Since $\gcd([L : \mathbb{Q}], [F : \mathbb{Q}]) = 2$, their intersection is precisely that field. Therefore K has the fibre-product group in (D.33). The same argument works for every S_3 extension with that quadratic sign field, proving the general full-image assertion.

We verify all local degrees. At 2, X^3-2 is Eisenstein and ζ_3 gives an unramified quadratic extension; thus L_2 has $(e,f)=(3,2)$. The polynomial $(X+1)^4-3$ is Eisenstein at 2; therefore $\mathbb{Q}_2(\sqrt[4]{3})$ is totally ramified of degree four. It cannot contain i: otherwise it would contain $\sqrt{-3}=i\sqrt{3}$, which generates the unramified quadratic extension of $\mathbb{Q}_2$. Thus F_2 has degree eight, residue degree two and ramification index four. The local intersection is the common quadratic field, so $[K_2:\mathbb{Q}_2]=24$. The ramification indices $3,4$ force $e\geq 12$, and the residue degree is at least two; hence $e=12, f=2$.

At 3, $(X-1)^3-2$ is Eisenstein. Adjoining ζ_3 adds a ramified quadratic extension, so L_3 has $e=6, f=1$. The field generated by $\sqrt[4]{3}$ is totally ramified of degree four, and i is unramified quadratic; thus F_3 has $e=4, f=2$. Again the intersection is the common quadratic field, so the compositum has degree twenty-four. The same bounds give $e=12, f=2$. Both primes have just one prime above them since these local degrees equal the global degree. All other primes are excluded by the two polynomial discriminants. Dividing the ramification and residue degrees by those of $\mathbb{Q}(i)/\mathbb{Q}$ gives the relative table. Every Eisenstein and quadratic residue calculation used here is recorded in `D_arithmetic.out`.

Finally decompose the full arithmetic representation as $\gamma\mapsto(h(\gamma),\chi_{-1}(\gamma))$ using the semidirect product. Its first coordinate satisfies

$$h(\gamma\gamma')=h(\gamma)\,\tau^{\,v(\gamma)}(h(\gamma')),\qquad (-1)^{v(\gamma)}=\chi_{-1}(\gamma). \tag{D.35}$$

This explains exactly the arithmetic twisting. On $G_{\mathbb{Q}(i)}$ the twist is trivial, and full image in G_{ar} implies full image Dic_3 on this subgroup. □

D.6 The generation remark

D.17. Exact modular indices and the failure of a general finite correction [Proved]

Take $\tau=\mathrm{diag}(-1,1)$. For every $n\geq 1$,

$$\langle\rho,\sigma\rangle=\mathrm{SL}_2(\mathbb{Z}/n),\qquad \langle\rho,\sigma,\tau\rangle=\overset{-1}{\det}(\{1,-1\})\subset\mathrm{GL}_2(\mathbb{Z}/n). \tag{D.36}$$

Consequently the index in $\mathrm{GL}_2(\mathbb{Z}/n)$ is one for $n=1,2$, and $\varphi(n)/2$ for $n>2$. The quotient is explicitly $(\mathbb{Z}/n)^\times/\{\pm1\}$. The program enumerated the generated matrix subgroups for every $n\leq 48$, obtaining the following indices:

n	1	2	3	4	5	6	7	8	9	10	11	12
index	1	1	1	1	2	1	3	2	3	2	5	2
n	13	14	15	16	17	18	19	20	21	22	23	24
index	6	3	4	4	8	3	9	4	6	5	11	4
n	25	26	27	28	29	30	31	32	33	34	35	36
index	10	6	9	6	14	4	15	8	10	8	12	6
n	37	38	39	40	41	42	43	44	45	46	47	48
index	18	9	12	8	20	6	21	10	12	11	23	8.

(D.37)

Topologically,

$$\overline{\langle\rho,\sigma\rangle}=\mathrm{SL}_2(\widehat{\mathbb{Z}}),\qquad \overline{\langle\rho,\sigma,\tau\rangle}=\overset{-1}{\det}(\{\pm1\}_{\mathrm{diag}})\subset\mathrm{GL}_2(\widehat{\mathbb{Z}}). \tag{D.38}$$

The second subgroup has infinite index and is not open. Thus scan 652 (§49, folio 628 verso, lower pagination 651) has no general finite-group correction: in the modular special-linear case the correction is trivial, whereas in the full general-linear case it is infinite. For $n>2$, τ itself is not in the determinant-one group, so a phrase $\langle\rho,\sigma,\tau\rangle\subset G^+$ requires a definition of G^+ containing orientation reversal.

Proof. The matrix $\sigma^{-1}\rho = \left(\begin{smallmatrix} 1 & -1 \\ 0 & 1 \end{smallmatrix}\right)$ is an elementary unipotent; its powers give every upper elementary matrix over $\mathbb{Z}/n$. Conjugating by σ gives every lower elementary matrix. These generate SL_2 over a local ring: in the first column of an invertible matrix one entry is a unit, interchange the rows if necessary, eliminate the lower entry, and reduce to a diagonal matrix $\mathrm{diag}(a, a^{-1})$. The latter is elementary because

$$\begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -a^{-1} & 1 \end{pmatrix} \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & a \\ -a^{-1} & 0 \end{pmatrix};$$

multiply by the inverse of the same expression at $a = 1$. For $\mathbb{Z}/n$, apply this argument to each prime-power factor and use the Chinese remainder theorem to choose elementary parameters supported on one factor. This proves the first equality in (D.36).

The determinant of every word in the three generators is ± 1; conversely multiplying a matrix of determinant -1 by τ gives determinant one. The determinant map from $\mathrm{GL}_2(\mathbb{Z}/n)$ onto $(\mathbb{Z}/n)^\times$ is split by $\mathrm{diag}(a, 1)$. This proves the second equality and index formula. The breadth-first enumeration in `D_finite.py` checked both generated group orders separately, using

$$|\,\mathrm{SL}_2(\mathbb{Z}/n)| = n^3 \prod_{p \mid n} (1 - p^{-2})$$

as the expected order; the output gives all forty-eight rows, not just the indices displayed here.

Finite-level surjectivity implies density in $\mathrm{SL}_2(\widehat{\mathbb{Z}})$, because congruence neighbourhoods are specified by one integer n. Adjoining τ gives exactly the two determinant fibres in (D.38). The quotient $\widehat{\mathbb{Z}}^\times/\{\pm 1\}_{\mathrm{diag}}$ is infinite; the indices (D.37), or primes p of arbitrarily large size, already show this. A closed finite-index subgroup of a profinite group is open, and this subgroup cannot be open since its finite-level indices are unbounded. The same obstruction persists in more general reductive groups with an additional torus on which the chosen generators act trivially. □

D.7 A precise motivic construction and its necessary extra data

For the question in §48, folio 547, scan 489, we use *elliptic motivic representation* to mean an object of the realization category MEM_* of Hain and Matsumoto (2020, Definition 6.1). Its data are a mixed Tate fibre at the specified cusp, its weight filtration M, a second filtration W, Betti monodromy, a filtered rational de Rham bundle with integrable logarithmic connection, and comparison maps. The residue is nilpotent, the connection preserves W and satisfies Griffiths transversality, the ℓ-adic monodromy is $G_{\mathbb{Q}}$-equivariant, and the W-graded pieces are sums of $\mathrm{Sym}^m H(r)$. Thus arithmetic compatibility is part of the data. An integral version additionally specifies compatible $\mathbb{Z}_\ell$-lattices; a rational realization alone does not specify finite torsion.

D.18. Specialization for elliptic torsion and unipotent fundamental groups [Known]

Let $E/\mathbb{Q}$ be an elliptic curve, with an actual rational base point in $E - \{0\}$ or a nonzero rational tangent vector at 0 whenever a fundamental group is used. The universal elliptic realization specializes to continuous representations

$$G_{\mathbb{Q}} \longrightarrow \mathrm{Aut}(E[n](\overline{\mathbb{Q}})) \simeq \mathrm{GL}_2(\mathbb{Z}/n), \qquad G_{\mathbb{Q}} \longrightarrow \mathrm{GL}(T_\ell E) \simeq \mathrm{GL}_2(\mathbb{Z}_\ell), \tag{D.39}$$

with determinant the cyclotomic character. Changing a basis conjugates these homomorphisms. The universal unipotent Lie algebra over $\mathcal{M}_{1,2}$, or its tangential restriction, is a pro-object of this realization category and specializes to the genuine continuous action on $\mathrm{Lie}\,\pi_1^{\mathrm{un}}(E - \{0\}, x)$. The relevant results are Hain and Matsumoto (2020, Example 6.6, Theorem 6.9, Corollary 6.10).

For $X = \mathbb{P}^1_{\mathbb{Q}} - \{0, 1, \infty\}$ with tangential base point $\overrightarrow{01}$, its geometric $\mathbb{Q}_\ell$-unipotent completion agrees with the rational unipotent completion of F_2 tensored with $\mathbb{Q}_\ell$. Writing $\mathfrak{p}_\ell$ for its Lie algebra and $L^\bullet$ for the lower central series,

$$W_{-2m}\mathfrak{p}_\ell = L^m\mathfrak{p}_\ell, \qquad \mathrm{Gr}^W_{2m+1}\,\mathfrak{p}_\ell = 0. \tag{D.40}$$

The Zariski closure of the Galois image is a negatively weighted extension of $\mathbb{G}_m$ by a prounipotent group; the weighted completion maps onto it. On positive graded Lie algebras its outer image is the rationalization of the corresponding Ihara filtration. These assertions are the reductions of Hain and Matsumoto (2003, Theorem A.6, Proposition 8.3, Theorem 8.4) given below, not an identification of a profinite group with its unipotent completion.

Proof. Let $\pi : \mathcal{E} \to \mathcal{M}_{1,1}$ be the universal elliptic curve. Over $\mathbb{Z}[1/n]$, multiplication by n has finite étale kernel of rank n^2. Its geometric fibre is $(\mathbb{Z}/n)^2$, so the finite étale sheaf $\mathcal{E}[n]$ is a representation of the arithmetic étale fundamental group of the stack. Pulling this sheaf back along the object $\operatorname{Spec}\mathbb{Q} \to \mathcal{M}_{1,1}$ defined by E gives exactly $E[n]$. On geometric torsion points this is the usual action obtained by applying Galois automorphisms to their coordinates, proving (D.39). The transition maps in $n = \ell^r$ commute with this pullback. Taking their inverse limit gives $T_\ell E$ and proves continuity.

The Weil pairing is functorial and identifies $\bigwedge^2 E[n]$ with μ_n. Applying $\gamma \in G_{\mathbb{Q}}$ to a symplectic basis multiplies its pairing by $\det(\gamma|E[n])$; applying it to the root of unity multiplies the exponent by the cyclotomic character. These exponents are equal. Passing to the inverse limit proves the determinant statement for $T_\ell E$. At primes dividing n, $\mathcal{E}[n]$ is finite flat, and the preceding *étale* assertion is deliberately restricted to $\mathbb{Z}[1/n]$; no equality of its special-fibre geometric point counts with n^2 is being asserted.

In the cited elliptic results, $H = R^1\pi_*\mathbb{Q}$ is the basic object, and the punctured-elliptic unipotent Lie algebra is the stated pro-object. For each finite lower-central quotient, pull back its compatible realizations along the chosen rational or tangential object. Functoriality of pullback preserves the bracket, filtrations, comparisons, and transition maps. The resulting finite-dimensional Galois actions are compatible, so their inverse limit is the action on the pro-Lie algebra. The rational base point gives an actual fibre functor and hence a genuine action. Without it the construction gives an outer action; an identification with a different fixed fibre requires a path and introduces precisely conjugation by that path's geometric monodromy. This explains the exact reduction from the universal theorem to the representations at issue.

For the genus-zero assertion, the units x and $1 - x$ give a $G_{\mathbb{Q}}$-invariant Kummer basis of

$$H^1_{\text{ét}}(X_{\overline{\mathbb{Q}}}, \mathbb{Q}_\ell(1)) = \mathbb{Q}_\ell^2. \tag{D.41}$$

Indeed the geometric first cohomology has rank two, and the two classes are independent by their residues at 0 and 1. The curve and the specified tangent vector have good integral models away from ℓ, so the action factors through $G_{\mathbb{Q},\{\ell\}}$, the maximal extension unramified outside ℓ and the real place. Equation (D.41) checks the precise hypothesis of Hain–Matsumoto Proposition 8.3, giving (D.40), the negative weights, and the surjection from weighted completion. Its Theorem 8.4 then identifies the associated positive graded outer image after tensoring the Ihara graded groups with $\mathbb{Q}_\ell$; the integral comparison has only finite kernel and cokernel at each degree as in its Proposition 8.2. Finally $\pi_1^{\text{top}}(X(\mathbb{C})) = F_2$ is finitely generated, which is exactly the hypothesis of Theorem A.6. It identifies the unipotent completion of its profinite or pro-ℓ completion with the rational unipotent completion after scalar extension.

As a finite check of the resulting free Lie algebra, `D_motives.py` computes the first six lower-central graded dimensions from the Witt formula: $2, 1, 2, 3, 6, 9$. These are dimensions of the Lie algebra of the geometric fundamental group, not dimensions of the unknown full Galois image. □

D.19. An arbitrary integral discrete representation is insufficient [Refuted]

There is no construction taking every integral representation of $\mathrm{SL}_2(\mathbb{Z})$ to an object of MEM_1 with that Betti monodromy. In addition, a geometric representation alone does not determine a unique arithmetic representation over $\mathbb{Q}$.

Proof. Use the discrete presentation $\mathrm{SL}_2(\mathbb{Z}) = \langle \rho, \sigma \mid \rho^3 = \sigma^2,\ \rho^6 = \sigma^4 = 1 \rangle$. The assignments

$$\rho \longmapsto B = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \qquad \sigma \longmapsto A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \tag{D.42}$$

give a homomorphism to $\mathrm{GL}_2(\mathbb{Z})$, since $B^3 = A^2 = I$. The cusp generator $\sigma^{-1}\rho$ has image

$$AB = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \det(TI - AB) = T^2 + T - 1. \tag{D.43}$$

One eigenvalue has absolute value greater than one, so no positive power is unipotent. Nilpotent logarithmic residue in the definition of MEM_1 forces unipotent cusp monodromy. Thus (D.42) cannot underlie such an object, even after a finite cover of the cusp. All matrix identities and the characteristic polynomial were executed in `D_motives.py`.

For arithmetic nonuniqueness take

$$E : y^2 = x^3 - x, \qquad E^{(2)} : y^2 = x^3 - 4x. \tag{D.44}$$

They are isomorphic over $\overline{\mathbb{Q}}$ and their ℓ-adic representations differ by χ_2. At 5, direct enumeration gives $|E(\mathbb{F}_5)| = 8$ and $|E^{(2)}(\mathbb{F}_5)| = 4$. Their Frobenius traces are therefore -2 and 2, so the arithmetic representations for $\ell \neq 5$ are not isomorphic, although the geometric fibres are. The same argument applies to a universal family and its constant quadratic twist: they have isomorphic geometric local systems but different arithmetic descent. `D_motives.out` records this count and five more primes, checking the twist relation at each one. We use this example over $\mathbb{Q}$; it does not assert that both twists belong to the everywhere integral category MEM_1. □

D.20. The extra data required for general arithmetic descent [Proved]

Fix a rational or tangential section of an arithmetic fundamental-group sequence, thereby writing its group as $\Pi = \Gamma \rtimes G_{\mathbb{Q}}$, with continuous action of $G_{\mathbb{Q}}$ on the geometric profinite group Γ. A continuous geometric representation $\rho : \Gamma \to \mathrm{GL}(V_\ell)$ extends to a continuous representation of Π if and only if one gives a continuous homomorphism $a : G_{\mathbb{Q}} \to \mathrm{GL}(V_\ell)$ satisfying

$$a(\gamma)\rho(h)a(\gamma)^{-1} = \rho(\gamma h) \quad (\gamma \in G_{\mathbb{Q}},\ h \in \Gamma). \tag{D.45}$$

Given one such a, the extensions up to conjugation by the centralizer $C = Z_{\mathrm{GL}(V_\ell)}(\rho(\Gamma))$ are classified by continuous nonabelian $H^1(G_{\mathbb{Q}}, C)$, where $G_{\mathbb{Q}}$ acts on C by conjugation through a.

A construction with the motivic meaning fixed above must add to this descent datum the mixed Tate fibre, de Rham and Hodge structures, boundary conditions, comparison maps, and, for torsion, stable integral lattices. D.18 supplies these in the two requested geometric cases. Neither the matrix equations of M nor a discrete integral monodromy representation alone provide them.

Proof. Restriction of an extension $\widetilde{\rho} : \Pi \to \mathrm{GL}(V_\ell)$ to the specified section gives a, and the conjugation relation in a semidirect product gives (D.45). Conversely $\widetilde{\rho}(h, \gamma) = \rho(h)a(\gamma)$

is continuous and multiplicative precisely because of (D.45). For another extension a', put $b(\gamma) = a'(\gamma)a(\gamma)^{-1}$. Equation (D.45) for both extensions says $b(\gamma) \in C$; multiplicativity says

$$b(\gamma\gamma') = b(\gamma)\, a(\gamma)b(\gamma')a(\gamma)^{-1}. \tag{D.46}$$

This is exactly the continuous cocycle equation. Conjugation by an element of C changes b by the usual coboundary, proving the classification. The additional realization requirements are those of the explicit definition above, not consequences of (D.45). D.19 proves that omitting them changes the assertion to a false one. □

E Sections, divided powers, and the envelope of §50

We use $\mathrm{Env}(\Gamma)$ for the inverse system of §50.4, and E_Γ for its affine inverse limit. We use $\underline{\Gamma}_M$ for the single finite type group scheme obtained from a chosen normal lattice $M \subset \Gamma$. These are different objects. The distinction is already explicit in the manuscript: §50.3, fols. 656–658, C4 scans 17–21; §50.4, fols. 659–662, C4 scans 23–29; and §51.1, fol. 665, C4 scan 35.

E.1 What "unipotent" means in the question

E.1. Two definitions that must not be identified [Proved]

In C4 scan 5, fol. 650bis, Grothendieck calls a geometric section $g \in G(\overline{k})$ unipotent when $g = \phi(1)$ for a homomorphism $\phi : \mathbb{G}_{a,\overline{k}} \to G_{\overline{k}}$. In characteristic $p > 0$, every such element satisfies $g^p = 1$. Consequently, an element of order p^2 in a unipotent algebraic group is a counterexample to the assertion that all conventionally unipotent elements arise this way, but is not a counterexample satisfying the manuscript's own definition. The quasi-unipotent condition in that definition is equivalent, on geometric fibres of positive characteristic, to finite order.

Proof. The multiplication-by-p map on $\mathbb{G}_a$ is zero in characteristic p, so $\phi(1)^p = \phi(p) = 1$. If a power g^n lies in such an image, then $g^{np} = 1$. Conversely, if g has finite order, a power is the identity and is the value of the zero homomorphism. This proves all assertions without assuming that the group itself is connected, smooth, or affine. □

E.2. The characteristic-zero assertion for linear targets [Proved]

Let A be a $\mathbb{Q}$-algebra and $G \hookrightarrow \mathrm{GL}_{r,A}$ a closed immersion of affine group schemes. Suppose that $g \in G(A)$ is fibrewise unipotent in the sense of E.1. There is a unique homomorphism of A-group schemes $\phi : \mathbb{G}_{a,A} \to G$ such that $\phi(1) = g$. In the specified representation,

$$N = g - I, \qquad X = \log(I + N), \qquad \phi(t) = \exp(tX),$$

where the two expressions are finite polynomials. No reducedness assumption on A is needed.

Proof. On every geometric fibre, N is nilpotent. The non-leading coefficients of its characteristic polynomial therefore belong to the nilradical of A. They generate a nilpotent ideal J: an ideal generated by finitely many nilpotent elements is nilpotent, by expanding a sufficiently long product. Cayley–Hamilton gives $N^r \in J\,\mathrm{End}(A^r)$, hence $N^{rq} = 0$ if $J^q = 0$. The finite formal identities for logarithm and exponential now apply in the nilpotent algebra generated by N, and give a homomorphism to $\mathrm{GL}_{r,A}$ with value g.

For every integer n, this homomorphism takes the value $g^n \in G(A)$. If F belongs to the defining ideal of G, its pullback is a polynomial $P(t) \in A[t]$ vanishing at all integers. For a

degree bound d, evaluation at $0, \dots, d$ has Vandermonde determinant $\prod_{i<j}(j-i)$, a unit in A. Thus $P = 0$, so the map factors through G.

Finally, a polynomial matrix homomorphism $P : \mathbb{G}_a \to \mathrm{GL}_r$ obeys $P'(t) = P(t)D$, where $D = P'(0)$. Comparing coefficients over a $\mathbb{Q}$-algebra gives $P(t) = \sum t^i D^i/i!$; polynomiality implies that D is nilpotent. Its value at 1 determines $D = \log P(1)$. This proves uniqueness. In particular, this proof justifies the descent argument of C4 scan 6 for linear targets. □

E.3. The exact integral divided-power criterion [Proved]

Let A be a torsion-free $\mathbb{Z}$-algebra, $G \hookrightarrow \mathrm{GL}_{r,A}$ a closed subgroup scheme, and $g \in G(A)$ such that $g - I$ is nilpotent. Put $X = \log g \in M_r(A \otimes \mathbb{Q})$. A homomorphism $\mathbb{G}_{a,A} \to G$ taking 1 to g exists if and only if

$$X^i/i! \in M_r(A) \quad \text{for every } i \geq 0.$$

When it exists it is unique. Thus the coefficient criterion (10) of §50.1, fol. 652, C4 scan 9, is correct, but the inference to it from (9) is false.

Proof. After tensoring with $\mathbb{Q}$, E.2 forces the polynomial $\exp(tX)$. Polynomial coefficients over the torsion-free ring A are uniquely determined in $A \otimes \mathbb{Q}$. This proves necessity and uniqueness. If all coefficients belong to A, the exponential defines a polynomial matrix with inverse $\exp(-tX)$; the addition identity is a polynomial identity over A, because it holds after tensoring with $\mathbb{Q}$. The same injectivity applied to each defining equation of G proves factorization through G. □

E.4. The actual exponential integrality implication fails in rank three [Refuted]

For the nilpotent Jordan matrix $J = E_{12} + E_{23} \in M_3(\mathbb{Z})$, set

$$X = J - \tfrac{1}{2}J^2 = \begin{pmatrix} 0 & 1 & -1/2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Then $X^3 = 0$, $\exp X = I + J$, and $\exp(-X) = I - J + J^2$ are integral, whereas X is not integral at 2. The assertion in C4 scan 9 that the $n = 3$ case of (9) implies (10) is false. The strongest exact replacement is E.3, and E.10 below proves the weaker assertion after taking a positive power.

Proof. Here $X^2 = J^2$, so the two exponential sums are exactly the displayed matrices. The $(1,3)$-entry of X is $-1/2$. In rank two, $X^2 = 0$ and $\exp X = I + X$, so this particular implication cannot fail in smaller rank. The exact rational identities and both inverse matrices are checked in `code/E_integrality.py` and recorded in `code/E_integrality.json`. Notice that $I + J$ modulo 2 has order 4. By E.1 this example alone does not answer the stronger fibrewise question; E.6 does. □

E.5. A counterexample to the literal signed reading of (11) [Refuted]

The literal implication (11)⇒(12) in C4 scans 9–10 is false for $X \in M_4(\mathbb{Q})$, even over $A = \mathbb{Z}_{(2)}$ (§50.1, fols. 652–652'). One counterexample is

$$X = \begin{pmatrix} 0 & 1/2 & 0 & 0 \\ 0 & 0 & 12 & 2 \\ 0 & 0 & 0 & 1/2 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad A_2 = \tfrac{1}{2}X^2 = 3E_{13} + 3E_{24} + \tfrac{1}{2}E_{14}, \quad A_3 = \tfrac{1}{6}X^3 = \tfrac{1}{2}E_{14}.$$

Thus $A_2 + A_3$ and $A_2 - A_3$ are integral, but neither A_2 nor A_3 is. If one adds the missing hypothesis $X \in M_4(A)$, for a torsion-free $\mathbb{Z}$-algebra A, the literal implication is true.

Proof. Direct multiplication gives the displayed formulas and $X^4 = 0$. The matrix $X_0 = 2X$ is integral and not divisible by 2; hence this also gives precisely the form requested in (11bis), with $\nu = 1$. It was the first counterexample after 168 candidates in the explicitly specified finite rational grid in `E_integrality.py`.

For the correction, write $C = X^2/2 + X^3/6$. Over $A[1/2]$, adding and subtracting the two signed expressions isolates both terms. Over $A[1/3]$, the matrix $3I + X$ is invertible with inverse the finite geometric series in $-X/3$, and

$$X^2/2 = 3C(3I + X)^{-1}$$

is integral. Subtracting from C gives $X^3/6$. The two open sets cover $\operatorname{Spec} A$, because $(2, 3) = A$; membership in $M_4(A)$ is local, so the result follows.

This reading must be kept distinct from E.4. The actual hypothesis $\exp(\pm X) \in M_4(A)$ contains $\pm X$, whereas (11) omits it. Visual inspection of scan 9 confirms that the omission is in the manuscript, not a silent correction supplied here. □

E.6. A genuinely fibrewise-unipotent integral section need not extend [Refuted]

Let $N \in M_5(\mathbb{Z})$ have superdiagonal $(1, 2, 3, 2)$ and all other entries zero. The section $g = I + N \in \mathrm{GL}_5(\mathbb{Z})$ is fibrewise unipotent in the precise sense of E.1, but is not $\phi(1)$ for any $\phi : \mathbb{G}_{a,\mathbb{Z}} \to \mathrm{GL}_{5,\mathbb{Z}}$. More generally, over $\mathbb{Z}_{(p)}$, the implication

$$N^p \in pM_r(\mathbb{Z}_{(p)}), \qquad N^r = 0 \quad \Longrightarrow \quad (I + N)^t \in M_r(\mathbb{Z}_{(p)}[t])$$

holds for $r \le p^2$, and the bound is optimal uniformly in N. The first counterexample from the weighted-shift family occurs in degree p^2, or rank $p^2 + 1$.

Proof. For the rank-five matrix, $N^2 \equiv 0 \pmod 2$, $N^3 \equiv 0 \pmod 3$, and $N^5 = 0$. Over a field of characteristic p, a matrix $I + \overline{N}$ with $\overline{N}^p = 0$ lies on a one-parameter subgroup: the truncated logarithm $D = \sum_{j=1}^{p-1} (-1)^{j+1} \overline{N}^j / j$ satisfies $D^p = 0$, and the truncated exponential $\sum_{j=0}^{p-1} t^j D^j / j!$ has the required group law and value. In characteristic zero use E.2. This treats every geometric fibre of the integral section.

Nevertheless,

$$X = \log(I + N) = \begin{pmatrix} 0 & 1 & -1 & 2 & -3 \\ 0 & 0 & 2 & -3 & 4 \\ 0 & 0 & 0 & 3 & -3 \\ 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \qquad X^4/4! = \tfrac{1}{2} E_{15}.$$

E.3 rules out the asserted homomorphism. All entries and identities are verified by the program.

For the general bound, expand $(I + N)^t = \sum_{k=0}^{r-1} \binom{t}{k} N^k$. The condition on N^p implies $N^k \in p^{\lfloor k/p \rfloor} M_r(\mathbb{Z}_{(p)})$. If $k < p^2$, Legendre's formula gives $v_p(k!) = \lfloor k/p \rfloor$. Since $k!\binom{t}{k}$ has integral coefficients, each summand is integral. The group law holds over $\mathbb{Q}$ and hence over the torsion-free coefficient ring.

For sharpness, set $d = p^2$ and give a rank-$d + 1$ shift the weights

$$w_i = i / p^{\max(v_p(i) - 1, 0)} \quad (1 \le i \le d).$$

Every product of p consecutive weights is divisible by p, whereas $\prod_{i=1}^{p^2} w_i = (p^2)!/p$. Consequently $N^p \equiv 0 \pmod p$, but the leading coefficient in $(I + N)^t$, namely $N^{p^2}/(p^2)!$, has

an entry $1/p$. For primes $q \neq p$ the weights have the same q-valuations as $1, \dots, p^2$, so all geometric fibres satisfy the required one-parameter condition. This also supplies genuine integral examples at every prime. The executed search through degree 12 gives first failing degrees 4 and 9 for $p = 2, 3$; no member fails for $p = 5$ in that range, as the proven threshold 25 predicts. □

E.7. Positive-characteristic and infinitesimal-base tests [Refuted]

For every prime p, the conventional unipotent group $W_2/\mathbb{F}_p$ contains $(1, 0)$ of order p^2, so it is not a value at 1 of a $\mathbb{G}_a$-homomorphism. More strongly, over $A = \mathbb{Z}/p^2$, the smooth affine group

$$G = \mathbb{A}^1_A, \qquad x * y = x + y + pC_p(x, y), \qquad C_p(x, y) = \frac{(x+y)^p - x^p - y^p}{p}$$

and section $g = 1$ satisfy the manuscript's fibrewise hypothesis, but admit no homomorphism $\mathbb{G}_{a,A} \to G$ with value g. For $p = 2$ the law is $x + y + 2xy$, so the smallest ring in this family is $\mathbb{Z}/4$. In every case g has order p, so this counterexample does not refute the positive-power assertion.

Proof. For Witt vectors the addition law is

$$(a_0, a_1) + (b_0, b_1) = \left(a_0 + b_0,\ a_1 + b_1 - \sum_{i=1}^{p-1} \frac{1}{p} \binom{p}{i} a_0^i b_0^{p-i} \right).$$

The standard identity $[p](a_0, a_1) = (0, a_0^p)$ follows by adding the ghost polynomials and reducing modulo p; a second multiplication by p gives zero. Hence $(1, 0)$ has order p^2. E.1 explains exactly which proposed implication this refutes. The program checks the group law on all triples for $p = 2, 3, 5$, respectively 64, 729, and 15,625 triples, and computes the three orders.

For the second example, C_p is an integral polynomial and an additive two-cocycle, being the divided coboundary of x^p. Because $p^2 = 0$ in A, changing an argument of C_p by a multiple of p does not change pC_p. The cocycle identity therefore proves associativity of the displayed law. The inverse is $-x - pC_p(x, -x)$. Thus the underlying smooth affine line does define a group. Summing the cocycle along the p copies of 1 gives $[p]g = p + p(p^p - p)/p = p^p = 0$ in A, while its reduction has exact order p. Its geometric fibre is $\mathbb{G}_a$, and the section reduces to 1, so the hypothesis of scan 6 holds. Suppose a polynomial $f(t) \in A[t]$ defined the claimed homomorphism. Then $f(0) = 0$, $f(1) = 1$, and

$$f(t + u) - f(t) - f(u) = pC_p(f(t), f(u)). \tag{E.7.1}$$

Modulo p, f is an additive polynomial. Let d be its largest degree with coefficient not divisible by p. It exists because $f(1) = 1$, and d is a power of p. The coefficient of $t^{(p-1)d}u^d$ on the right of (E.7.1) is $pa_d^p \neq 0$ modulo p^2. On the left it is $\binom{pd}{d} a_{pd}$. The coefficient a_{pd} is divisible by p by the choice of d, and $\binom{pd}{d}$ is divisible by p (indeed its valuation is 1). The left coefficient is therefore zero modulo p^2, a contradiction. This proves failure in every degree. Independently, the program tests all 1,024 candidate polynomials over $\mathbb{Z}/4$ of degree at most 6 with $f(0) = 0$, $f(1) = 1$, using the polynomial identity rather than only its values on the 16 pairs of $\mathbb{Z}/4$-points; none succeeds. It also checks the deformed group laws on all triples of points for $p = 2, 3, 5$, computes the orders of g, and verifies the leading-term obstruction for $d = 1, p, p^2, p^3$. □

E.2 The doubtful lemma and the positive-power theorem

E.8. Integer-valued polynomials and the finite-difference correction [Proved]

A polynomial $P \in \mathbb{Q}[t]$ of degree at most d is integer-valued on $\mathbb{Z}$ if and only if

$$P(t) = \sum_{j=0}^{d} c_j \binom{t}{j}, \qquad c_j \in \mathbb{Z}.$$

Integer values, even at every integer, do not imply integral monomial coefficients. For any commutative ring A, a polynomial $P \in A[t_1, \dots, t_r]$ which vanishes on every integral tuple has all its coefficients annihilated by some positive integer. If in addition $P(0) = 0$, some substitution $t_i \mapsto Dt_i$ makes it identically zero.

Proof. Repeated forward differences give the Newton expansion $c_j = \Delta^j P(0)$. Thus integral values imply integral coefficients in the binomial basis; the converse follows because all binomial polynomials take integral values at positive and negative integers. This is the $\mathbb{Z}$ case discussed in the supplied Bhargava (2009, §1, p. 964); the preceding argument proves the case we use directly, without invoking a general P-ordering theorem. The polynomial $\binom{t}{2} = (t^2 - t)/2$ proves the warning.

For the assertion over A, choose an individual degree bound d in every variable. Evaluation on $\{0, \dots, d\}^r$ is the tensor product of integral Vandermonde matrices. Multiplication by the adjugate shows that a nonzero integer annihilates every coefficient of a polynomial in its kernel. Choose one such integer D. After replacing every variable by Dt_i, each nonconstant coefficient acquires a factor divisible by D; the constant coefficient was zero. This proves the last assertion.

The degree-12 computation records every binomial basis element and its monomial denominator. It also checks the two specified matrix families $X = \log(1 + J_{d+1})$ and the weighted family of E.6 for each $1 \le d \le 12$ and $p = 2, 3, 5$. This is an exhaustive search of those 72 cases, not an assertion of exhaustion of all rational matrices. The counterexamples and the proofs E.4–E.6 settle the implications for which the search was used.

Two further readings on C4 scan 13, fol. 654, can be tested exactly. In (20), the reading $X \in M_5(A)$ makes the displayed conclusion $X \in M_5(A)$ tautological. With the alternative initial reading $X \in M_5(K)$, the conclusion is false: take $A = \mathbb{Z}_{(2)}$, $K = \mathbb{Q}$, and $X = \frac{1}{2}E_{12} \in M_5(K)$. All its powers of degree at least two vanish, so both signed higher-power combinations are integral, although X is not. This is a test of the literal alternatives, not a proposed final transcription. The cubic denominator 3! in the checked transcription is forced by the exponential addition law; the first-pass 6! alternative gives coefficient $-119/240$ of $t^2 u E_{14}$ in $P(t + u) - P(t)P(u)$, for a rank-four Jordan block. Both calculations are included in `E_integrality.py`. Finally, the phrase "car. 0 i.e. a $\mathbb{Q}$-algebra" on that page cannot be an equivalence of ring-theoretic hypotheses: $\mathbb{Z}$ is a counterexample. Our E.2 uses the stated $\mathbb{Q}$-algebra hypothesis, and E.6 shows why it cannot simply be discarded. □

E.9. The affine dilation proposition of scans 14–16 [Proved]

The proposition of §50.2, fols. 654'–655', C4 scans 14–16, has the following precise affine form. Let A be a ring such that $A \to A[1/n]$ is injective, and let $G = \operatorname{Spec} B$ be an affine group scheme of finite type over A. Every homomorphism $\phi : \mathbb{G}_{a,A[1/n]} \to G_{A[1/n]}$ extends after precomposition with multiplication by n^v, for some $v \ge 0$, to a homomorphism $\psi : \mathbb{G}_{a,A} \to G$. More generally, for an affine finite type A-scheme Y, a morphism $\mathbb{A}^1_{A[1/n]} \to Y_{A[1/n]}$ admits such an extension if and only if its value at zero extends to $Y(A)$.

Proof. Choose algebra generators $b_1, \dots, b_s$ of B. Their images are polynomials $\sum_j a_{ij} t^j \in A[1/n][t]$. Their constant coefficients are in A, because a group homomorphism takes zero to the identity. One exponent v makes $n^{vj} a_{ij} \in A$ for all $j \geq 1$ and all i. The substituted polynomials define the required algebra map: every relation vanishes after localization and therefore vanishes in $A[t]$, by the injectivity assumption. The two group-law maps agree after localization, so the same argument in $A[t, u]$ proves that the extension is a group homomorphism. For Y, the same argument applies to its coordinate ring once its constant terms come from the specified zero section; evaluating at zero proves necessity. Finite affine covers give the corresponding statement for a quasi-compact base with schematically dense $D(n)$, since the extensions are unique and hence glue.

This proves the proposition actually written in C4 scans 14–16. The substitution is $t \mapsto n^v t$, not $t \mapsto t^{n^v}$. The latter is not additive in characteristic zero. The correction is supported both by the type of the map and by formula (22) in scan 14. □

E.10. A complete positive-power theorem for linear group schemes [Proved]

Let A be any commutative ring and $G \hookrightarrow \mathrm{GL}_{r,A}$ a closed immersion of affine group schemes of finite presentation over A. A section $g \in G(A)$ is fibrewise quasi-unipotent in the manuscript's sense if and only if there exist an integer $m > 0$ and a homomorphism $\phi : \mathbb{G}_{a,A} \to G$ with $\phi(1) = g^m$. This proves the "Théorème (!)" of fol. 652', C4 scans 10–11, under the specified linearity hypothesis, including all linear examples over $\mathbb{Z}$, $\mathbb{Z}_p$, and $\mathbb{Z}/p^2$.

Proof. Only necessity needs proof. For each prime ideal of A, a positive power of the matrix g is unipotent in the residue field. The sets

$$V\big(\text{entries of } (g^a - I)^r\big), \qquad a = 1, 2, \dots,$$

cover $\operatorname{Spec} A$. Each is open and closed in the constructible topology, which is compact; alternatively, apply compactness to the Boolean algebra of finitely generated radical conditions. For clarity, this compactness holds for arbitrary A: an ultrafilter $\mathcal{U}$ on $\operatorname{Spec} A$ defines the prime ideal $\mathfrak{p} = \{f : V(f) \in \mathcal{U}\}$. The identities for $V(fg)$, and the inclusion $V(f) \cap V(g) \subset V(f+g)$, prove primality and additivity; the ultrafilter converges to $\mathfrak{p}$ for the topology generated by the clopen sets $V(f), D(f)$. Thus this topology is compact. A finite subcover therefore exists. Taking the least common multiple of its exponents shows that $N = g^a - I$ is nilpotent on every residue field. As in the Cayley–Hamilton argument of E.2, the finitely many characteristic coefficients generate a nilpotent ideal, and $N^b = 0$ for some b.

Choose $d = (b-1)!$, with $d = 1$ if $b = 1$. Every polynomial $\binom{dt}{j}$, $1 \leq j < b$, has integral coefficients: its constant term is zero, and each other coefficient contains a factor d, which is divisible by $j!$. Thus

$$P(t) = \sum_{j=0}^{b-1} \binom{dt}{j} N^j$$

belongs to $M_r(A[t])$. The binomial identity proves $P(t+u) = P(t)P(u)$ over the universal torsion-free ring $\mathbb{Z}[z]/(z^b)$, and hence after every base change. Its inverse is $P(-t)$, and $P(n) = g^{adn}$ for every integer n. It therefore defines a homomorphism to GL_r.

There are finitely many defining equations for G inside GL_r. Their pullbacks by $P(t)$, including the inverse-determinant coordinate, vanish at every integer and at zero. E.8 supplies one further dilation $t \mapsto Dt$ which makes every pullback identically zero. The resulting homomorphism factors through G, and its value at 1 is g^{adD}. Sufficiency is the definition of quasi-unipotence. □

E.11. The remaining non-linear scope of scans 6 and 10 [Open]

We have not proved the following full-source statement: for every quasi-compact scheme S, every separated group scheme G/S locally of finite presentation, and every fibrewise quasi-unipotent $g \in G(S)$, a positive power of g is the value at 1 of a homomorphism $\mathbb{G}_{a,S} \to G$. Likewise E.2 has not established the assertion for all group schemes over all $\mathbb{Q}$-algebras without a linearity hypothesis.

Our proved cases are E.2, E.3, E.9, and E.10. The executed matrix and polynomial tests are E.4–E.8, with their complete outputs. The precise missing step is extension from a general affine or non-affine group scheme to a faithful finite locally free linear representation, or a substitute for that step. We do not assume such a representation exists. A first remaining quantified test in base dimension zero is $S = \operatorname{Spec} \mathbb{Q}[\epsilon]/(\epsilon^2)$, with arbitrary separated affine finite presentation G/S and no assumed closed linear embedding; the analogous integral test is over $\mathbb{Z}/4$. These are cases not covered by the proof, not claims that the corresponding questions are open in the literature. The example E.7 does not obstruct the positive-power theorem, because its section has order p. This distinction is essential to the status assigned to the full manuscript assertion.

E.3 The envelope as it is actually defined

E.12. The terminology and the finite-stage pushout [Proved]

The "métabélien" groups of §50.4, fol. 660, C4 scan 25, are finitely generated virtually abelian groups. Let Γ be such a group and $M \simeq \mathbb{Z}^r$ a normal subgroup of finite index, with $H = \Gamma/M$. Choose a set section $s : H \to \Gamma$, $s(1) = 1$, and write its action and cocycle as

$$a_h(m) = s(h)ms(h)^{-1}, \qquad c(h,k) = s(h)s(k)s(hk)^{-1} \in M.$$

There is a smooth affine $\mathbb{Z}$-group scheme

$$\underline{\Gamma}_M = \coprod_{h \in H} \mathbb{A}^r_{\mathbb{Z}}, \qquad \mathcal{O}(\underline{\Gamma}_M) = \prod_{h \in H} \mathbb{Z}[x_1, \dots, x_r],$$

whose multiplication on the (h,k) block is

$$(m,h)(n,k) = (m + a_h n + c(h,k), hk). \tag{E.12.1}$$

It lies in an exact sequence of fppf group sheaves $1 \to \mathbb{G}_a^r \to \underline{\Gamma}_M \to H_{\mathbb{Z}} \to 1$, and $\Gamma \to \underline{\Gamma}_M(\mathbb{Z})$ is an isomorphism. The construction is independent, up to the unique coordinate change specified by the two sections, of the chosen section.

Proof. The identity $c(h,k) + c(hk,l) = a_h c(k,l) + c(h,kl)$ follows by associating $s(h)s(k)s(l)$ in two ways. It proves associativity of (E.12.1); the identity is $(0,1)$, and solving the same formula gives polynomial inverses. Each block is an affine space over $\mathbb{Z}$, proving the stated geometric properties. Projection to H has a section on every component as a morphism of schemes, so the sequence is exact even on sections over an arbitrary base scheme, with locally constant H-labels. Replacing $s(h)$ by $b(h)s(h)$, $b(h) \in M$, replaces the vector coordinate by $m - b(h)$; this explicitly identifies the two laws.

For a disconnected test scheme T, the label is a locally constant function $T \to H$, not necessarily one fixed element of H. This is exactly the correction made by Zariski sheafification in C4 scans 18–20. Formula (E.12.1) is also the pushout of the extension $1 \to M \to \Gamma \to H \to 1$ along $M_{\mathbb{Z}} \to \mathbb{G}_a^r$, with the prescribed conjugation action. This proves its universal property among maps of such extensions directly from the cocycle formula. □

E.13. Coordinate ring and fibres of the full envelope [Proved]

Set $M_n = nM$, $H_n = \Gamma/nM$, and $E_\Gamma = \varprojlim_n \underline{\Gamma}_{nM}$, where indices are ordered by divisibility. Then

$$\mathcal{O}(E_\Gamma) = \varinjlim_n \prod_{h \in H_n} \mathbb{Z}[x_1, \dots, x_r]. \tag{E.13.1}$$

For $n \mid n'$, the coordinate changes on each component are $x \mapsto (n'/n)x + b$, with explicitly determined $b \in \mathbb{Z}^r$. The construction is independent of the chosen normal lattice. Its kernel over the proconstant group $\widehat{\Gamma}_\mathbb{Z} = \varprojlim H_{n,\mathbb{Z}}$ is

$$E_\Gamma^0 \simeq \operatorname{Spec}\bigl(\mathbb{Z} + (x_1, \dots, x_r)\mathbb{Q}[x_1, \dots, x_r]\bigr). \tag{E.13.2}$$

For every prime p, there is a canonical isomorphism

$$(E_\Gamma)_{\mathbb{F}_p} \simeq \widehat{\Gamma}_{\mathbb{F}_p}. \tag{E.13.3}$$

In particular these fibres are reduced, of Krull dimension zero, and generally not of finite type. The finite-stage scheme of E.12 instead has $|H|$ smooth components of dimension r in every characteristic.

Proof. The inclusions $n'M \hookrightarrow nM$ and identity of Γ give the transition maps by E.12. They are affine, so the inverse limit is represented by the indicated direct limit of rings. The subgroups nM are cofinal among all finite-index subgroups of M, and among normal sublattices used in the construction. Any two normal lattices have a common finite-index intersection; this proves independence. Every finite-index normal subgroup of Γ contains some nM, so $\varprojlim H_n$ really is the full profinite completion.

On the identity components the transition is multiplication by n'/n. Its coordinate-ring limit is the union of $\mathbb{Z}[x_1/n, \dots, x_r/n]$ in $\mathbb{Q}[x_1, \dots, x_r]$. This union is precisely (E.13.2): clear the finitely many rational denominators of every nonconstant monomial; the constant term must be integral. This proves the kernel assertion as a statement about affine group schemes.

In rank one its points are coherent sequences (a_n), $a_n = ma_{mn}$, as in scan 29, fol. 662, formula (56). If an integer $q > 0$ annihilates A, every $a_n = qa_{qn} = 0$. If every residue characteristic is positive, such a q does exist: otherwise $A \otimes \mathbb{Q} \neq 0$ has a prime ideal, producing a characteristic-zero residue field of A. For a $\mathbb{Q}$-algebra the sequence is uniquely $a_n = a_1/n$, so the point group is $(A, +)$. Characteristic zero alone is insufficient: for $A = \mathbb{Z}$ every coherent sequence is zero. Also, "all residue characteristics divide q" does not imply $qA = 0$ for that same q, as $A = \mathbb{Z}/4$, $q = 2$, shows. These give the precise corrections to the bracketed remarks in scan 29.

After reduction modulo p, the transition from the np-stage to the n-stage has zero linear part. Hence every polynomial at stage n becomes a locally constant function at stage np. Conversely all locally constant functions on each finite H_n are already in the stage coordinate ring. Therefore its direct limit is exactly $\varinjlim_n \mathbb{F}_p^{H_n}$, the coordinate ring of the proconstant group. It is a filtered union of products of fields, so is reduced and absolutely flat; every prime is maximal.

For $\Gamma = \mathbb{Z}$, (E.13.1) can also be described without chosen coordinates: it is the ring of integer-valued rational quasipolynomials on $\mathbb{Z}$, namely functions which are rational polynomials on the classes of one finite modulus. An integer-valued polynomial on a class has integral monomial coefficients after refining the modulus enough to clear all nonconstant denominators; its constant term is its integral value at the chosen representative. This proves the description in both directions. It contains nontrivial periodic idempotents, which the divided-power polynomial algebra alone does not contain. □

E.14. The universal property for linear targets, with the correct colimit [Proved]

For any ring A, any finitely generated virtually abelian Γ, and any closed finite presentation subgroup scheme $G \hookrightarrow \mathrm{GL}_{r,A}$, restriction along $\Gamma_A \to (E_\Gamma)_A$ is a bijection

$$\mathrm{Hom}_{A\text{-grp}}((E_\Gamma)_A, G) \simeq \{f : \Gamma \to G(A) \mid f \text{ is fibrewise quasi-unipotent}\}. \tag{E.14.1}$$

The left side is $\varinjlim_n \mathrm{Hom}_{A\text{-grp}}((\underline{\Gamma}_{nM})_A, G)$, not the inverse limit printed in formula (58), C4 scan 31. This proves the universal assertion of §50.5, fol. 663, scan 32, for the stated targets.

Proof. An affine morphism from a filtered inverse limit to a finitely presented affine scheme factors through a finite stage: choose images for finitely many algebra generators and then a later stage where their finitely many relations vanish. Applied also to the multiplication and inverse equations, this gives the displayed colimit of group homomorphisms. This elementary algebra argument explains the direction of the limit correction.

Let f be on the right of (E.14.1). Choose a basis of the normal lattice M. Applying E.10 to its finitely many images and taking a common multiple of the resulting exponents gives one-parameter homomorphisms with values $f(ne_i)$ at 1. We take the explicit matrix polynomials of E.10, which are polynomials in powers of $f(e_i)$. Since M is abelian, these homomorphisms commute with each other and with every $f(m)$, $m \in M$, over A itself. Their product is a vector-group homomorphism $v : (nM) \otimes \mathbb{G}_a \to G$ extending $f|_{nM}$.

For each of the finitely many coset representatives of Γ/M, the required identity between conjugation of v and the prescribed linear action on its source is again a finite set of polynomial identities. They hold on every integral tuple. Another application of E.8, and a further common dilation, make them identities over A. The cocycle formula E.12 now extends f to $\underline{\Gamma}_{nM}$.

If two extensions agree on Γ, pass to a common stage. On each of its finitely many affine-space components their coordinate differences vanish on the integral lattice. E.8 kills every such difference after a common refinement, proving uniqueness in the colimit. Finally, any map from a stage has quasi-unipotent restriction: a positive power of each element of Γ lies in the lattice, and its image lies on the image of the line through the corresponding integral lattice vector. This proves the reverse inclusion and the bijection. □

For the full reading "G affine over S" or the stronger reading "affine over S_0" in scan 32, E.11 records the remaining linearization issue. The computations E.4–E.8 are the attack on the section-extension step, and E.14 explains exactly which proved correction is sufficient for the universal property. We do not turn that unproved generality into a claim about all affine group schemes.

E.4 The explicit modular example and its points

E.15. The lattice and quotient in the modular example [Proved]

Fix

$$\rho = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, \qquad \sigma = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \lambda_0 = -\sigma\rho\sigma\rho, \quad \lambda_1 = -\rho\sigma\rho\sigma.$$

Then $\Pi_0 = \langle \lambda_0, \lambda_1 \rangle$ is a normal free subgroup of rank two and index 12 in $\mathrm{SL}_2(\mathbb{Z})$. For $\mathfrak{S} = \mathrm{SL}_2(\mathbb{Z})/[\Pi_0, \Pi_0]$,

$$1 \longrightarrow M = \Pi_0^{\mathrm{ab}} \simeq \mathbb{Z}^2 \longrightarrow \mathfrak{S} \longrightarrow H \longrightarrow 1, \quad H = \langle r, s \mid r^6 = 1,\ s^2 = r^3,\ srs^{-1} = r^{-1} \rangle$$

has $|H| = 12$. The action on the basis (λ_0, λ_1) is

$$R = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \qquad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Moreover $\mathfrak{S}^{\mathrm{ab}} \simeq \mathbb{Z}/12$.

Proof. Matrix multiplication gives

$$\rho\lambda_0\rho^{-1} = \lambda_1, \qquad \rho\lambda_1\rho^{-1} = (\lambda_1\lambda_0)^{-1}, \qquad \sigma\lambda_0\sigma^{-1} = \lambda_1, \qquad \sigma\lambda_1\sigma^{-1} = \lambda_0.$$

Thus the subgroup generated by the two λ_i is normal. Using the amalgam presentation $\mathrm{SL}_2(\mathbb{Z}) = C_4 *_{C_2} C_6$, imposing $\lambda_0 = 1$ gives $\sigma\rho\sigma^{-1} = \rho^{-1}$; $\lambda_1 = 1$ is its conjugate. The resulting presentation is the one displayed for H. Every word reduces to $r^i s^j$, $0 \le i < 6$, $0 \le j < 2$, so its order is at most 12. The calculation in `E_envelope.py` realizes 12 distinct classes as

$$H \simeq \mathrm{SL}_2(\mathbb{Z}/4)/\langle \overline{\lambda}_0, \overline{\lambda}_1 \rangle;$$

the denominator is a normal Klein four group. This proves the order and the index without a coset-order guess.

The subgroup Π_0 intersects each conjugate of the amalgam factors trivially, because the factors of orders 4 and 6 inject into H. It therefore acts freely on the amalgam tree. Its quotient graph has $12/4 + 12/6 = 5$ vertices and $12/2 = 6$ edges, and is connected. Its fundamental group is free of rank $6 - 5 + 1 = 2$. The two generating elements therefore give a basis of its abelianization. Abelianizing the four conjugation formulas gives R, S. Finally, quotienting by $[\Pi_0, \Pi_0]$, which is contained in the commutator subgroup of $\mathrm{SL}_2(\mathbb{Z})$, does not change its abelianization. Abelianizing the amalgam presentation gives $\langle r, s \mid 6r = 0, 4s = 0, 3r = 2s \rangle \simeq \mathbb{Z}/12$. These are the discrete groups in §51.1, fol. 664, C4 scan 33; the word "profinis" in that passage is incompatible with the uncapped lattices in the same displayed sequence. If the deleted hats are retained instead, the sequence must be read as one of profinite completions; it is then a different typed assertion. Our input is the discrete group specified in the question. □

E.16. An explicit integral coordinate ring for the modular finite stage [Proved]

The group scheme $\underline{\mathfrak{S}}_M$ of §51.1 has

$$\mathcal{O}(\underline{\mathfrak{S}}_M) = \prod_{h \in H} \mathbb{Z}[x, y]. \tag{E.16.1}$$

Its multiplication is (E.12.1) with the following completely specified cocycle. Take the twelve representatives

$$1, r, s, r^2, rs, sr, s^2, r^2 s, rs^2, sr^2, s^3, r^2 s^2.$$

Associate to a word its linear action obtained from R, S, and its translation obtained by composing

$$r \mapsto (R, 0), \qquad s \mapsto (S, (1/3, -1/3)^t).$$

If t_h is that translation, set

$$c(h, k) = t_h + a_h t_k - t_{hk}. \tag{E.16.2}$$

Every $c(h, k)$ is integral. The model is smooth and flat of relative dimension two over $\mathbb{Z}$, with twelve reduced geometrically connected components in each fibre, including 2 and 3. Its number of $\mathbb{F}_p$-points is $12p^2$.

Proof. The affine transformations assigned to ρ, σ satisfy $\rho^6 = \sigma^4 = 1$, $\rho^3 = \sigma^2$, and send λ_0, λ_1 to translations by $(1, 0)$ and $(0, 1)$. Hence the translation of a word in Π_0 is precisely its abelianized lattice coordinate. The difference word defining $c(h, k)$ lies in Π_0; this proves integrality and shows that this is the cocycle of the specified discrete extension. Thus E.12 proves the coordinate ring and geometric claims. The complete multiplication table, all 144

cocycle values, and all 1,728 cocycle identities are independently computed in `E_envelope.json`; the program also verifies the four integral matrix conjugation identities from E.15.

For $p = 2, 3, 5, 7, 11, 13$, the point counts are respectively $48, 108, 300, 588, 1452, 2028$. The discrete quotients $\mathfrak{S}/nM$, generated by the images of ρ, σ, have computed orders $12, 48, 108, 192, 768, 972$ for $n = 1, 2, 3, 4, 8, 9$, agreeing with $12n^2$. The denominators 3 in the auxiliary translations do not introduce singularities: the group law itself uses only the integral cocycle (E.16.2). □

E.17. Integral, local, and profinite-ring points of the envelope [Proved]

For every finitely generated virtually abelian Γ,

$$E_\Gamma(\mathbb{Z}) = \Gamma, \qquad E_\Gamma(\mathbb{Z}_p) = \widehat{\Gamma}, \qquad E_\Gamma(\widehat{\mathbb{Z}}) = \prod_p \widehat{\Gamma}. \tag{E.17.1}$$

The local equality is an isomorphism of topological groups, with the inverse-limit topology on the left. For the finite-stage model one instead has

$$\underline{\Gamma}_M(\widehat{\mathbb{Z}}) = \prod_p \underline{\Gamma}_M(\mathbb{Z}_p), \qquad \underline{\Gamma}_M^{\Delta}(\widehat{\mathbb{Z}}) \simeq \widehat{\Gamma}, \tag{E.17.2}$$

where Δ means that all local images in H coincide. These statements apply in particular to $\mathfrak{S}$ of E.15.

Proof. The map to the first stage identifies a $\mathbb{Z}$-point with an element of Γ. At stage n, compatibility says that its lattice coordinate equals a chosen residue modulo nM plus n times an integral coordinate. This determines both the residue and the coordinate, so no extra point lies over a given element of Γ. The natural map from Γ supplies all the points.

The kernel (E.13.2) has no nonzero $\mathbb{Z}_p$-points: its coordinates form systems $x_n = mx_{mn}$, so each fixed x_n is divisible by every power of p and is zero. Each finite-stage group $\underline{\Gamma}_{nM}(\mathbb{Z}_p)$ is profinite, being a finite disjoint union of copies of $\mathbb{Z}_p^r$. The map from Γ to it extends uniquely and continuously to $\widehat{\Gamma}$. These extensions are compatible in n and give a map $\widehat{\Gamma} \to E_\Gamma(\mathbb{Z}_p)$ inverse to projection on the proconstant quotient: their composite is the identity on the dense subgroup Γ, and the other composite differs from the identity by the zero kernel just calculated.

For an affine scheme $X = \operatorname{Spec} B$, $\operatorname{Hom}(B, \prod_p \mathbb{Z}_p) = \prod_p \operatorname{Hom}(B, \mathbb{Z}_p)$. This gives the third equality of (E.17.1) and the first in (E.17.2), without a connectedness assumption on $\operatorname{Spec} \widehat{\mathbb{Z}}$. In the finite-stage model, requiring a common component $h \in H$ leaves exactly one copy of $M \otimes \widehat{\mathbb{Z}}$ above every h, with the same cocycle as the profinite completion of Γ. The topology on M induced from Γ is its full profinite topology because all nM are normal finite-index subgroups. This proves (E.17.2).

Thus the Δ in C4 scan 35, formula (13), is necessary. For the full envelope, requiring a common value in every finite quotient across all primes similarly cuts out the diagonal $\widehat{\Gamma} \subset \prod_p \widehat{\Gamma}$. An assertion of equality with all $\widehat{\mathbb{Z}}$-points would be false. □

E.5 Comparisons with the proposed hulls

E.18. Lead 3 does not identify the envelope with a pro-algebraic or Malcev hull [Refuted]

Let $P_{\mathbb{Q}}(\Gamma)$ denote the pro-algebraic group universal for all finite-dimensional rational representations of a discrete group Γ, and let $U_{\mathbb{Q}}(\Gamma)$ denote its unipotent completion. For finitely generated virtually abelian Γ, there are canonical quotient maps

$$P_{\mathbb{Q}}(\Gamma) \longrightarrow (E_\Gamma)_{\mathbb{Q}} \longrightarrow U_{\mathbb{Q}}(\Gamma), \tag{E.18.1}$$

defined by the natural maps from Γ. Both can be strict. For $\Gamma = \mathbb{Z}$,

$$(E_{\mathbb{Z}})_{\mathbb{Q}} \simeq \mathbb{G}_{a,\mathbb{Q}} \times \widehat{\mathbb{Z}}_{\mathbb{Q}}, \qquad U_{\mathbb{Q}}(\mathbb{Z}) = \mathbb{G}_{a,\mathbb{Q}}.$$

The character $n \mapsto 2^n$ belongs to the first theory and does not factor through $(E_{\mathbb{Z}})_{\mathbb{Q}}$. For $\Gamma = \mathfrak{S}$, the unipotent completion is trivial, but its envelope retains the quotient $\mathfrak{S}_3$.

Proof. Every representation of the envelope restricts to a quasi-unipotent representation by E.14. Conversely every such representation extends by the same theorem. This identifies its representation category as the full tensor subcategory of quasi-unipotent representations of Γ, giving the first map in (E.18.1) by the defining universal property of $P_{\mathbb{Q}}$. Its coordinate ring is generated by matrix coefficients of these representations, so the map is a pro-algebraic quotient. Unipotent representations are quasi-unipotent, giving the second map in exactly the same way.

At the n-stage for $\mathbb{Z}$, a point in residue class a has additive coordinate $a + nt$. Over $\mathbb{Q}$ this gives an isomorphism with $\mathbb{G}_a \times (\mathbb{Z}/n)$, and the transition preserves the first coordinate and reduces the second. Taking limits proves the displayed decomposition. Its finite-dimensional representations have only roots of unity as semisimple eigenvalues of the distinguished generator; 2 is not one.

For $\mathfrak{S}$, its finite abelianization gives $\mathrm{Hom}(\mathfrak{S}, \mathbb{G}_a(\mathbb{Q})) = 0$. If a homomorphism to a unipotent group had nontrivial Zariski closure, that nontrivial unipotent group in characteristic zero would have a nontrivial vector-group abelianization. Composing with a nonzero linear functional would contradict the displayed vanishing. This can also be seen directly by taking the first nonzero superdiagonal in an upper-unitriangular realization. Thus the unipotent completion is trivial. Reduction modulo 2 of the quotient H supplies the nontrivial finite quotient $\mathfrak{S}_3$ of the envelope.

The integral Hall model of the rank-one nilpotent lattice is $\mathbb{G}_{a,\mathbb{Z}}$; the Lazard model of its additive pro-p completion is the additive $\mathbb{Z}_p$-Lie lattice. On this abelian example there are no Baker–Campbell–Hausdorff denominators. The maps to these models are the first-stage projection $E_{\mathbb{Z}} \to \mathbb{G}_a$ and its base change or map on $\mathbb{Z}_p$-points. They are not isomorphisms: the envelope has every finite constant cyclic quotient, whereas $\mathbb{G}_a$ has no nontrivial morphism to a finite constant group over $\mathbb{Q}$. This is a separating example for the proposed identification with an integral Malcev or Lazard–Hall form, without pretending that those constructions are a single canonical integral functor on all virtually abelian groups.

There is also a direct coordinate-ring distinction from divided powers. In rank one the usual divided-power polynomial algebra is $D = \bigoplus_{j \geq 0} \mathbb{Z}\, t^j/j! \subset \mathbb{Q}[t]$. Its multiplication and coproduct are

$$t^{[i]}t^{[j]} = \binom{i+j}{i} t^{[i+j]}, \qquad \Delta(t^{[n]}) = \sum_{i+j=n} t^{[i]} \otimes t^{[j]}.$$

It is not $\mathcal{O}(E^0_{\mathbb{Z}}) = \mathbb{Z} + t\mathbb{Q}[t]$: the latter contains $t/2$, whereas the degree-one part of D is $\mathbb{Z}t$. Inclusion gives a specified morphism $E^0_{\mathbb{Z}} \to \operatorname{Spec} D$. Their characteristic-$p$ fibres also differ: the first is a point, while the class of t is nonzero in D/pD and has $t^p = 0$. Thus neither the coordinate ring of the vector group nor its familiar divided-power distribution algebra can silently be identified with the envelope's kernel. E.3 gives the exact role divided powers do play in the section-extension problem.

For Hain's completion relative to a finite quotient $\rho : \Gamma \to H$, with Γ virtually abelian, there is a canonical compatible map $(E_\Gamma)_{\mathbb{Q}} \to \mathcal{R}_\rho(\Gamma)$. Indeed every representation of the latter has the image of some power of each γ in a unipotent group; E.14 extends every finite-dimensional quotient representation, and uniqueness makes the extensions compatible. For general reductive quotients such a comparison can fail in both directions. Take $\Gamma = \mathbb{Z}$ and $\rho(n) = 2^n \in \mathbb{G}_m(\mathbb{Q})$.

Its relative completion is $\mathbb{G}_a \times \mathbb{G}_m$, with $n \mapsto (n, 2^n)$. To prove this, the closure of a cyclic image in any candidate extension is commutative. Multiplicative Jordan decomposition splits it into a diagonalizable part and a unipotent part. The former maps isomorphically onto $\mathbb{G}_m$, since its kernel must also be unipotent; the latter is generated by one logarithm, so is the image of $\mathbb{G}_a$. This gives the required unique map. Uniqueness can also be checked by density of $\{(n, 2^n) : n \in \mathbb{Z}\}$: in a nonzero Laurent polynomial $\sum_j P_j(n)2^{jn}$, the largest j with nonzero P_j dominates for positive integers n, so the expression cannot vanish identically.

There is no compatible map from $(E_{\mathbb{Z}})_{\mathbb{Q}}$ to this relative completion, because its multiplicative coordinate would extend the excluded character 2^n. There is no compatible map in the other direction either: the connected group $\mathbb{G}_a \times \mathbb{G}_m$ has no nontrivial morphism to a constant finite group, whereas the envelope's quotient $\mathbb{Z}/2$ is nontrivial on the distinguished discrete subgroup. Both receive the canonical quotient map from the full pro-algebraic completion. □

E.19. The relative completion with abelian kernel is the finite stage [Proved]

Let $\Gamma_0 = \mathrm{SL}_2(\mathbb{Z})$, with the specified quotient $\Gamma_0 \to H$ of order 12 and kernel Π_0. In characteristic zero, the universal pro-algebraic extension of H by a commutative unipotent group receiving this map is $(\underline{\mathfrak{S}}_M)_{\mathbb{Q}}$. Equivalently, if $\mathcal{R}$ is relative unipotent completion for $\Gamma_0 \to H$, and $\mathcal{U}$ its unipotent kernel, then

$$\mathcal{R}/\overline{[\mathcal{U},\mathcal{U}]} \simeq (\underline{\mathfrak{S}}_M)_{\mathbb{Q}}. \tag{E.19.1}$$

This group is not $(E_{\mathfrak{S}})_{\mathbb{Q}}$: its component group is the finite group H, while the latter has the infinite proconstant quotient $\widehat{\mathfrak{S}}$. The same statement holds for $\mathrm{GL}_2(\mathbb{Z})$ and its quotient of order 24, with the orientation-reversing action included.

Proof. We use the universal property of relative completion formulated in Hain (1998, §1, pp. 47–48); the abelian-kernel calculation can be proved directly. A map to an extension of H by a commutative unipotent group sends Π_0 to that commutative kernel, so it kills $[\Pi_0, \Pi_0]$. Over $\mathbb{Q}$, a commutative unipotent group is a vector group: finite logarithm and exponential identify it with its Lie algebra. The induced homomorphism from $M = \Pi_0^{\mathrm{ab}}$ therefore extends uniquely and linearly to $M \otimes \mathbb{G}_a$. The conjugation and cocycle identities extend by linearity. E.12 consequently extends the original map uniquely to $(\underline{\mathfrak{S}}_M)_{\mathbb{Q}}$.

Conversely that finite-stage group is itself an extension of H by a vector group and receives the prescribed map. Thus it has exactly the universal property of the quotient on the left of (E.19.1). This argument also works for inverse limits of vector groups, so it gives the pro-algebraic assertion. The difference of component groups proves non-isomorphism with the full envelope. The proof for the order-24 quotient is identical: its normal lattice is still Π_0^{ab}, but the additional generator $\tau' = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ conjugates ρ, σ to their inverses and sends $\lambda_0 \mapsto \lambda_1^{-1}$, $\lambda_1 \mapsto \lambda_0^{-1}$. Its action on M is $-S$. The program verifies these identities and the order $96/4 = 24$ of the corresponding quotient of $\mathrm{GL}_2(\mathbb{Z}/4)$. □

E.20. The remaining comparison with the named automorphism schemes [Open]

E.15–E.19 compute the group $\mathfrak{S}$, its finite-stage model, its full envelope, their fibres, their points, and the relative completion with abelian kernel. They do not identify either of these group schemes with all versions of $M_{\rho,\sigma}$ or $N_{\rho,\sigma}$ in §49–§51. E.24 below computes the fine parameter scheme of C4 scan 41 over $\mathbb{Z}$, and identifies the actual automorphism functor with both conjugator signs positive over $\mathbb{Q}$. E.25 disproves the unqualified assertion in scan 36 that the two generator images determine an automorphism over every base. The exact remaining task is to compare every signed version of $M_{\rho,\sigma}$ and $N_{\rho,\sigma}$ with the functor of automorphisms of $\underline{\mathfrak{S}}_M$ satisfying §51.2, conditions (15)–(23), specifying whether linearity on $\underline{M}$ is imposed, whether the conjugators and their signs are remembered, and which fppf quotient removes

redundant parameters. The group-scheme half of the After BISITE Fermat-tower follow-up also asks for comparisons relative to $(\mathbb{Z}/n)^2$ and $\mathfrak{S}_3 \times \widehat{\mathbb{Z}}^*$. The order-12/order-24 calculation in E.19 does not establish these comparisons.

The first still unproved identification with the named signed functors in the requested residue-characteristic list is $A = \mathbb{F}_2$, with both locally existing conjugators and the sign data included. We do not claim this case is open in the literature. The input data are fully explicit: the integral cocycle (E.16.2), the matrices R, S, and the rank-two vector group. Besides the finite quotient and cocycle checks in `E_envelope.py`, the program `E_automorphisms.py` computes the integral parameter scheme, its points in characteristics 2 and 3 and over $\mathbb{Z}/4, \mathbb{Z}/8, \mathbb{Z}/9$, all sign alternatives in scan 41, the generic automorphism composition, and the infinitesimal obstruction. The source itself introduces $\underline{Z}_{\rho,\sigma}$ as a subgroup of $\underline{\mathrm{Aut}}(\underline{\mathcal{G}}^+)$ in scan 36; replacing that automorphism construction by the underlying group $\underline{\mathfrak{S}}_M$ would change the question. We leave this comparison explicitly at that point.

For a precise first finite deck-quotient test, choose the free group $F_2 = \langle x, y\rangle$ and the surjection

$$q_2 : F_2 \longrightarrow (\mathbb{Z}/2)^2, \qquad x \longmapsto (1,0), \qquad y \longmapsto (0,1).$$

More generally let q_n be the same abelianization map modulo n, and let $\mathcal{R}_{q_n}(F_2)$ denote its rational relative completion over the constant finite group $((\mathbb{Z}/n)^2)_{\mathbb{Q}}$, in the universal-property sense of E.19. Neither a comparison of $\mathcal{R}_{q_2}(F_2)$, nor a comparison of its quotient obtained by abelianizing the unipotent kernel, with a specified version of $M_{\rho,\sigma,\mathbb{Q}}$ or $N_{\rho,\sigma,\mathbb{Q}}$ is proved here. The comparison morphism, compatibility with the chosen group map and deck quotient, and its image and kernel remain to be constructed. This is the first nontrivial n in this deck-quotient family; it is distinct from the signed $\mathbb{F}_2$-base question above.

For the target $\mathfrak{S}_3 \times \widehat{\mathbb{Z}}^*$, the corresponding group-scheme question must specify the actual tower fundamental group and its continuous quotient map, and either the proconstant target over $\mathbb{Q}$ or the compatible system of constant finite targets $\mathfrak{S}_3 \times (\mathbb{Z}/m)^*$. No such tower-source comparison is constructed here, and the map q_n is not asserted to provide it. The proved modular comparison, the finite-stage cocycle computations, and E.24–E.25 are the partial results for this group-scheme follow-up; no unrestricted identification of Fermat-tower completions is claimed.

E.6 The covering test and the two rational theories

E.21. The Stallings lead, with the precise completions [Proved]

Let $F = \langle x, y\rangle$ be the discrete free group, use $[x, y] = xyx^{-1}y^{-1}$, and define $f(x) = x$, $f(y) = y[x, y]$. For every finite c, f induces an automorphism of $F/\gamma_{c+1}F$. It induces continuous automorphisms of every pro-p completion and of the rational prounipotent completion. Nevertheless the action $x \mapsto (12), y \mapsto (13)$ on three letters is transitive, while its pullback along f has orbits $\{1, 2\}, \{3\}$.

Proof. The induced map on F^{ab} is the identity. The group $\gamma_i F/\gamma_{i+1}F$ is generated by iterated commutators of weight i in x, y, and replacing either generator by itself times a commutator changes such an expression only in weight at least $i + 1$. Thus f is the identity on each successive quotient. Induction in the central extensions

$$1 \to \gamma_c F/\gamma_{c+1}F \to F/\gamma_{c+1}F \to F/\gamma_c F \to 1$$

proves that it is an automorphism at every stage. The inverses are compatible; passing to rational unipotent versions of these nilpotent quotients and then taking limits gives the asserted

prounipotent automorphism. This proves the needed special case of the Stallings conclusion directly.

For a pro-p group generated by x, y, their images under f generate its Frattini quotient, since they agree there with x, y. They topologically generate the group: otherwise their closed subgroup would be contained in an open subgroup of index p, contradicting that assertion. Thus the induced endomorphism is surjective. A finitely generated profinite group is Hopfian. Indeed it has only finitely many open normal subgroups of any fixed index; inverse image under a surjective endomorphism is an injective self-map of each such finite set and hence a permutation. Its kernel must therefore lie in every open normal subgroup, and is trivial. This proves the pro-p assertion for every prime.

For $a = (12)$, $b = (13)$, direct permutation multiplication gives $b[a, b] = a$. Hence the pulled-back action has both generators acting by (12). The program computes the two orbit decompositions exactly. This also shows why neither the individual pro-p quotients nor the inverse system of nilpotent quotients can recover this particular finite action: $\mathfrak{S}_3$ is not nilpotent. No assertion is made here that every discrete feature of F is lost, or that this was Grothendieck's historical reason for introducing §50. □

E.22. The modular envelope detects the three-letter covering [Proved]

The quotient $\mathfrak{S} \to \mathfrak{S}_3$ induced by $\mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{F}_2)$ extends to both $\underline{\mathfrak{S}}_M$ and $E_{\mathfrak{S}}$ as a morphism to the constant finite $\mathbb{Z}$-group scheme $(\mathfrak{S}_3)_{\mathbb{Z}}$. Consequently these envelopes retain the associated three-sheeted covering. The unipotent completion of $\mathfrak{S}$ does not.

Proof. Both λ_0 and λ_1 are the identity modulo 2, so the reduction map kills Π_0 and factors through $H = \mathfrak{S}/M$. Its image is all of $\mathrm{SL}_2(\mathbb{F}_2) \simeq \mathfrak{S}_3$: the reductions of ρ and σ have orders 3 and 2 and generate the six matrices. Compose the component projection of the finite-stage model, or the proconstant projection of the envelope, with this quotient. The three-letter permutation representation therefore factors through either scheme. E.18 shows that the unipotent completion of $\mathfrak{S}$ is trivial, so this nontrivial action cannot factor through it.

For the free group in E.21, its free metabelian quotient in the modern sense also retains the action because $\mathfrak{S}_3'' = 1$. This does not identify that free metabelian quotient with an object to which §50 applies: it is not virtually abelian. Indeed it surjects onto the two-generated metabelian group $\mathbb{Z} \wr \mathbb{Z}$. In that group, if an abelian finite-index subgroup existed, it would contain a nonzero power of the shift and a nonzero power of the lamp at the origin; these do not commute, since the shift moves that lamp to a different position. A quotient of a virtually abelian group is virtually abelian, giving the contradiction. The manuscript's terminology and the reviewer's modern use of "metabelian" have different scopes. □

E.23. What the two rational theories retain on this example [Known]

For the pointed classifying space $B\mathfrak{S}$, fibrewise rationalization retains the discrete fundamental group $\mathfrak{S}$ and has no higher homotopy groups. Toën's schematization has fundamental group $P_{\mathbb{Q}}(\mathfrak{S})$. In this example it is $BP_{\mathbb{Q}}(\mathfrak{S})$. The envelope of §50 is instead the integral group scheme $E_{\mathfrak{S}}$, whose rational representations are exactly the quasi-unipotent rational representations of $\mathfrak{S}$. The full pro-algebraic completion is strictly larger, even for this particular group.

Proof. Rivera et al. (2021, Theorem 3; repeated as Theorem 16) identify fibrewise rational equivalences by the unchanged discrete π_1 and the rationalized higher homotopy groups. Applying that statement to the Eilenberg–Mac Lane space $B\mathfrak{S}$, whose higher homotopy groups vanish, gives the first assertion with no nilpotence assumption. The same conclusion follows

directly from its loop algebra $\mathbb{Q}[\mathfrak{S}]$ and its category of all derived local systems; here arbitrary, not only finite-dimensional, modules are allowed.

Toën (2006, Corollary 3.3.7) identify the fundamental group of schematization with the full pro-algebraic completion. For completeness we check that no higher schematic homotopy is needed for this example. A finitely generated free abelian group M is algebraically good over $\mathbb{Q}$: on each finite-dimensional representation, its group cohomology is computed by the Koszul complex of the commuting operators $g_i - 1$. After extending scalars to an algebraic closure, decompose into simultaneous generalized characters. If some character value is not 1, the corresponding operator is invertible and contracts the complex. On the unipotent block, $g_i - 1$ differs from $\log g_i$ by a commuting invertible factor, so the complex is the Lie-algebra cohomology complex of the additive vector group. These are exactly the cohomology complexes for the pro-algebraic completion of M. Explicitly, over $\overline{\mathbb{Q}}$ this completion is $\mathbb{G}_a^r \times D((\overline{\mathbb{Q}}^*)^r)$: simultaneous multiplicative Jordan decomposition gives the diagonalizable factor, and logarithms of the commuting unipotent parts give the additive factor. Its diagonalizable factor has exact invariants, by the direct grading decomposition, so precisely the trivial-character block remains in cohomology. Every finite collection of algebraic characters descends to a number field and then, by restriction of scalars, to a rational representation; hence this description applies to the base change of the rational completion. Equality of the cohomology comparisons descends by faithful scalar extension.

The finite extension $1 \to M \to \mathfrak{S} \to H \to 1$ gives the corresponding extension of pro-algebraic groups. One can construct it by the pushout of the cocycle of E.12 along $M \to P_{\mathbb{Q}}(M)$; density proves its universal property for all representations. In both the discrete and algebraic Hochschild–Serre spectral sequences, taking H-invariants is exact because $|H| = 12$ is invertible in $\mathbb{Q}$. The cohomology comparison for M therefore gives the comparison for $\mathfrak{S}$, for every finite dimensional local system. The criterion of Toën (2006, Lemma 3.3.2 and Definition 3.4.1) now gives $(B\mathfrak{S} \otimes \mathbb{Q})^{\mathrm{sch}} \simeq BP_{\mathbb{Q}}(\mathfrak{S})$.

Finally, induce the character $M \to \mathbb{Q}^*$, $\lambda_0 \mapsto 2$, $\lambda_1 \mapsto 1$, from the index-12 subgroup M to $\mathfrak{S}$. This gives a 12-dimensional rational representation. The eigenvalues of λ_0 include 2, so it is not quasi-unipotent and cannot factor through the envelope by E.14. The matrices, the defining group relations, and all twelve eigenvalues are computed in `E_envelope.py`. This proves strictness on the requested example, rather than importing the unrelated counterexample used in the earlier GALOIS package.

None of these comparisons says that the envelope kills a nontrivial element of $\mathfrak{S}$. That group is residually finite: an element outside M is detected in H, and a nonzero element of M is detected in some M/nM. Thus its canonical map to the envelope, and consequently to the full pro-algebraic completion, is injective on the discrete group. What differs is the universal representation problem and the resulting target, not an elementwise kernel on this particular input. □

E.7 The automorphism parameters in the first Fermat example

E.24. The fine parameter scheme and the rational automorphism group [Proved]

Put $e_0 = (1,0)^t$, $e_1 = (0,1)^t$, $A_0 = I - R$, and $B_0 = I - S$, with the integral action matrices of E.15. In §51.2, fols. 667–668, C4 scans 39–41, fix the reading supported by (21), (28), and (29):

$$\nu = 2\nu' + \nu_2, \qquad A_0\alpha - \nu_3 e_1 = B_0\beta + \nu e_1, \qquad \mu = 2\nu + 1 \in \mathbb{G}_m, \quad \nu_2, \nu_3 \in \{0,1\}. \tag{E.24.1}$$

For each of the four sign pairs, the resulting fine parameter scheme is the smooth relative three-dimensional scheme

$$\operatorname{Spec} \mathbb{Z}[a, b, k, (12k + 6\nu_2 + 4\nu_3 + 1)^{-1}]. \tag{E.24.2}$$

It is smooth also at 2 and 3. This assertion concerns the parameter scheme with conjugators retained.

For every $\mathbb{Q}$-algebra, the functor of automorphisms of $\underline{\mathfrak{S}}_M$ preserving $\underline{M}$, whose two conjugators lie in $\underline{M}$ and whose images satisfy the loop relation in (17), is represented by $\mathbb{G}_a \rtimes \mathbb{G}_m$. Its parameters and action are

$$\begin{aligned} &u_{t,\nu}(\rho) = (t, \nu - t)\rho, \qquad u_{t,\nu}(\sigma) = (t, -t)\sigma, \qquad \mu = 1 + 2\nu, \\ &u_{t,\nu}|_{\underline{M}} = \mu I, \qquad (t,\mu)(t',\mu') = (t + \mu t', \mu\mu'). \end{aligned} \tag{E.24.3}$$

This connected rational group differs from the finite-stage envelope, which has twelve geometric connected components.

Proof. Write $\alpha = (a_0, a_1)$, $\beta = (b_0, b_1)$, and $w = \nu'$. The two equations in (E.24.1) reduce to

$$b_0 = a_0 + a_1 + b_1, \qquad 3a_1 - 2w = \nu_2 + \nu_3 =: d.$$

The complete integral parametrization is

$$a_0 = a, \quad b_1 = b, \quad a_1 = d + 2k, \quad w = d + 3k, \quad b_0 = a + d + 2k + b;$$

its inverse has $k = w - a_1$. The unit to invert is $\mu = 12k + 6\nu_2 + 4\nu_3 + 1$, proving (E.24.2). The coefficient matrix on (α, β) has maximal-minor gcd 3, exactly as the marginal calculation on scan 41 suggests. After the ν' column is included, that gcd is 1. Thus the failure of surjectivity of the former linear map at 3 is not a singularity of the latter parameter scheme. At 2 and 3 the unit μ is automatically nonzero on every fibre; each fixed sign component has p^3 points. For $p > 3$ it has $p^2(p-1)$ points. For $\mathbb{Z}/q$, $q = 4, 8, 9$, each component has q^3 points. The program `E_automorphisms.py` verifies these counts by enumerating the pairs (a_1, w), with the two free coordinates included, for every requested prime up to 13 and each of these moduli.

We distinguish the other legible sign alternatives. Replacing the signs in the two formulas by $\nu = 2\nu' + s_2\nu_2$ and $A_0\alpha + s_3\nu_3 e_1 = B_0\beta + \nu e_1$, where $s_2, s_3 \in \{\pm 1\}$, changes only the integral constant in $3a_1 - 2\nu' = s_2\nu_2 - s_3\nu_3$. Every such parameter scheme remains an open subscheme of $\mathbb{A}^3_{\mathbb{Z}}$, but its fibres need not agree: when $s_3 = +1$ and $\nu_3 = 1$, its characteristic-three fibre is empty because $\mu \equiv 0$. The matrix $\tau = \operatorname{diag}(-1, 1)$ gives $\tau(\rho)\rho^{-1} = \lambda_1^{-1}$, $\tau(\sigma)\sigma^{-1} = \omega$, and acts as $-I$ on M. Its parameters $\alpha = \beta = 0$, $\nu_2 = \nu_3 = 1$, $\nu = -1$ agree with the minus sign in (28), and give a point also in characteristic three. This is mathematical evidence for that sign with these fixed conventions, not a final palaeographical decision. The signs $s_2 = \pm 1$ are related by translating ν' by ν_2; formula (21) fixes the plus convention. The alternative middle "=" in (32), if read as a chain of equalities, is not equivalent to (31): take $\alpha = \beta = e_0$, $\nu = \nu_2 = \nu_3 = 0$. Then $A_0\alpha = B_0\beta = (1, -1)^t \neq 0$, whereas their difference is zero. All four coherent sign alternatives are computed at $p = 2, 3, 5$.

Over a $\mathbb{Q}$-algebra, A_0 is invertible and $\operatorname{Im} B_0 = \{(t, -t)\}$. Consequently the conjugacy and loop conditions are exactly $\eta = (t, -t)$ and $\zeta = \eta + \nu e_1 = (t, \nu - t)$. Using the affine realization in E.16, direct multiplication of the images of ρ, σ gives

$$u(\rho)^3 = u(\sigma)^2 = \omega, \qquad u(\lambda_0) = \mu e_0, \qquad u(\lambda_1) = \mu e_1.$$

These equalities include their finite components. The first ones give the order relations, and the last two commute. Thus the presentation in E.15 gives a homomorphism on the discrete group. Its restriction to M extends by the linear map μI; that map commutes with R, S. The pushout construction E.12 therefore gives a group-scheme endomorphism. Conversely every additive polynomial over a $\mathbb{Q}$-algebra is linear: in the equation $f(x + y) = f(x) + f(y)$, the coefficient of $x^{n-1}y$ annihilates na_n, so $a_n = 0$ for $n > 1$. The images of the basis of

M therefore determine its restriction to $\underline{M}$, and the finite component representatives then determine the whole homomorphism.

Composition on the generator images gives $t'' = t + \mu t'$ and $\nu'' = \nu + \mu\nu'$, whence $\mu'' = \mu\mu'$. The inverse has parameters $(-t/\mu, -\nu/\mu)$. This proves all functorial assertions in (E.24.3), as well as the precise difference from the underlying finite-stage group scheme. The symbolic relations and the composition identity are checked in the same program. □

E.25. Generator images do not determine the integral automorphism functor [Refuted]

The assertion in §51.2, fol. 665 verso, C4 scan 36, formula (16), that an automorphism is determined by its values on ρ, σ is false for the finite-stage model over all bases. For every prime p, let $A = \mathbb{F}_p[\epsilon]/(\epsilon^2)$. In the integral cocycle coordinates of E.16, the formula

$$(v, h) \longmapsto (T(v), h), \qquad T(v) = v + \epsilon\big(v^{[p]} - v\big) \tag{E.25.1}$$

defines a nonidentity automorphism of $(\underline{\mathfrak{S}}_M)_A$ which fixes every element of the discrete group $\mathfrak{S}$, and preserves the loop subgroup $\mathbb{G}_a\lambda_0$. Here $v^{[p]}$ means coordinatewise Frobenius. In particular it satisfies the positive-sign existential conjugator conditions with $\alpha = \beta = 0$, yet is invisible to the two-image parameters. The smallest example in this family is the four-element ring $\mathbb{F}_2[\epsilon]/(\epsilon^2)$. Requiring a scalar action on the entire loop subgroup would exclude this example, but is stronger than the manuscript's condition that this subgroup be preserved.

Moreover the proposed $\varepsilon_3(u)$ in scan 37, fol. 666, formula (18), need not be determined by u even over $\mathbb{Q}$. In the order-24 ambient model, the identity automorphism has both the conjugator $\alpha = 1$ and the oppositely oriented conjugator

$$\alpha = (-1/3, 1/3)\tau.$$

Consequently retaining the conjugator sign defines additional parameter data, rather than a character on the unqualified automorphism subgroup.

A correct replacement over an arbitrary base is to impose that the automorphism preserve $\underline{M}$ and restrict to an A-linear map there. Within that subfunctor the two generator images do determine the automorphism. Over $\mathbb{Q}$-algebras linearity is automatic, as in E.24.

Proof. In characteristic p, T is additive. The polynomial $U(x) = x - \epsilon(x^p - x)$ is its inverse: expanding $T(U(x))$ and $U(T(x))$ uses $\epsilon^2 = 0$ and $\epsilon^p = 0$, and gives x in both cases. It is nonidentity because the coefficient of x^p is the nonzero element ϵ. It commutes with every matrix over $\mathbb{F}_p$, since Frobenius fixes their entries. In particular it commutes with every action matrix A_h, and it fixes every integral cocycle value $c(h, k)$, whose coordinates reduce to $\mathbb{F}_p$. Applying it to

$$(v, h)(w, k) = (v + A_h w + c(h, k), hk)$$

therefore proves the group identity for (E.25.1) as a polynomial identity over A. Each discrete lattice coordinate also lies in the prime field and is fixed. The coordinate line generated by e_0 is preserved by both T and U, proving the loop-subgroup assertion. The program reduces symbolic polynomials modulo (p, ϵ^2), checks additivity, both inverse identities, all twelve action matrices, and all 144 cocycle values, for $p = 2, 3, 5$. These are polynomial checks, so the counterexample does not rely on equality only on the finite set of A-valued points.

For the sign assertion, E.24 gives $\tau\rho\tau^{-1}\rho^{-1} = \lambda_1^{-1}$. If $\alpha = \alpha_0\tau$, its commutator with ρ is therefore $(I - R)\alpha_0 - e_1$ in the additive normal subgroup. Taking $\alpha_0 = (-1/3, 1/3)$ makes this zero. The orientation of $\alpha_0\tau$ is negative, whereas the orientation of the identity conjugator is positive. Both are allowable conjugators in $\{1, \tau\}\underline{M}$ and give the identical value $u(\rho) = \rho$. The exact matrix identity and rational vector computation are checked in `E_automorphisms.py`.

For the replacement, agreement on ρ, σ implies agreement on their words λ_0, λ_1. Two A-linear maps on $\underline{M}$ agreeing on its basis agree identically. Each component of the model is a translate of $\underline{M}$ by a word in ρ, σ, so the two morphisms agree on every component. This proves the replacement without reducedness or characteristic restrictions. It also explains why finite-stage automorphisms and the full envelope's universal property must be kept separate: the further transition indexed by pM kills the nonconstant correction in (E.25.1), in agreement with the colimit uniqueness of E.14. □

References

Bhargava, M. (2009). On P-orderings, rings of integer-valued polynomials, and ultrametric analysis. *Journal of the American Mathematical Society*, 22(4):963–993. verified: supplied PDF opened; Section 1, especially p. 964, checked. The elementary integer-valued-polynomial case used here is proved in E.8.

Brylinski, J.-L. and Deligne, P. (2001). Central extensions of reductive groups by K_2. *Publications mathématiques de l'IHÉS*, 94:5–85. verified: attached PDF opened; Introduction and Theorem 7.2 read. Used to distinguish its central-extension problem from the noncentral kernel of the literal scan-336 presentation.

Conrad, K. (n.d.). $\mathrm{SL}_2(\mathbb{Z})$. verified: attached PDF opened and text read, including Theorems 1.1, 3.2 and C.1; author-hosted PDF also opened on 12 November 2026. The manuscript normalization is explicitly reduced to these statements.

Farb, B. and Margalit, D. (2012). *A Primer on Mapping Class Groups*, volume 49 of *Princeton Mathematical Series*. Princeton University Press. verified: primary book text, version 5.0, opened independently; Sections 2.2.1–2.2.2, Proposition 2.3, Theorem 2.5 and the following once-punctured-torus paragraph, and Section 8.1 checked (version-5.0 pages 51–57 and 230–231). Author's bibliography independently confirms the 2012 print edition.

Grothendieck, A. (1981). La longue marche à travers la théorie de Galois. Verified against the supplied manuscript scans and checked transcriptions, cote 140-3, sections 48–49. Part D uses scans 479–481, 489, 546, 596–611, 629–634, and 652. Unpublished manuscript; dual pagination is retained in the discussion.

Hain, R. and Matsumoto, M. (2003). Weighted completion of Galois groups and Galois actions on the fundamental group of $\mathbb{P}^1 - \{0, 1, \infty\}$. *Compositio Mathematica*, 139(2):119–167. Verified: supplied arXiv:math/0006158v2 text opened, Theorem A.6, Propositions 8.2–8.3, and Theorem 8.4 read. The cited numbering refers to that version. Journal metadata verified on the Cambridge University Press article page. The BibTeX key retains the 2001 preprint-version year.

Hain, R. and Matsumoto, M. (2020). Universal mixed elliptic motives. *Journal of the Institute of Mathematics of Jussieu*, 19:663–766. Verified: arXiv:1512.03975v4 (December 22, 2017), Definition 6.1, Example 6.6, Example 6.8, Theorem 6.9, and Corollary 6.10 opened and read. The cited numbering is that version. Journal metadata verified on the arXiv abstract page.

Hain, R. M. (1998). The Hodge de Rham theory of relative Malcev completion. *Annales scientifiques de l'École Normale Supérieure*, 31(1):47–92. verified: original article opened; the universal property in Section 1, pp. 47–48, was checked. E.19 proves the abelian-kernel calculation directly; no de Rham or Hodge assertion is imported.

Herfort, W. and Ribes, L. (1984). On free pro-$\mathcal{C}$-products of pro-$\mathcal{C}$-groups. *Anzeiger der Österreichischen Akademie der Wissenschaften, Mathematisch-Naturwissenschaftliche Klasse*, 121:31–34. Theorems 1 and 2, p. 31, with hypotheses and proof outline on pp. 31–34. Primary text verified directly; theorem reductions in B.16.

Milne, J. S. (2020). Class field theory. Version 4.03, August 6, 2020. Verified: supplied PDF opened and text read; Chapter I, Theorem 4.16 (global Kronecker–Weber), Chapter III, Theorem 4.4 (Hilbert symbol, especially the norm criterion).

Rivera, M., Wierstra, F., and Zeinalian, M. (2021). Rational homotopy equivalences and singular chains. *Algebraic & Geometric Topology*, 21(3):1535–1552. verified: author-hosted revised article and arXiv manuscript opened; Theorem 3, repeated with proof as Theorem 16, and Corollary 21 checked. Reduction to the classifying space in this paper is supplied in E.23.

Schneps, L. (2012). An introduction to profinite Grothendieck–Teichmüller theory: Lecture 3a, braided tensor categories and geometry. MIT, November 5–9, 2012. Verified: author's lecture PDF opened; page 19 gives precisely the conventions for relations (I), (II), and (III) used in D.5. Authorship and lecture-series metadata checked on the author's MIT lecture page.

Serre, J.-P. (1969). Algèbre et géométrie: Cours de 1968–1969. Collège de France, annual course report. verified: official five-page PDF opened; sections 1.1, 1.2 and 2.2 read. Section 1.2 is the precise tree-amalgam theorem used; the lattice stabilizers and arithmetic reduction for Z[1/p] are proved in C.9.

Toën, B. (2006). Champs affines. *Selecta Mathematica, New Series*, 12(1):39–134. verified: April 2006 arXiv version 6 and author-hosted text opened; Lemma 3.3.2, Theorem 3.3.4, Corollary 3.3.7, Definition 3.4.1 and its following examples checked. Publisher page independently confirms pages 39–134; E.23 supplies the algebraic-goodness reduction for the specified virtually abelian group.